STABILITY OF THE FABER-KRAHN INEQUALITY FOR BERGMAN SPACE AND LOG-SUBHARMONIC FUNCTIONS

JAIME GÓMEZ, DAVID KALAJ, PETAR MELENTIJEVIĆ, AND JOÃO P. G. RAMOS

Abstract. We prove quantitative stability results for sharp hyperbolic concentration (Faber–Krahn type) inequalities in Bergman-type settings associated with the automorphism group of the unit ball. In the weighted $\mathcal{M}$–Bergman monoid $\mathbf{B}^2_\alpha$ (based on the invariant measure $d\tau$ and the radial weight Φ_n^α), we show that if a pair (g,Ω) nearly saturates the sharp inequality, then g must be close (in normalized Bergman norm) to a single extremal profile $c\,g_a$ generated by ball automorphisms. More precisely, the distance to the extremal manifold is bounded by an explicit power of the deficit $\delta(g;\Omega)$ multiplied by an explicit amplification factor depending only on $\tau(\Omega)$, with rate

$$\big(C(\tau(\Omega))\,\delta(g;\Omega)\big)^{1/(n+2)}.$$

In the planar case $n=2$ this yields the exponent $1/4$ for log-subharmonic data; we believe this exponent is not optimal, and the evidence in Section 3.3 points instead to the sharp exponent $1/2$.

In contrast, in a genuinely holomorphic vector-valued Bergman setting on the disk we obtain the sharp stability exponent $1/2$ with constants independent of the target dimension. This shows that the loss from $1/2$ to $1/4$ reflects the larger log-subharmonic class rather than the hyperbolic geometry. We also establish analogous stability statements in Gaussian phase space (Fock-type settings), including higher-dimensional versions with the natural exponent $1/(m+2)$. Our approach combines hyperbolic rearrangements, differential inequalities for decreasing rearrangements, and quantitative reproducing-kernel/RKHS estimates that control the distance to extremizers through the peak size of suitable normalized densities.

1. Introduction

1.1. **Rigidity and stability for concentration inequalities.** Sharp concentration inequalities in time–frequency analysis and complex function spaces typically exhibit a strong rigidity phenomenon: equality is attained by a low-dimensional family of extremizers obtained from a single profile by the symmetries of the ambient geometry, and *near*-equality forces the data to lie quantitatively close to that extremal manifold. This principle has been developed in the Euclidean phase-space setting for the short-time Fourier transform and related transforms, and in holomorphic Bergman-type spaces; see, for instance, [5, 4] and the references therein.

The guiding point of this paper is that the stability exponent depends strongly on the analytic structure of the admissible class. In the holomorphic regime one often expects the sharp exponent $1/2$, while for larger classes such as log-subharmonic functions the best possible exponent may drop. We illustrate this dichotomy in two complementary settings: (i) vector-valued holomorphic Bergman data on the unit disk, where we recover the sharp $1/2$ stability rate (uniformly in the target dimension), and (ii) log-subharmonic/log-$\mathcal{M}$-subharmonic data in hyperbolic geometries, where we obtain the exponent $1/(n+2)$ (in particular $1/4$ in the planar case), which is the exponent delivered by the present rearrangement-and-peak argument. In the planar log-subharmonic case we identify below a sharpness mechanism suggesting that the optimal exponent should be $1/2$.

1.2. **Vector-valued holomorphic stability in the disk.** We begin with the vector-valued holomorphic case on the unit disk $\mathbb{D}$. Assuming the scalar stability estimates in $\mathbf{B}^2_\alpha(\mathbb{D})$ proved in [4], we show that the *same* constants yield stability for $\mathbb{C}^N$-valued holomorphic functions in $\mathbf{B}^2_\alpha(\mathbb{D};\mathbb{C}^N)$, uniformly in N (Theorem 2.2). In particular, the sharp exponent $1/2$ persists in the vector-valued setting: small deficit implies closeness to an extremizer of the form $\mathbf{f}_\omega(\cdot)\,v$ with rate $\delta^{1/2}$. We also derive two useful corollaries with the same stability estimate: a modulus stability statement for $|f|$, and a "square-sum" stability statement for Hilbert envelopes $G(z)=\big(\sum_{j=1}^N |f_j(z)|^2\big)^{1/2}$.

The vector-valued proof follows the strategy of [4]: a rearrangement reduction at the level of the scalar density $\|F\|^2_{\mathbb{C}^N}$, combined with a quantitative reproducing-kernel (RKHS) argument to pass from peak control to norm distance.

1.3. **Log-subharmonic stability in hyperbolic geometry.** We then turn to stability in hyperbolic (ball) geometry for a larger class of functions than holomorphic ones. We work on the unit ball $\mathbb{B}^n$ endowed with its hyperbolic structure and with the conformal automorphism group $\mathcal{M}$. Following [6, 7], we consider the monoid

of real-analytic functions whose logarithm is $\mathcal{M}$-subharmonic, and the associated weighted $\mathcal{M}$–Bergman spaces $\mathbf{B}^p_\alpha$ defined using the invariant measure $d\tau$ and the radial weight Φ^α_n. In this setting one has a sharp Faber–Krahn type inequality: among all measurable sets of fixed hyperbolic measure, hyperbolic balls maximize the normalized concentration functional, and the extremal functions are generated by the action of $\mathcal{M}$ on a single radial profile [13, 7].

Theorem 3.3 quantifies the corresponding rigidity. Let $\Omega \subset \mathbb{B}^n$ be measurable with $0 < \tau(\Omega) < \infty$, and let $g \in \mathbf{B}^2_\alpha \setminus \{0\}$ be log-$\mathcal{M}$-subharmonic. Writing $\delta(g;\Omega)$ for the deficit in the sharp concentration inequality, we prove a power-law stability estimate

$$\inf_{\substack{|c|=\|g\|_{2,\alpha}\\ |a|<1}} \frac{\|g - cg_a\|_{2,\alpha}}{\|g\|_{2,\alpha}} \lesssim \Big(C(\tau(\Omega))\,\delta(g;\Omega)\Big)^{\frac{1}{n+2}},$$

where $C(\tau(\Omega)) = (1-\theta(\tau(\Omega)))^{-1}$ is an explicit amplification factor depending only on the hyperbolic size of Ω. In particular, in the planar case $n = 2$ we obtain the exponent $1/4$ (Theorem 3.4). Unlike the holomorphic case, we do not know how to improve $1/4$ for log-subharmonic data, and we believe this exponent is sharp.

1.4. **Gaussian analogues in Fock space.** Beyond the hyperbolic ball, we prove stability analogues in Gaussian phase space. For real-analytic log-subharmonic functions in the complex Fock space we obtain an inequality of the form

$$\min_{\substack{c=\|F\|_{\mathcal{F}^2}\\ z_0\in\mathbb{C}}} \frac{\|F - cg_{z_0}\|_{\mathcal{F}^2}}{\|F\|_{\mathcal{F}^2}} \lesssim \big(e^{|\Omega|}\,\delta(F;\Omega)\big)^{1/4},$$

and we extend this to $\mathbb{R}^m$ with exponent $1/(m+2)$. The proofs follow the rearrangement-and-peak-control strategy, combined with Gaussian mean-value estimates for log-subharmonic functions (Lemma 4.3 and Lemma 4.4), in the spirit of [5].

Remark 1.1 (Background and related results)**.** *The terminology "Faber–Krahn type inequality" used in this paper is motivated by the classical principle that balls are extremal for spectral, isoperimetric, and concentration-type problems. In spaces of constant negative curvature, sharp isoperimetric inequalities go back to Schmidt's work on hyperbolic space* [1]*. Quantitative stability versions of hyperbolic isoperimetric inequalities were later studied, for example, by Bögelein, Duzaar and Scheven* [3]*. Although the inequalities considered here are functional concentration inequalities rather than perimeter inequalities, the same geometric principle is present: extremal sets are hyperbolic balls, and extremal functions are generated from a canonical profile by the symmetry group.*

The proof also relies on rearrangement ideas, such as the Hardy–Littlewood principle, decreasing rearrangements, and comparison of superlevel sets. Standard background on symmetrization and rearrangement methods may be found in Baernstein's monograph [2]*; for general analytic inequalities and functional-analytic background we also refer to Lieb and Loss* [10]*. In the Gaussian and Fock-space parts, the relevant rearrangement estimates are closely related to monotonicity principles for subharmonic functions. In particular, the differential inequalities for decreasing rearrangements used here are in the spirit of the monotonicity theorem of Kulikov, Nicola, Ortega-Cerdà and Tilli* [9]*. These principles appear, for instance, in inequalities of the form*

$$(u^*)'(s) + u^*(s) \geq 0$$

in the planar Gaussian setting, and in their higher-dimensional analogues.

Finally, the Bergman-space part of the paper is connected with recent stability results for sharp concentration inequalities in holomorphic function spaces. Besides the uniform stability results from [4]*, related sharp stability phenomena for convex functionals on weighted Bergman spaces were studied by Melentijević* [11]*. These results form part of the holomorphic background for the contrast emphasized in the present work: in the holomorphic Bergman setting one expects the sharp deficit exponent* $1/2$*, whereas in the larger log-subharmonic class the exponent obtained here is* $1/(n+2)$*, and in particular* $1/4$ *in the planar case.*

1.5. **Ideas of the proof and organization.** Across the different settings, the arguments share three main ingredients. First, we reduce arbitrary sets to extremal (superlevel) sets via Hardy–Littlewood rearrangement. Second, we convert deficit control into a quantitative bound on the peak value of a normalized density (hyperbolic: $u = \Phi^\alpha_n|g|^2$, Gaussian: $u = |F|^2 e^{-\pi|\cdot|^2}$), using a clipped comparison functional and, in higher dimension, differential inequalities for decreasing rearrangements. Third, we pass from peak control to distance-to-extremals by a quantitative reproducing-kernel/RKHS estimate (cf. [8]).

The paper is organized as follows. In Section 2 we treat the vector-valued holomorphic Bergman case in the unit disk, proving stability with the sharp exponent 1/2 and deriving modulus and square-sum corollaries, following the approach of [4]. In Section 3 we establish stability for log-subharmonic and log-$\mathcal{M}$-subharmonic data in hyperbolic geometry, proving Theorem 3.3 and the explicit planar 1/4 estimate. Section 4 develops the Fock-space analogues in $\mathbb{C}$ and $\mathbb{R}^m$.

2. Stability of vector-valued holomorphic functions

Fix $\alpha > -1$. We write

$$dA_\alpha(z) := (1-|z|^2)^\alpha\, dz, \qquad d\mu(z) := (1-|z|^2)^{-2}\, dz, \qquad s := \mu(\Omega) \in (0,\infty).$$

Let $\mathbf{B}^2_\alpha(\mathbb{D})$ be the weighted Bergman space with norm $\|f\|^2_{\mathbf{B}^2_\alpha} := \int_{\mathbb{D}} |f|^2\, dA_\alpha$. For $\omega \in \mathbb{D}$, let the normalized Bergman extremals be

$$\mathbf{f}_\omega(z) = \sqrt{\frac{\alpha+1}{\pi}}\, \frac{(1-|\omega|^2)^{\frac{\alpha+2}{2}}}{(1-\overline{\omega}z)^{\alpha+2}}, \qquad \|\mathbf{f}_\omega\|_{\mathbf{B}^2_\alpha} = 1.$$

We also recall the sharp concentration profile (for hyperbolic balls of μ-measure s)

$$\theta_\alpha(s) := 1 - \Big(1 + \frac{s}{\pi}\Big)^{-(\alpha+1)}.$$

Given $f \in \mathbf{B}^2_\alpha(\mathbb{D})$ and $\Omega \subset \mathbb{D}$ with $\mu(\Omega) = s$, define the deficit

$$\delta(f;\Omega,\alpha) := 1 - \frac{\int_\Omega |f(z)|^2\, dA_\alpha(z)}{\|f\|^2_{\mathbf{B}^2_\alpha}\, \theta_\alpha(s)} \in [0,1]. \tag{2.1}$$

(For the vector-valued extension below, we replace $|f|^2$ by $\|F\|^2_{\mathbb{C}^N}$.)

We shall use the following explicit stability result, proved in our setting in [4, Theorem 1.2].

Theorem 2.1 (Bergman stability)**.** *Let $\alpha > -1$, let $\Omega \subset \mathbb{D}$ satisfy $\mu(\Omega) = s > 0$, and let $f \in \mathbf{B}^2_\alpha(\mathbb{D})$. Then there exists an absolute, computable constant $C > 0$ such that*

$$\inf_{\substack{|c| = \|f\|_{\mathbf{B}^2_\alpha} \\ \omega \in \mathbb{D}}} \frac{\|f - c\,\mathbf{f}_\omega\|_{\mathbf{B}^2_\alpha}}{\|f\|_{\mathbf{B}^2_\alpha}} \le M(s,\alpha)\, \delta(f;\Omega,\alpha)^{1/2}, \tag{2.2}$$

where

$$M(s,\alpha) := C\left(1 + \frac{\alpha+2}{\alpha+1}\left[\Big(1+\frac{s}{\pi}\Big)^{\alpha+1} - 1\right]\right)^{1/2}, \tag{2.3}$$

and where $\mathbf{f}_\omega$ is the normalized reproducing kernel of $\mathbf{B}^2_\alpha(\mathbb{D})$, given explicitly by

$$\mathbf{f}_\omega(z) = \sqrt{\frac{\alpha+1}{\pi}}\, \frac{(1-|\omega|^2)^{\frac{\alpha+2}{2}}}{(1-\overline{\omega}z)^{\alpha+2}}.$$

Moreover, there exists an explicit constant $K(s,\alpha) > 0$ such that

$$\mathcal{A}_{\mathbb{D}}(\Omega) \le K(s,\alpha)\, \delta(f;\Omega,\alpha)^{1/2}. \tag{2.4}$$

Vector-valued extension (kernel times a vector). For $N \in \mathbb{N}$, define the vector-valued Bergman space $\mathbf{B}^2_\alpha(\mathbb{D};\mathbb{C}^N)$ with norm $\|F\|^2_{\mathbf{B}^2_\alpha} := \int_{\mathbb{D}} \|F(z)\|^2_{\mathbb{C}^N}\, dA_\alpha(z)$. Its reproducing kernel is the scalar Bergman kernel times I_N, so the extremals are exactly the functions $z \mapsto \mathbf{f}_\omega(z)\, v$ with $\omega \in \mathbb{D}$ and $v \in \mathbb{C}^N$.

Define the vector deficit (depending only on $\|F\|^2$) by

$$\delta(F;\Omega,\alpha) := 1 - \frac{\int_\Omega \|F(z)\|^2_{\mathbb{C}^N}\, dA_\alpha(z)}{\|F\|^2_{\mathbf{B}^2_\alpha}\, \theta_\alpha(s)}. \tag{2.5}$$

Theorem 2.2 (Vector-valued stability: same constants)**.** *For every $N \in \mathbb{N}$, every $F \in \mathbf{B}^2_\alpha(\mathbb{D};\mathbb{C}^N)$, and every $\Omega \subset \mathbb{D}$ with $\mu(\Omega) = s$,*

$$\inf_{\substack{v \in \mathbb{C}^N,\ \omega \in \mathbb{D} \\ \|v\|_{\mathbb{C}^N} = \|F\|_{\mathbf{B}^2_\alpha}}} \frac{\|F - \mathbf{f}_\omega\, v\|_{\mathbf{B}^2_\alpha}}{\|F\|_{\mathbf{B}^2_\alpha}} \le \mathsf{M}(s,\alpha)\, \delta(F;\Omega,\alpha)^{1/2}, \tag{2.6}$$

$$\mathcal{A}_{\mathbb{D}}(\Omega) \le \mathsf{K}(s,\alpha)\, \delta(F;\Omega,\alpha)^{1/2}. \tag{2.7}$$

In particular, the constants are independent of N.

Proof. By homogeneity we may assume $F \not\equiv 0$ and normalize

$$\|F\|_{\mathbf{B}^2_\alpha} = 1.$$

All estimates below are invariant under the scaling $F \mapsto \lambda F$, and we rescale at the end.

Define

$$u_F(z) := \|F(z)\|^2_{\mathbb{C}^N}(1-|z|^2)^{\alpha+2}, \qquad \rho_F(t) := \mu(\{z \in \mathbb{D} : \ u_F(z) > t\}),$$

and set

$$a_0^2 := \frac{\pi}{\alpha+1} \max_{z\in\mathbb{D}} u_F(z) \in (0,1], \qquad \delta_0^2 := 1 - a_0^2.$$

Choose $\omega \in \mathbb{D}$ such that $u_F(\omega) = \max_{\mathbb{D}} u_F$. Since

$$K(\omega,\omega) = \frac{\alpha+1}{\pi}(1-|\omega|^2)^{-(\alpha+2)},$$

we obtain

$$\|F(\omega)\|^2_{\mathbb{C}^N} K(\omega,\omega)^{-1} = \frac{\pi}{\alpha+1} u_F(\omega) = a_0^2, \tag{2.8}$$

and therefore

$$\Delta_F(\omega) := 1 - \|F(\omega)\|^2_{\mathbb{C}^N} K(\omega,\omega)^{-1} = 1 - a_0^2 = \delta_0^2. \tag{2.9}$$

We first establish the set estimate (2.7). The scalar set-stability argument depends only on the distribution function of the nonnegative quantity

$$u(z) = |f(z)|^2(1-|z|^2)^{\alpha+2}$$

and on the associated rearrangement inequalities over sets of fixed μ-measure. In the present vector-valued setting, the concentration functional and the deficit depend on F only through the scalar function u_F. Thus the same rearrangement argument applies verbatim with u replaced by u_F, and yields

$$\mathcal{A}_{\mathbb{D}}(\Omega) \le \mathsf{K}(s,\alpha)\, \delta(F;\Omega,\alpha)^{1/2}.$$

In particular, the constant $\mathsf{K}(s,\alpha)$ is independent of N.

Next we control the peak value of u_F in terms of the deficit. By Lemma 2.8,

$$1 - a_0^2 \le M_\alpha(s)\, \delta(F;\Omega,\alpha). \tag{2.10}$$

The proof of that lemma is the same as in the scalar case, with

$$u(z) = |f(z)|^2(1-|z|^2)^{\alpha+2}$$

replaced by

$$u_F(z) = \|F(z)\|^2_{\mathbb{C}^N}(1-|z|^2)^{\alpha+2}.$$

It uses only the superlevel-set estimate from Lemma 2.5, the rearrangement-gap lower bound from Lemma 2.7, and the ratio-monotonicity coming from the log-subharmonicity of u_F; the latter holds because $\log \|F\|_{\mathbb{C}^N}$ is subharmonic. Hence the function $M_\alpha(s)$ is also dimension-free.

We now pass from the pointwise deficit at ω to a norm-distance estimate. Applying Lemma 2.4 to the RKHS $\mathcal{H} = \mathbf{B}^2_\alpha(\mathbb{D};\mathbb{C}^N)$ at the point ω, and using (2.9), we find a unit vector $e \in \mathbb{C}^N$ and a scalar $c \in \mathbb{C}$ with $|c| = \|F\|_{\mathbf{B}^2_\alpha} = 1$ such that

$$\|F - c\,\mathbf{f}_\omega e\|_{\mathbf{B}^2_\alpha} \le \sqrt{2}\,\Delta_F(\omega)^{1/2} = \sqrt{2}\,(1-a_0^2)^{1/2}.$$

Combining this with (2.10) gives

$$\inf_{\substack{\|v\|_{\mathbb{C}^N}=1 \\ \omega\in\mathbb{D}}} \|F - \mathbf{f}_\omega v\|_{\mathbf{B}^2_\alpha} \le \sqrt{2}\, M_\alpha(s)^{1/2}\, \delta(F;\Omega,\alpha)^{1/2}.$$

Thus, with

$$\mathsf{M}(s,\alpha) := \sqrt{2}\, M_\alpha(s)^{1/2},$$

we obtain (2.6) in the normalized case $\|F\|_{\mathbf{B}^2_\alpha} = 1$.

Finally, rescaling from $F/\|F\|_{\mathbf{B}^2_\alpha}$ back to F yields the stated infimum condition

$$\|v\|_{\mathbb{C}^N} = \|F\|_{\mathbf{B}^2_\alpha},$$

and proves (2.6) in full generality. Since all constants involved come from Lemmas 2.5, 2.7, 2.8, and 2.4, all of which are dimension-free, the constants $\mathsf{M}(s,\alpha)$ and $\mathsf{K}(s,\alpha)$ are independent of N. □

Corollary 2.3. *Fix $\alpha > -1$ and $\Omega \subset \mathbb{D}$ with $\mu(\Omega) = s > 0$.*

(A) Modulus stability (scalar). *For every $f \in \mathbf{B}^2_\alpha(\mathbb{D})$,*

$$\inf_{\substack{c>0\\ \omega\in\mathbb{D}}} \frac{\big\||f| - c\,|\mathbf{f}_\omega|\big\|_{\mathbf{B}^2_\alpha}}{\|f\|_{\mathbf{B}^2_\alpha}} \;\le\; \mathsf{M}(s,\alpha)\,\delta(f;\Omega,\alpha)^{1/2}. \tag{2.11}$$

Moreover, $\mathcal{A}_{\mathbb{D}}(\Omega) \le \mathsf{K}(s,\alpha)\,\delta(f;\Omega,\alpha)^{1/2}$.

(B) Square-sum stability (Hilbert envelope). *Let $N \in \mathbb{N}$, $f_1, \dots, f_N \in \mathbf{B}^2_\alpha(\mathbb{D})$, and set*

$$G(z) := \Big(\sum_{j=1}^{N} |f_j(z)|^2\Big)^{1/2}.$$

Let $F := (f_1, \dots, f_N) \in \mathbf{B}^2_\alpha(\mathbb{D};\mathbb{C}^N)$, so that $G(z) = \|F(z)\|_{\mathbb{C}^N}$. Then

$$\inf_{\substack{c>0\\ \omega\in\mathbb{D}}} \frac{\|G - c\,|\mathbf{f}_\omega|\|_{\mathbf{B}^2_\alpha}}{\|G\|_{\mathbf{B}^2_\alpha}} \;\le\; \mathsf{M}(s,\alpha)\,\delta(F;\Omega,\alpha)^{1/2}, \tag{2.12}$$

and $\mathcal{A}_{\mathbb{D}}(\Omega) \le \mathsf{K}(s,\alpha)\,\delta(F;\Omega,\alpha)^{1/2}$.

Proof. **(A)** For any $c \in \mathbb{C}$ and $\omega \in \mathbb{D}$ we have the pointwise inequality

$$\big||f(z)| - |c|\,|\mathbf{f}_\omega(z)|\big| \le |f(z) - c\,\mathbf{f}_\omega(z)|.$$

Integrating against dA_α yields $\||f| - |c||\mathbf{f}_\omega|\|_{\mathbf{B}^2_\alpha} \le \|f - c\mathbf{f}_\omega\|_{\mathbf{B}^2_\alpha}$. Taking the infimum over c, ω and using (2.2) gives (2.11). The set bound is (2.4).

(B) Let $F = (f_1, \dots, f_N)$ and $G(z) = \|F(z)\|_{\mathbb{C}^N}$. Fix $\omega \in \mathbb{D}$, $c > 0$ and a unit vector $e \in \mathbb{C}^N$. By the reverse triangle inequality in $\mathbb{C}^N$,

$$\big|\|F(z)\|_{\mathbb{C}^N} - \|c\,\mathbf{f}_\omega(z)\,e\|_{\mathbb{C}^N}\big| \le \|F(z) - c\,\mathbf{f}_\omega(z)\,e\|_{\mathbb{C}^N}.$$

Since $\|c\,\mathbf{f}_\omega(z)\,e\| = c\,|\mathbf{f}_\omega(z)|$, squaring and integrating gives

$$\|G - c|\mathbf{f}_\omega|\|_{\mathbf{B}^2_\alpha} \le \|F - c\,\mathbf{f}_\omega e\|_{\mathbf{B}^2_\alpha(\mathbb{D};\mathbb{C}^N)}.$$

Taking infima over $c > 0$, $\omega \in \mathbb{D}$ and $\|e\| = 1$, and applying Lemma 2.2, we obtain (2.12) and the corresponding set estimate.

□

Lemma 2.4 (Quantitative RKHS estimate, vector-valued)**.** *Let $\mathcal{H}$ be a reproducing kernel Hilbert space of $\mathbb{C}^N$-valued functions on X whose reproducing kernel is $K(\cdot,\cdot)I_N$, and let $F \in \mathcal{H}$. For $x \in X$ define*

$$\Delta_F(x) := \frac{\|F\|^2_{\mathcal{H}} - \|F(x)\|^2_{\mathbb{C}^N}\,K(x,x)^{-1}}{\|F\|^2_{\mathcal{H}}}, \qquad \text{so that} \qquad \|F(x)\|^2_{\mathbb{C}^N}\,K(x,x)^{-1} \le \|F\|^2_{\mathcal{H}}.$$

Then for every $x \in X$ there exist $c \in \mathbb{C}$ and a unit vector $e \in \mathbb{C}^N$ with $|c| = \|F\|_{\mathcal{H}}$ such that

$$\frac{\|F - cf_x e\|_{\mathcal{H}}}{\|F\|_{\mathcal{H}}} \le \sqrt{2}\,\Delta_F(x)^{1/2}, \tag{2.13}$$

where $f_x = K(x,x)^{-1/2}K_x$ is the normalized kernel in the scalar space.

Proof. Fix $x \in X$. If $F(x) \neq 0$, choose $e := F(x)/\|F(x)\|_{\mathbb{C}^N}$ and $c := \|F\|_{\mathcal{H}}$ (so $|c| = \|F\|_{\mathcal{H}}$). Using the reproducing property,

$$\langle F, f_x e\rangle_{\mathcal{H}} = \frac{\langle F(x), e\rangle_{\mathbb{C}^N}}{\sqrt{K(x,x)}} = \frac{\|F(x)\|_{\mathbb{C}^N}}{\sqrt{K(x,x)}} \in \mathbb{R}_+.$$

Hence

$$\begin{aligned}
\|F - cf_x e\|^2_{\mathcal{H}} &= \|F\|^2_{\mathcal{H}} + |c|^2\|f_x e\|^2_{\mathcal{H}} - 2\,\mathrm{Re}\langle F, cf_x e\rangle_{\mathcal{H}} \\
&= 2\|F\|^2_{\mathcal{H}} - 2\|F\|_{\mathcal{H}}\frac{\|F(x)\|_{\mathbb{C}^N}}{\sqrt{K(x,x)}} \\
&\le 2\|F\|^2_{\mathcal{H}} - 2\frac{\|F(x)\|^2_{\mathbb{C}^N}}{K(x,x)} = 2\|F\|^2_{\mathcal{H}}\,\Delta_F(x),
\end{aligned}$$

where in the inequality we used $\|F(x)\|_{\mathbb{C}^N}/\sqrt{K(x,x)} \le \|F\|_{\mathcal{H}}$. This gives (2.13).

If $F(x) = 0$, then $\Delta_F(x) = 1$ and taking any unit e and $c = \|F\|_{\mathcal{H}}$ yields $\|F - cf_x e\|^2_{\mathcal{H}} = \|F\|^2_{\mathcal{H}} + |c|^2 = 2\|F\|^2_{\mathcal{H}}$, so (2.13) holds as well. □

Lemma 2.5 (Super-level set estimate for vector-valued Bergman data)**.** *Let $\alpha > -1$ and let $F \in \mathbf{B}^2_\alpha(\mathbb{D}; \mathbb{C}^N)$ satisfy $\|F\|_{\mathbf{B}^2_\alpha(\mathbb{D};\mathbb{C}^N)} = 1$. Define*

$$u_F(z) := \|F(z)\|^2_{\mathbb{C}^N}(1 - |z|^2)^{\alpha+2}, \qquad \rho_F(t) := \mu(\{z \in \mathbb{D} : \ u_F(z) > t\}),$$

and set

$$a_0^2 := \frac{\pi}{\alpha+1} \max_{z\in\mathbb{D}} u_F(z) \in (0, 1], \qquad \delta_0^2 := 1 - a_0^2, \qquad T := \frac{\alpha+1}{\pi} a_0^2.$$

Then there exists $\widetilde{c}_0 \in (0, 1)$ such that for every $c_0 \geq \widetilde{c}_0$ there exist $C_0 \in (c_0, 1)$ and $C > 0$ with the following property: if $a_0^2 \geq C_0$, then for every

$$t \in \Big(\frac{\alpha+1}{\pi} c_0, \ \frac{\alpha+1}{\pi} a_0^2\Big)$$

one has

$$\rho_F(t) \leq \pi\Big(\frac{1 + K_0(1 - a_0^2)}{a_0^2}\Big)\Big(\Big(\frac{\frac{\alpha+1}{\pi} a_0^2}{t}\Big)^{\frac{1}{\alpha+2}} - 1\Big), \qquad K_0 = \frac{C}{c_0^3}. \tag{2.14}$$

Moreover, the constants can be chosen independently of N.

Proof. The proof follows the scalar argument of Lemma 3.1 [4] (and the corresponding level-set analysis in [4]), with one additional estimate for the components orthogonal to the maximizer direction. We indicate the modifications and keep the rest identical.

Let z_* maximize u_F. By composing with a disk automorphism and multiplying by a unitary matrix (and a unimodular scalar), we may assume (see [4, Lemma 3.1]) that

$$F(0) = \sqrt{T}\, e_1, \qquad u_F(0) = \max_{\mathbb{D}} u_F, \qquad \langle F'(0), F(0)\rangle_{\mathbb{C}^N} = 0,$$

so in particular the first component $f := \langle F, e_1\rangle$ satisfies

$$f(0) = \sqrt{T}, \qquad f'(0) = 0. \tag{2.15}$$

Decompose

$$F(z) = f(z)e_1 + H(z), \qquad H(z) \in e_1^\perp, \quad \langle H(z), e_1\rangle = 0, \quad H(0) = 0,$$

so that

$$\|F(z)\|^2_{\mathbb{C}^N} = |f(z)|^2 + \|H(z)\|^2_{\mathbb{C}^N}, \qquad z \in \mathbb{D}. \tag{2.16}$$

Write

$$\frac{f(z)}{\sqrt{T}} = 1 + R(z).$$

By (2.15), the Taylor series of R starts at order ≥ 2, exactly as in the scalar proof. Writing $f(z) = \sum_{k\geq 0} a_k z^k/\sqrt{c_k}$ in the usual orthonormal Bergman basis (for the scalar space), one has $a_1 = 0$ and

$$\sum_{k\geq 2} |a_k|^2 \leq \sum_{k\geq 1} |a_k|^2 \leq 1 - a_0^2 = \delta_0^2.$$

Thus, setting $\delta^2 := \delta_0^2/T$, Cauchy–Schwarz yields the same bounds as in the scalar proof:

$$|R(z)| \leq \delta\, g_1(|z|)^{1/2}, \qquad |R'(z)| \leq \delta\, g_2(|z|)^{1/2}, \qquad |R''(z)| \leq \delta\, g_3(|z|)^{1/2}, \tag{2.17}$$

with the same explicit functions g_1, g_2, g_3 as in Lemma 3.1 in [4].

Let $h(z) := 2\,\mathrm{Re}\, R(z)$; then h is harmonic and (2.17) implies the same bounds as [4, eq. 3.4] in the scalar proof.

It remains only to estimate the orthogonal component H. Expand each component of H in the same orthonormal basis. Since $H(0) = 0$, all constant coefficients vanish. Moreover,

$$\|H\|^2_{\mathbf{B}^2_\alpha(\mathbb{D};\mathbb{C}^N)} \leq \|F\|^2 - \|f\|^2 \leq 1 - a_0^2 = \delta_0^2.$$

Hence, by Cauchy–Schwarz and the diagonal kernel identity

$$\sum_{k\geq 0} \frac{|z|^{2k}}{c_k} = K(z, z) = \frac{\alpha+1}{\pi}(1 - |z|^2)^{-(\alpha+2)},$$

we obtain the pointwise estimate

$$\frac{\|H(z)\|^2_{\mathbb{C}^N}}{T} \le \frac{\delta_0^2}{T}\sum_{k\ge 1}\frac{|z|^{2k}}{c_k} = \delta^2\,\widehat{g}_1(|z|), \qquad \widehat{g}_1(r) := \frac{\alpha+1}{\pi}\big[(1-r^2)^{-(\alpha+2)}-1\big]. \tag{2.18}$$

Compared to the scalar proof, $\widehat{g}_1$ is the same kernel series but without subtracting the $k=1$ term; this is exactly where the possible linear terms of H are absorbed.

Now fix $z = re^{i\theta}$ and assume $u_F(z) > t$. Using (2.16) we get

$$\frac{t}{T}(1-r^2)^{-(\alpha+2)} < \frac{|f(z)|^2+\|H(z)\|^2}{T} = |1+R(z)|^2 + \frac{\|H(z)\|^2}{T} = 1 + h(z) + |R(z)|^2 + \frac{\|H(z)\|^2}{T}.$$

Using (2.17) to bound $|R(z)|^2 \le \delta^2 g_1(r)$ and then (2.18), we obtain

$$\frac{t}{T}(1-r^2)^{-(\alpha+2)} < 1 + \delta^2\big(g_1(r)+\widehat{g}_1(r)\big) + h(re^{i\theta}). \tag{2.19}$$

Define the modified radial term

$$g_1^*(r) := g_1(r) + \widehat{g}_1(r) = \frac{\alpha+1}{\pi}\big[(1-r^2)^{-(\alpha+2)} - 1 - (\alpha+2)r^2\big] + \frac{\alpha+1}{\pi}\big[(1-r^2)^{-(\alpha+2)}-1\big],$$

and repeat the scalar construction with

$$g_\theta(r,\sigma) := \frac{t}{T}(1-r^2)^{-(\alpha+2)} - \delta^2 g_1^*(r) - \sigma h(re^{i\theta}).$$

Then (2.19) implies $g_\theta(r,1) < 1$ and the star-shaped sets E_σ are defined exactly as in the scalar proof.

All subsequent steps—boundedness and star-shapedness of E_σ for $t > \frac{\alpha+1}{\pi}c_0$, the Taylor expansion of $F(\sigma) = \mu(E_\sigma)$, the estimates on r'_σ, r''_σ, the comparison of $F(1)$ and $F(0)$, and the final estimate for $\mu(E_1)$—use only: (i) the harmonicity bounds for h coming from (2.17), and (ii) the fact that the radial error term is of size δ^2 times an explicit function of r. Replacing g_1 by the larger g_1^* changes only universal constants (and yields the same structure $K_0 = C/c_0^3$ after the same bookkeeping), while remaining independent of N.

Therefore the scalar argument yields (2.14). □

Remark 2.6. *The threshold $\widetilde{c}_0$ arises from a purely one-dimensional inequality comparing explicit bounds on r_0 (as in the scalar proof); the same comparison works here after enlarging the radial term to g_1^*. Thus one may keep the same qualitative statement (and in practice the same numerical value, up to re-optimizing constants). Since the paper does not depend on optimizing $\widetilde{c}_0$, we do not track the best value.*

Lemma 2.7 (Lower bound on the rearrangement gap: vector-valued case)**.** *Let $\alpha > -1$ and let $F \in \mathbf{B}^2_\alpha(\mathbb{D};\mathbb{C}^N)$ with $\|F\|_{\mathbf{B}^2_\alpha(\mathbb{D};\mathbb{C}^N)} = 1$. Set*

$$u_F(z) := \|F(z)\|^2_{\mathbb{C}^N}(1-|z|^2)^{\alpha+2}, \qquad a_0^2 := \frac{\pi}{\alpha+1}\max_{\mathbb{D}} u_F, \qquad \delta_0^2 := 1 - a_0^2,$$

and let u_F^* be the decreasing rearrangement of u_F with respect to μ. Let $v^*(s) := \frac{\alpha+1}{\pi}\Big(1+\frac{s}{\pi}\Big)^{-(\alpha+2)}$. Let $s^* > 0$ be the minimal solution of $u_F^*(s) = v^*(s)$.

Assume Lemma 2.5 holds for u_F (with constants c_0, C_0, K_0 as there). Then there exists an absolute constant $C > 0$ such that

$$C\,\frac{\alpha+1}{\alpha+2}\,(1-a_0^2) \;\le\; \int_0^{s^*}\big(v^*(s) - u_F^*(s)\big)\,ds. \tag{2.20}$$

Moreover, the constant C can be chosen independently of N.

Proof. This is the scalar proof with u replaced by u_F and ρ replaced by ρ_F. We sketch the two places where an input is used.

Let $t^* := v^*(s^*) = u_F^*(s^*)$. First, one shows that

$$t^* \le \frac{\alpha+1}{\pi}\,c$$

for some absolute $c \in (0,1)$: if $t^* \le \frac{\alpha+1}{\pi}c_0$ we are done; otherwise, apply Lemma 2.5 at $t = t^*$ and use $\rho_F(t^*) = s^* = \pi\big((\frac{\alpha+1}{\pi}\frac{}{t^*})^{\frac{1}{\alpha+2}} - 1\big)$ to obtain the analogue of [4, eq. 3.16] and hence a uniform upper bound on t^* (exactly as in your scalar argument).

Next, let $\omega(t)$ denote the right-hand side of the superlevel estimate from Lemma 2.5, and let ν be the inverse of v^*,

$$\nu(t) = \pi\Big(\Big(\frac{\frac{\alpha+1}{\pi}}{t}\Big)^{\frac{1}{\alpha+2}} - 1\Big).$$

Then the rearrangement identity and the inequality $\rho_F(t) \le \omega(t)$ on the admissible range yield

$$\int_0^{s^*} (v^* - u_F^*)\, ds \;\ge\; \int_{\tau_1}^{T} (\nu(t) - \omega(t))\, dt, \qquad T = \frac{\alpha+1}{\pi} a_0^2,$$

for a suitable τ_1 (chosen as in the scalar proof so that $\tau_1 > \frac{\alpha+1}{\pi} c_0$). Carrying out the same change of variables $t = \frac{\alpha+1}{\pi} r$ then gives the same one-dimensional estimate in r, with the same elementary positivity argument for the first term in the expansion (your $f_1(r)$ bound). This yields a lower bound of the form

$$\int_0^{s^*} (v^* - u_F^*)\, ds \;\gtrsim\; \frac{\alpha+1}{\alpha+2}\, \delta_0^2,$$

i.e. (2.20). All constants remain independent of N because the only nontrivial input is the dimension-free superlevel estimate (Lemma 2.5). □

Lemma 2.8 (Pre-RKHS bound: vector-valued case)**.** *Let $\alpha > -1$, $\Omega \subset \mathbb{D}$ with $s = \mu(\Omega) > 0$, and let $F \in \mathbf{B}^2_\alpha(\mathbb{D}; \mathbb{C}^N)$. Define the deficit (depending only on $u_F = \|F\|^2(1-|z|^2)^{\alpha+2}$) by*

$$\delta(F;\Omega,\alpha) := \frac{\|F\|^2\, \theta_\alpha(s) - \int_\Omega u_F\, d\mu}{\|F\|^2\, \theta_\alpha(s)}, \qquad \theta_\alpha(s) := 1 - \Big(1 + \frac{s}{\pi}\Big)^{-(\alpha+1)}.$$

Then there exists a function $M_\alpha(s)$ (the same one as in the scalar proof) such that

$$1 - a_0^2 \;\le\; M_\alpha(s)\, \delta(F;\Omega,\alpha), \tag{2.21}$$

where $a_0^2 = \frac{\pi}{\alpha+1} \max_{\mathbb{D}} u_F$. Moreover, $M_\alpha(s)$ is independent of N.

One may take for instance

$$M_\alpha(s) = C\Big(1 + \frac{\alpha+2}{\alpha+1}\Big[\Big(1+\frac{s}{\pi}\Big)^{\alpha+1} - 1\Big]\Big),$$

with $C > 0$ an absolute constant.

Proof. This is again the scalar proof with u replaced by u_F. Write $s_0 := \mu(\Omega)$ and let $A := \{u_F > u_F^*(s_0)\}$ be the superlevel set of u_F of measure s_0. Set $\delta_{s_0} := \delta(F; A, \alpha) \le \delta(F;\Omega,\alpha)$ and define

$$\varepsilon := \int_0^{s_0} (v^*(s) - u_F^*(s))\, ds = \delta_{s_0} \int_0^{s_0} v^*(s)\, ds.$$

A key input is the monotonicity of the ratio

$$r(s) := \frac{u_F^*(s)}{v^*(s)}.$$

Since F is holomorphic, $\log \|F\|_{\mathbb{C}^N}$ is subharmonic (with $\log 0 = -\infty$), hence $\|F\|^2_{\mathbb{C}^N}$ is log–subharmonic. Therefore monotonicity extends verbatim to the $\mathbb{C}^N$–valued setting; see [13, 4]. From here, the two case splits ($s_0 > s^*$ and $s_0 < s^*$) proceed verbatim, yielding the analogues of [4, eq.3.21-eq.3.24], where the lower bound (2.20) is used in place of [4, Lemma 3.2]. Combining those estimates gives (2.21) with

$$M_\alpha(s) = C\Big(1 + \frac{\alpha+2}{\alpha+1}\big[(1+s/\pi)^{\alpha+1} - 1\big]\Big),$$

exactly as in the scalar Step III. All constants are independent of N because (i) the ratio-monotonicity depends only on log-subharmonicity of u_F, and (ii) Lemma 2.7 is dimension-free. □

Remark 2.9. *The following examples and propositions show that the desired stability estimate with deficit exponent $1/2$ can be established for several special classes of log-subharmonic mappings and functions; see, for instance, Examples 2.10 and 2.11, as well as Lemma 2.12 and Propositions 2.13, 2.14, and 2.15. These results cover, in particular, certain radial classes and functions admitting Hilbert-valued holomorphic factorizations or additional monotonicity/unimodality structure. However, we were not able to prove a corresponding general result for arbitrary log-subharmonic functions in which the right-hand side is controlled by the deficit with exponent $1/2$. In the next section, we establish a weaker statement for general log-subharmonic functions, where the deficit appears with exponent $1/4$.*

Example 2.10. Let $a \in (0,1)$. Define holomorphic functions $a_k : \mathbb{D} \to \mathbb{C}$ by

$$a_{n+1}(z) := \sqrt{n+1}\, a^{n/2}\, z^n, \qquad n = 0, 1, 2, \dots.$$

Then for every $z \in \mathbb{D}$ we have

$$\frac{1}{1 - a|z|^2} = \sqrt{\sum_{k=1}^{\infty} |a_k(z)|^2}.$$

In particular, the map $A : \mathbb{D} \to \ell^2$ given by

$$A(z) = (a_k(z))_{k \geq 1}$$

is holomorphic and satisfies

$$\|A(z)\|_{\ell^2} = \frac{1}{1 - a|z|^2} \qquad \text{for all } z \in \mathbb{D}.$$

Proof. For $|x| < 1$ we have the power series identity

$$\frac{1}{(1-x)^2} = \sum_{n=0}^{\infty} (n+1) x^n.$$

Substituting $x = a|z|^2$ (which satisfies $|x| < 1$ for $z \in \mathbb{D}$) gives

$$\frac{1}{(1 - a|z|^2)^2} = \sum_{n=0}^{\infty} (n+1) a^n |z|^{2n}.$$

By definition,

$$|a_{n+1}(z)|^2 = (n+1) a^n |z|^{2n},$$

and therefore

$$\sum_{k=1}^{\infty} |a_k(z)|^2 = \sum_{n=0}^{\infty} (n+1) a^n |z|^{2n} = \frac{1}{(1 - a|z|^2)^2}.$$

Taking square roots completes the proof. □

Example 2.11 (Hilbert–valued holomorphic factorization for radial functions)**.** Let $u : \mathbb{D} \to (0, \infty)$ be radial, $u(z) = U(|z|)$, and assume that the function

$$F(r) := U(r)^2$$

admits a (real-analytic) power series expansion in r^2 with *nonnegative* coefficients,

$$F(r) = \sum_{n=0}^{\infty} b_n\, r^{2n}, \qquad b_n \geq 0, \qquad 0 \leq r < 1. \tag{2.22}$$

Then there exist holomorphic functions $a_k : \mathbb{D} \to \mathbb{C}$ such that for every $z \in \mathbb{D}$,

$$u(z) = \Big(\sum_{k=0}^{\infty} |a_k(z)|^2 \Big)^{1/2}. \tag{2.23}$$

More precisely, one may take

$$a_n(z) := \sqrt{b_n}\, z^n, \qquad n = 0, 1, 2, \dots, \tag{2.24}$$

so that $\sum_{n \geq 0} |a_n(z)|^2 = \sum_{n \geq 0} b_n |z|^{2n} = F(|z|) = u(z)^2$.

Moreover, defining the partial sums

$$u_N(z) := \Big(\sum_{n=0}^{N} |a_n(z)|^2 \Big)^{1/2},$$

we have $u_N(z) \uparrow u(z)$ pointwise for all $z \in \mathbb{D}$, and the convergence is uniform on compact subsets of $\mathbb{D}$.

If, in addition, u is C^2 on $\mathbb{D}$ and each a_n is holomorphic, then the right-hand side of (2.23) is log-subharmonic; in particular, any u represented as in (2.23) is log-subharmonic.

Proof. Assumption (2.22) implies $F(|z|) = \sum_{n\geq 0} b_n|z|^{2n}$ for all $z \in \mathbb{D}$. With a_n as in (2.24), each a_n is holomorphic and

$$\sum_{n=0}^{\infty} |a_n(z)|^2 = \sum_{n=0}^{\infty} b_n|z|^{2n} = F(|z|) = u(z)^2,$$

which gives (2.23). Monotone convergence of u_N is immediate since $b_n \geq 0$. Uniform convergence on compacta follows from uniform convergence of the power series for $F(r)$ on $[0, r_0]$ for any $r_0 < 1$.

For the final claim, note that $z \mapsto \sum_{n=0}^{N} |a_n(z)|^2$ is subharmonic as a finite sum of squared moduli of holomorphic functions, hence $\log \sum_{n=0}^{N} |a_n(z)|^2$ is subharmonic wherever the sum is positive. Passing to $N \to \infty$ preserves log-subharmonicity under locally uniform limits (or via standard monotone limit arguments), yielding that u is log-subharmonic. □

For $\alpha > -1$, set

$$dA_\alpha(z) := \frac{\alpha+1}{\pi}(1-|z|^2)^\alpha \, dA(z), \qquad d\mu(z) := \frac{dA(z)}{(1-|z|^2)^2}.$$

For $s > 0$, let $r = r(s) \in (0,1)$ be determined by

$$\mu(\{|z| < r\}) = s \qquad \Longleftrightarrow \qquad r^2 = \frac{s}{s+\pi}.$$

Define

$$\theta_\alpha(s) := \int_{|z|<r} dA_\alpha = 1 - (1-r^2)^{\alpha+1} = 1 - \Big(1 + \frac{s}{\pi}\Big)^{-(\alpha+1)},$$

$$C_\alpha(s) := \frac{1}{(1-r^2)^{\alpha+1}} = \Big(1 + \frac{s}{\pi}\Big)^{\alpha+1}.$$

Lemma 2.12 (Centered-disk radial stability)**.** *Let $\alpha > -1$ and let $g : \mathbb{D} \to [0,\infty)$ be radial and nondecreasing in $|z|$. Assume*

$$\|g\|_{2,\alpha}^2 = \int_{\mathbb{D}} g(z)^2 \, dA_\alpha(z) = 1.$$

Fix $r \in (0,1)$ and write

$$W := 1 - (1-r^2)^{\alpha+1}, \qquad S := (1-r^2)^{\alpha+1}.$$

Define the centered-disk deficit

$$\delta_{r,\alpha}(g) := 1 - \frac{1}{W}\int_{|z|<r} g(z)^2 \, dA_\alpha(z) \in [0,1].$$

Then

$$\|g-1\|_{2,\alpha}^2 \leq \frac{3}{S}\,\delta_{r,\alpha}(g) = \frac{3}{(1-r^2)^{\alpha+1}}\,\delta_{r,\alpha}(g).$$

Proof. Set $u := g^2$ and $\delta := \delta_{r,\alpha}(g)$. Since $\int_{\mathbb{D}} u \, dA_\alpha = 1$,

$$\int_{|z|<r} u \, dA_\alpha = W(1-\delta), \qquad \int_{r<|z|<1} u \, dA_\alpha = S + W\delta.$$

Because u is radial and nondecreasing,

$$u(\rho) \geq u(r) \quad (\rho \in [r,1)),$$

hence

$$u(r)\, S \leq \int_{r<|z|<1} u \, dA_\alpha = S + W\delta,$$

so

$$u(r) \leq 1 + \frac{W}{S}\delta.$$

Likewise, on $[0,r]$ one has $u(\rho) \leq u(r)$, hence

$$\int_{|z|<r} u \, dA_\alpha \leq u(r)\, W,$$

which gives

$$u(r) \geq 1 - \delta.$$

Thus, with $\kappa := W/S$,

$$\sqrt{1-\delta} \leq g(r) \leq \sqrt{1+\kappa\delta}.$$

Now split the $L^1(dA_\alpha)$-mass of g into inner and outer parts.

On the outer annulus, monotonicity gives

$$\int_{r<|z|<1} g\,dA_\alpha \geq S\,g(r) \geq S\sqrt{1-\delta}.$$

On the inner disk, since $g(\rho) \leq g(r)$ for $\rho < r$, we have pointwise

$$u = g^2 \leq g(r)\,g,$$

hence

$$\int_{|z|<r} g\,dA_\alpha \geq \frac{\int_{|z|<r} u\,dA_\alpha}{g(r)} \geq \frac{W(1-\delta)}{\sqrt{1+\kappa\delta}}.$$

Therefore

$$\int_{\mathbb{D}} g\,dA_\alpha \geq S\sqrt{1-\delta} + W\,\frac{1-\delta}{\sqrt{1+\kappa\delta}} =: \Psi(\delta).$$

Since $\|g\|_{2,\alpha} = \|1\|_{2,\alpha} = 1$,

$$\|g-1\|_{2,\alpha}^2 = 2\Big(1 - \int_{\mathbb{D}} g\,dA_\alpha\Big) \leq 2(1-\Psi(\delta)).$$

Now use the elementary bounds

$$1-\sqrt{1-\delta} \leq \frac{\delta}{2}, \qquad 1 - \frac{1-\delta}{\sqrt{1+t}} \leq \delta + \frac{t}{2} \qquad (0 \leq \delta \leq 1,\ t \geq 0),$$

with $t = \kappa\delta$. Then

$$1-\Psi(\delta) \leq S\frac{\delta}{2} + W\Big(\delta + \frac{\kappa\delta}{2}\Big) = \Big(\frac{S}{2} + W\Big)\delta + \frac{W^2}{2S}\delta.$$

Since $\frac{S}{2} + W \leq 1$ and $W \leq 1$,

$$1-\Psi(\delta) \leq \delta + \frac{1}{2S}\delta \leq \frac{3}{2S}\delta.$$

Hence

$$\|g-1\|_{2,\alpha}^2 \leq 2\cdot\frac{3}{2S}\delta = \frac{3}{S}\delta.$$

□

Proposition 2.13 (Arbitrary sets under monotone bathtub ratio)**.** *Let $\alpha > -1$ and let $g : \mathbb{D} \to [0,\infty)$ be radial and log-subharmonic. Assume*

$$\|g\|_{2,\alpha} = 1, \qquad u(r) := g(r)^2, \qquad \Phi_\alpha(r) := u(r)(1-r^2)^{\alpha+2}.$$

Let $\Omega \subset \mathbb{D}$ be measurable with $\mu(\Omega) = s \in (0,\infty)$, and define

$$\delta(g;\Omega,\alpha) := 1 - \frac{1}{\theta_\alpha(s)}\int_\Omega g(z)^2\,dA_\alpha(z).$$

Assume that Φ_α is nonincreasing on $[0,1)$. Then

$$\delta(g;\Omega,\alpha) \geq 0 \quad \textit{and} \quad \|g-1\|_{2,\alpha}^2 \leq 3\,C_\alpha(s)\,\delta(g;\Omega,\alpha).$$

Proof. Since g is radial log-subharmonic, it is radial subharmonic, hence nondecreasing in r.

Write the angular profile of Ω as

$$m(\rho) := \frac{1}{2\pi}\int_0^{2\pi} \mathbf{1}_\Omega(\rho e^{i\theta})\,d\theta \in [0,1].$$

Then

$$\int_\Omega u(|z|)\,dA_\alpha(z) = 2(\alpha+1)\int_0^1 u(\rho)(1-\rho^2)^\alpha \rho\,m(\rho)\,d\rho = 2(\alpha+1)\int_0^1 \Phi_\alpha(\rho)\,m(\rho)\,d\nu(\rho),$$

where

$$d\nu(\rho) := \frac{\rho}{(1-\rho^2)^2}\,d\rho.$$

Also

$$\mu(\Omega) = 2\pi\int_0^1 m(\rho)\,d\nu(\rho) = s.$$

Thus maximizing $\int_\Omega u\,dA_\alpha$ among all sets with fixed μ-mass s is exactly a bathtub problem for the weight Φ_α on $([0,1), d\nu)$. By the Bathtub principle, an optimizer is a superlevel set of Φ_α. Since Φ_α is nonincreasing, its superlevel sets are initial intervals $[0,R)$, so the maximizer is the centered disk $D_r := \{|z| < r\}$ with $\mu(D_r) = s$.

Therefore

$$\int_\Omega u\,dA_\alpha \le \int_{D_r} u\,dA_\alpha.$$

Because $\theta_\alpha(s) = \int_{D_r} dA_\alpha$, this is equivalent to

$$\delta(g;\Omega,\alpha) \ge \delta_{r,\alpha}(g) \ge 0.$$

Applying Lemma 2.12 at radius r yields

$$\|g-1\|_{2,\alpha}^2 \le \frac{3}{(1-r^2)^{\alpha+1}}\delta_{r,\alpha}(g) \le 3\,C_\alpha(s)\,\delta(g;\Omega,\alpha).$$

□

Proposition 2.14 (Disk regime for strictly unimodal bathtub ratio)**.** *Let $\alpha > -1$ and let $g : \mathbb{D} \to [0,\infty)$ be radial and log-subharmonic. Assume*

$$\|g\|_{2,\alpha} = 1, \qquad u(r) := g(r)^2, \qquad \Phi_\alpha(r) := u(r)(1-r^2)^{\alpha+2}.$$

Assume that Φ_α is strictly unimodal: there exists $r_0 \in (0,1)$ such that Φ_α is strictly increasing on $[0,r_0]$ and strictly decreasing on $[r_0,1)$, and also $\Phi_\alpha(1^-) = 0$.

Let $r_* \in (r_0,1)$ be the unique point such that

$$\Phi_\alpha(r_*) = \Phi_\alpha(0),$$

and set

$$s_* := \mu(\{|z| < r_*\}).$$

If $\Omega \subset \mathbb{D}$ satisfies $\mu(\Omega) = s \ge s_*$, then

$$\delta(g;\Omega,\alpha) \ge 0 \quad \text{and} \quad \|g-1\|_{2,\alpha}^2 \le 3\,C_\alpha(s)\,\delta(g;\Omega,\alpha).$$

Proof. As above, maximizing $\int_\Omega u\,dA_\alpha$ under the constraint $\mu(\Omega) = s$ is a bathtub problem for the weight Φ_α on $([0,1), d\nu)$. Hence an optimizer is a superlevel set $\{\Phi_\alpha > \lambda\}$.

Because Φ_α is strictly unimodal, each superlevel set is either

1. a centered disk $[0,R)$ if $\lambda \le \Phi_\alpha(0)$, or
2. an annulus $(R_-(\lambda), R_+(\lambda))$ if $\lambda > \Phi_\alpha(0)$.

Now suppose $\lambda > \Phi_\alpha(0)$. Since $\Phi_\alpha(r_*) = \Phi_\alpha(0)$ and Φ_α is strictly decreasing on $[r_0,1)$, one has

$$R_+(\lambda) < r_*.$$

Thus the annulus $\{\Phi_\alpha > \lambda\}$ is contained in $\{|z| < r_*\}$, so

$$\mu(\{\Phi_\alpha > \lambda\}) < \mu(\{|z| < r_*\}) = s_*.$$

Hence no annular optimizer can have μ-mass $s \ge s_*$.

Therefore, for every $s \ge s_*$, the optimizer must be the centered disk D_r with $\mu(D_r) = s$. Consequently,

$$\int_\Omega u\,dA_\alpha \le \int_{D_r} u\,dA_\alpha,$$

which is equivalent to

$$\delta(g;\Omega,\alpha) \ge \delta_{r,\alpha}(g) \ge 0.$$

Since radial log-subharmonic functions are nondecreasing, Lemma 2.12 applies and gives

$$\|g-1\|_{2,\alpha}^2 \le \frac{3}{(1-r^2)^{\alpha+1}}\delta_{r,\alpha}(g) \le 3\,C_\alpha(s)\,\delta(g;\Omega,\alpha).$$

□

Proposition 2.15 (A single absolute comparison constant, $\alpha = 0$)**.** *Let*

$$u(r) = 1 + \sum_{k=1}^{n} a_k r^k, \qquad a_k \geq 0,$$

and define

$$g(z) := \sqrt{u(|z|)}, \qquad \widetilde{g} := \frac{g}{\|g\|_{2,0}}, \qquad dA_0(z) := \frac{1}{\pi}\, dA(z).$$

Let

$$d\mu(z) := \frac{dA(z)}{(1-|z|^2)^2}, \qquad \mu(\Omega) = s > 0, \qquad r = r(s) \text{ by } \mu(\{|z| < r\}) = s,$$

so that

$$r^2 = \frac{s}{s+\pi}, \qquad \theta_0(s) = \frac{s}{s+\pi} = r^2, \qquad C_0(s) = \frac{1}{1-r^2} = 1 + \frac{s}{\pi}.$$

Define

$$\delta(\widetilde{g}; \Omega, 0) := 1 - \frac{1}{\theta_0(s)} \int_\Omega \widetilde{g}(z)^2\, dA_0(z),$$

and similarly

$$\delta(\widetilde{g}; D_r, 0) := 1 - \frac{1}{r^2} \int_{|z|<r} \widetilde{g}(z)^2\, dA_0(z).$$

Then

$$\delta(\widetilde{g}; D_r, 0) \leq K_*\, \delta(\widetilde{g}; \Omega, 0), \qquad K_* := \frac{25\sqrt{5}}{25\sqrt{5}-24}. \tag{2.25}$$

Consequently,

$$\|\widetilde{g} - 1\|_{2,0}^2 \leq 3K_*\, C_0(s)\, \delta(\widetilde{g}; \Omega, 0), \tag{2.26}$$

and hence

$$\|\widetilde{g} - 1\|_{2,0} \leq C_*\, C_0(s)^{1/2}\, \delta(\widetilde{g}; \Omega, 0)^{1/2}, \qquad C_* := \big(3K_*\big)^{1/2} = \Big(3\,\frac{25\sqrt{5}}{25\sqrt{5}-24}\Big)^{1/2}.$$

Proof. Set

$$\widetilde{u}(r) := \widetilde{g}(r)^2 = \frac{u(r)}{\|g\|_{2,0}^2}, \qquad \varphi(r) := \widetilde{u}(r)(1-r^2)^2.$$

Because $a_k \geq 0$, the function u is radial and nondecreasing, hence so is $\widetilde{g}$. Therefore Lemma 2.12 with $\alpha = 0$ gives

$$\|\widetilde{g} - 1\|_{2,0}^2 \leq 3\, C_0(s)\, \delta(\widetilde{g}; D_r, 0). \tag{2.27}$$

So it remains only to compare the disk deficit to the arbitrary-set deficit.

First, since $dA_0 = \frac{1}{\pi} dA$ and $d\mu = \frac{dA}{(1-|z|^2)^2}$,

$$\int_\Omega \widetilde{u}(|z|)\, dA_0(z) = \frac{1}{\pi} \int_\Omega \varphi(|z|)\, d\mu(z).$$

Hence, for every measurable Ω with $\mu(\Omega) = s$,

$$\int_\Omega \widetilde{u}\, dA_0 \leq \frac{s}{\pi} \max_{0 \leq t < 1} \varphi(t).$$

Since $\theta_0(s) = \frac{s}{s+\pi}$, this implies

$$\frac{1}{\theta_0(s)} \int_\Omega \widetilde{u}\, dA_0 \leq \max \varphi,$$

and therefore

$$\delta(\widetilde{g}; \Omega, 0) \geq 1 - \max \varphi. \tag{2.28}$$

On the other hand, because $\widetilde{u}$ is radial and nondecreasing,

$$\frac{1}{r^2} \int_{|z|<r} \widetilde{u}\, dA_0 \geq \widetilde{u}(0),$$

hence

$$\delta(\widetilde{g}; D_r, 0) \leq 1 - \widetilde{u}(0). \tag{2.29}$$

We now estimate $\max\varphi$ in terms of $\widetilde{u}(0)$. Write

$$N := \|g\|_{2,0}^2 = \int_{\mathbb{D}} u(|z|)\,dA_0 = 1 + \sum_{k=1}^{n} \frac{2a_k}{k+2}.$$

Then

$$\widetilde{u}(0) = \frac{1}{N}, \qquad 1 - \widetilde{u}(0) = \sum_{k=1}^{n} \frac{2a_k}{(k+2)N}.$$

Also,

$$\widetilde{u}(r) = \frac{1}{N} + \sum_{k=1}^{n} \frac{a_k}{N} r^k,$$

so

$$\varphi(r) = \widetilde{u}(r)(1-r^2)^2 \le \widetilde{u}(0) + \sum_{k=1}^{n} \frac{a_k}{N} r^k (1-r^2)^2.$$

Rewrite

$$\frac{a_k}{N} = \frac{2a_k}{(k+2)N} \cdot \frac{k+2}{2},$$

to get

$$\varphi(r) \le \widetilde{u}(0) + \sum_{k=1}^{n} \frac{2a_k}{(k+2)N} \Big(\frac{k+2}{2} r^k (1-r^2)^2 \Big).$$

Hence

$$\max\varphi \le \widetilde{u}(0) + \sum_{k=1}^{n} \frac{2a_k}{(k+2)N} M_k,$$

where

$$M_k := \max_{0\le r<1} \frac{k+2}{2}\, r^k (1-r^2)^2.$$

A direct calculation shows that M_k is decreasing in $k \ge 1$, so

$$M_k \le M_1.$$

Since

$$M_1 = \max_{0\le r<1} \frac{3}{2}\, r(1-r^2)^2 = \frac{24}{25\sqrt{5}},$$

it follows that

$$\max\varphi \le \widetilde{u}(0) + \frac{24}{25\sqrt{5}}\,(1 - \widetilde{u}(0)).$$

Therefore

$$1 - \max\varphi \ge \Big(1 - \frac{24}{25\sqrt{5}}\Big)(1 - \widetilde{u}(0)).$$

Combining this with (2.28) and (2.29), we obtain

$$\delta(\widetilde{g}; D_r, 0) \le \frac{1 - \widetilde{u}(0)}{1 - \max\varphi}\, \delta(\widetilde{g}; \Omega, 0) \le \frac{1}{1 - \frac{24}{25\sqrt{5}}}\, \delta(\widetilde{g}; \Omega, 0),$$

which is exactly (2.25).

Finally, insert (2.25) into (2.27) to get

$$\|\widetilde{g} - 1\|_{2,0}^2 \le 3K_*\, C_0(s)\, \delta(\widetilde{g}; \Omega, 0).$$

Taking square roots gives the last claim. □

3. Stability of concentration estimates for $\mathcal{M}$-logsubharmonic functions

Let $\mathbb{B} = \mathbb{B}^n$ be the unit ball in $\mathbb{R}^n$. A mapping $u \in C^2(\mathbb{B}^n, \mathbb{C})$ or more generally $u \in C^2(\mathbb{B}^n, \mathbb{R}^k)$ is said to be *hyperbolic harmonic* if u (see Rudin [14] and Stoll [15]) satisfies the hyperbolic Laplace equation

$$\Delta_h u(x) = (1-|x|^2)^2 \Delta u(x) + 2(n-2)(1-|x|^2) \sum_{i=1}^n x_i \frac{\partial u}{\partial x_i}(x) = 0, \tag{3.1}$$

where Δ denotes the usual Laplacian in $\mathbb{R}^n$. We call Δ_h the *hyperbolic Laplacian operator*. For $x \in \mathbb{B}$ we define the area element of the unit ball by

$$d\tau(x) = \frac{1}{(1-|x|^2)^n}\, dV(x). \tag{3.2}$$

We will also sometimes denote hyperbolic measure of the set A by $|A|_h = \tau(A)$. The Poisson kernel for Δ_h is defined by

$$P_h(x,\zeta) = \frac{(1-|x|^2)^{n-1}}{|x-\zeta|^{2n-2}}, \qquad (x,\zeta) \in \mathbb{B} \times \mathbb{S}.$$

Let $\mathcal{M}$ be the group of conformal automorphisms of the unit ball onto itself. Then for fixed ζ, $x \mapsto P_h(x,\zeta)$ is $\mathcal{M}$-harmonic, and, given a map $f \in \mathcal{L}(\mathbb{S})$, the function

$$u(x) = P_h[f](x) := \int_{\mathbb{S}} P_h(x,\zeta) f(\zeta)\, d\sigma(\zeta)$$

is the Poisson extension of f: it is $\mathcal{M}$-harmonic in $\mathbb{B}$, and agrees with f on $\mathbb{S}$.

We say that a smooth, real function u is $\mathcal{M}$-subharmonic if $\Delta_h u(x) \geq 0$[1]. Notice further that

$$(\Delta_h u)(m(x)) = \Delta_h (u \circ m)(x), \tag{3.3}$$

for every Möbius transformation $m \in \mathcal{M}$ of the unit ball onto itself. For this reason, we shall also call hyperbolic harmonic (resp. hyperbolic subharmonic) functions $\mathcal{M}$-harmonic (resp. $\mathcal{M}$-subharmonic) functions. Note that, for $n = 2$, the $\mathcal{M}$-harmonic and $\mathcal{M}$-subharmonic functions coincide with the usual Euclidean harmonic and subharmonic functions.

It follows from (3.3), by putting $u = \mathrm{Id}$, that

$$\Delta_h m = 2(n-2)(1-|m|^2)m.$$

Thus, Möbius transformations are (considered as vectorial functions) hyperbolic harmonic only in the case $n = 2$.

3.1. **Appropriate weights.** In order to construct the weights used in our definition of Bergman spaces, we need to define a couple of preliminary concepts. Thus, we define the hypergeometric functions F by

$$F\left[\begin{array}{c} a,b,c \\ u,v \end{array}; t\right] := \sum_{n=0}^{\infty} \frac{(a)_n (b)_n (c)_n}{n!\,(u)_n (v)_n} t^n, \qquad |t| < 1,$$

and

$$F\left[\begin{array}{c} a,b,c \end{array}; t\right] := \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{n!\,(c)_n} t^n, \qquad |t| < 1,$$

and by analytic continuation elsewhere. Here $(a)_n$ denotes the rising factorial, $(a)_n = a(a+1)\cdots(a+n-1)$.

Then one solution to the equation $\Delta_h \log v = -4(n-1)^2$ is given (see [6, Proposition 6.1]) by

$$v = \Phi_n(r) = \exp\left\{ \frac{(n-1)(2-n)r^2}{n} F\left[\begin{array}{c} 1,1,2-\frac{n}{2} \\ 2,1+\frac{n}{2} \end{array}; r^2\right]\right\} (1-r^2)^{n-1}. \tag{3.4}$$

We will sometimes write $\Phi_n(x)$ instead of $\Phi_n(|x|)$. Note that $\Phi_n(r) \leq (1-r^2)^{n-1}$, with strict inequality for $n > 2$ and $r > 0$. If $n = 2$, then $\Phi_n(|x|) = 1 - |x|^2$ and this coincides with the case treated in [13]. Observe, moreover, that $E_n(1-r^2)^{n-1} < \Phi_n(r)$, where

$$E_n = \exp\left\{ \frac{(n-1)(2-n)}{n} F\left[\begin{array}{c} 1,1,2-\frac{n}{2} \\ 2,1+\frac{n}{2} \end{array}; 1\right]\right\}.$$

[1]This definition can be extended to the case of upper semicontinuous functions, by using the so-called invariant mean value property [15]

This shows that the Bergman norm induced by Φ_n is equivalent to that induced by the weight $(1-r^2)^{n-1}$. The use of these weights appears for the first time in [6].

3.2. **Admissible monoid.** We define $\mathfrak{E}$ to be the set of real-analytic functions g in $\mathbb{B}$ so that $f := \log|g|$ is $\mathcal{M}-$subharmonic. For $n=2$, this class of mappings contains holomorphic functions in the unit disk, however in higher-dimensional case, this is not the case, see [6][Remark 1.2.] for more details on this issue. In other words, for $n = 2m \geq 4$, the $\mathfrak{E}$ contains very few holomorphic mappings.

Let $\mathfrak{E}_+ = \{f \in \mathfrak{E} : f > 0\}$. Observe that $\mathfrak{E}$ is a monoid where the operation is simply the multiplication of two functions. Observe that $1 = e^0$, so $1 \in \mathfrak{E}$. This monoid also contains the Abelian group $\mathcal{G} = \{e^f : \Delta_h f = 0\}$. Fos some additional properties of this monoid which is also a convex cone see [6] and [7].

Definition 3.1. For $0 < p < \infty$ and $\beta > 1$ we say that a smooth function f in $\mathbb{B}$ belongs to the $\mathcal{M}-$Bergman space B^p_β if

$$\|f\|^p_{B^p_\beta} = c(\beta)\int_{\mathbb{B}} |f(x)|^p \Phi^\beta_n(|x|)d\tau(x) < \infty,$$

where

$$\frac{1}{c(\beta)} = \left(\int_{\mathbb{B}} \Phi^\beta_n(x)(1-|x|^2)^{-n}\frac{dV(x)}{\omega_n}\right).$$

Definition 3.2. For $0 < p < \infty$ and $\beta > 1$ we define the $\mathcal{M}-$Bergman monoid $\mathbf{B}^p_\beta$ consisting of functions f in $B^p_\beta \cap \mathfrak{E}$ having harmonic majorant $F_f \in B^p_\beta$.

We now state the main stability estimate in the hyperbolic setting.

Theorem 3.3. *There exists an explicitly computable constant $C_{n,\beta} > 0$, depending only on n and β, such that for every measurable set $\Omega \subset \mathbb{B}^n$ with $0 < \tau(\Omega) < \infty$ and every log-$\mathcal{M}$-subharmonic function $g \in \mathbf{B}^2_\beta \setminus \{0\}$, one has*

$$\inf_{\substack{|c|=\|g\|_{2,\beta}\\ a\in\mathbb{B}^n}} \frac{\|g - cg_a\|_{2,\beta}}{\|g\|_{2,\beta}} \leq C_{n,\beta}\Big(C(\tau(\Omega))\,\delta(g;\Omega,\beta)\Big)^{\frac{1}{n+2}},$$

where

$$C(\sigma) = \left(1 - c_\beta\int_0^\sigma v^*(s)\,ds\right)^{-1} = \big(1-\theta(\sigma)\big)^{-1},$$

and

$$\theta(\sigma) = nc_\beta\int_0^{r(\sigma)} \rho^{n-1}\Phi^\beta_n(\rho)(1-\rho^2)^{-n}\,d\rho.$$

Theorem 3.4 (Stability in dimension $n = 2$ (parameter $\alpha = \beta - 2$))**.** *Fix $\alpha > -1$ and let $\mathbb{D} = \mathbb{B}^2 \subset \mathbb{R}^2$. Let τ be the hyperbolic measure on $\mathbb{D}$,*

$$d\tau(x) = \frac{1}{(1-|x|^2)^2}\,dA(x).$$

There exists an explicitly computable constant $C_{2,\alpha}$ such that for every measurable set $\Omega \subset \mathbb{D}$ with $0 < \tau(\Omega) < \infty$ and every log-subharmonic function $g \in \mathbf{B}^2_\alpha \setminus \{0\}$ we have

$$\inf_{\substack{|c|=\|g\|_{2,\alpha}\\ a\in\mathbb{D}}} \frac{\|g - c\,g_a\|_{2,\alpha}}{\|g\|_{2,\alpha}} \leq C_{2,\alpha}\Big(C(\tau(\Omega))\,\delta(g;\Omega)\Big)^{\frac{1}{4}}.$$

Here the norm is the (normalized) weighted Bergman L^2 norm

$$\|h\|^2_{2,\alpha} := \frac{\alpha+1}{\pi}\int_{\mathbb{D}} |h(z)|^2(1-|z|^2)^\alpha\,dA(z),$$

and the deficit is

$$\delta(g;\Omega) := 1 - \frac{\displaystyle\int_\Omega |g(z)|^2(1-|z|^2)^\alpha\,dA(z)}{\theta_\alpha(\tau(\Omega))\displaystyle\int_{\mathbb{D}} |g(z)|^2(1-|z|^2)^\alpha\,dA(z)} \in [0,1].$$

The extremal family is given by the normalized kernels (up to unimodular constants):

$$g_a(z) = \frac{(1-|a|^2)^{\frac{\alpha+2}{2}}}{|1-\overline{a}z|^{\alpha+2}}, \qquad a \in \mathbb{D}, \qquad \|g_a\|_{2,\alpha} = 1.$$

Finally, the amplification factor is $C(\sigma) = (1-\theta_\alpha(\sigma))^{-1}$, *where*

$$\theta_\alpha(\sigma) = 1 - \big(1 - r(\sigma)^2\big)^{\alpha+1}, \qquad \tau(\{|z| < r(\sigma)\}) = \sigma.$$

In dimension 2 *one has explicitly*

$$\tau(\{|z| < r\}) = \pi\,\frac{r^2}{1-r^2}, \qquad \text{so} \qquad r(\sigma)^2 = \frac{\sigma}{\sigma+\pi},$$

and therefore

$$C(\sigma) = \big(1-\theta_\alpha(\sigma)\big)^{-1} = \big(1-r(\sigma)^2\big)^{-(\alpha+1)} = \left(\frac{\sigma+\pi}{\pi}\right)^{\alpha+1}.$$

We will start from the sharp pointwise estimate for Bergman spaces of logsubharmonic functions. The following function has been maximizes in [13] for $n = 2$ and in [7] for $n > 2$.

$$R_n(f,\Omega) = \frac{c(\beta)\int_\Omega |f(x)|^2 \Phi_n^\beta(|x|)\,d\tau(x)}{\|f\|^2_{\mathbf{B}^2_\beta}} \tag{3.5}$$

over all $f \in \mathbf{B}^2_\beta$ and $\Omega \subset \mathbb{B}$ with $\tau(\Omega) = s$.

Theorem 3.5. [13, 7] *Let* $\beta > 1$, *and* $s > 0$ *be fixed. Among all functions* $f \in \mathbf{B}^2_\beta$ *and among all measurable sets* $\Omega \subset \mathbb{B}$ *such that* $\tau(\Omega) = s$, *the quotient* $R_n(f,\Omega)$ *as defined in* (3.5) *satisfies the inequality*

$$\delta(f,\Omega) := 1 - \frac{R_n(f,\Omega)}{R_n(1,\mathbb{B}_s)} \geq 0, \tag{3.6}$$

where $\mathbb{B}_s$ *is a ball centered at the origin with* $\tau(\mathbb{B}_s) = s$. *Moreover, there is equality in* (3.6) *if and only if* f *belongs to the extremal set and* Ω *is a ball such that* $\tau(\Omega) = s$. *The extremal set consists of functions* $g(x) = \frac{\Phi_n^{\beta/2}(|m(x)|)}{\Phi_n^{\beta/2}(|x|)}$, *where* $m \in \mathcal{M}$ *is a Möbius transformation of the unit ball onto itself.*

Proof of Theorem 3.3 for $n = 2$. Write $\mathbb{D} = \mathbb{B}^2$ and

$$d\tau(z) = \frac{1}{(1-|z|^2)^2}\,dA(z), \qquad \Phi_2(z) = 1 - |z|^2.$$

Fix $\alpha > -1$ and let $\Omega \subset \mathbb{D}$ be measurable with $0 < \tau(\Omega) =: s_0 < \infty$. By homogeneity we may assume $\|g\|_{2,\alpha} = 1$; then

$$\inf_{\substack{|c|=\|g\|_{2,\alpha}\\ |a|<1}} \frac{\|g - cg_a\|_{2,\alpha}}{\|g\|_{2,\alpha}} = \inf_{|a|<1} \|g - g_a\|_{2,\alpha}.$$

1) Rearrangements and the level-set deficit. Set

$$u(z) := (1-|z|^2)^{\alpha+2}|g(z)|^2, \qquad v(z) := (1-|z|^2)^{\alpha+2}.$$

Let u^*, v^* be the decreasing rearrangements of u, v with respect to τ. Define the superlevel set $\Omega_{s_0} := \{u > u^*(s_0)\}$, so that $\tau(\Omega_{s_0}) = s_0$ and

$$\int_\Omega u\,d\tau \leq \int_{\Omega_{s_0}} u\,d\tau \overset{\tau(\Omega_{s_0})=s_0}{=} \int_0^{s_0} u^*(s)\,ds \qquad \text{(Hardy–Littlewood)}.$$

Let

$$\delta_{s_0} := 1 - \frac{\displaystyle\int_{\Omega_{s_0}} u\,d\tau}{\displaystyle\int_{\mathbb{B}_{s_0}} v\,d\tau} = 1 - \frac{\displaystyle\int_0^{s_0} u^*(s)\,ds}{\displaystyle\int_0^{s_0} v^*(s)\,ds},$$

where $\mathbb{B}_{s_0}$ denotes the hyperbolic disk centered at the origin with $\tau(\mathbb{B}_{s_0}) = s_0$. Since $\delta(g;\Omega)$ is defined by

$$\delta(g;\Omega) = 1 - \frac{\displaystyle\int_\Omega u\,d\tau}{\displaystyle\int_{\mathbb{B}_{s_0}} v\,d\tau} \geq 1 - \frac{\displaystyle\int_{\Omega_{s_0}} u\,d\tau}{\displaystyle\int_{\mathbb{B}_{s_0}} v\,d\tau} \overset{\text{def}}{=} \delta_{s_0},$$

the previous inequality implies

$$\delta_{s_0} \leq \delta(g;\Omega). \tag{3.7}$$

2) Explicit formula for v^* and θ_α. For $0 < r < 1$ one computes

$$\tau(\{|z| < r\}) = \int_0^{2\pi} \int_0^r \frac{1}{(1-\rho^2)^2}\, \rho\, d\rho\, d\theta = \pi\, \frac{r^2}{1-r^2}.$$

Solving

$$s = \pi\, \frac{r^2}{1-r^2} \quad \Longrightarrow \quad s(1-r^2) = \pi r^2 \quad \Longrightarrow \quad s = r^2(s+\pi),$$

gives

$$r^2 = \frac{s}{s+\pi}, \qquad 1 - r^2 = \frac{\pi}{s+\pi} = \Big(1 + \frac{s}{\pi}\Big)^{-1},$$

and hence

$$v^*(s) = (1-r^2)^{\alpha+2} = \Big(1 + \tfrac{s}{\pi}\Big)^{-(\alpha+2)}, \qquad s > 0. \tag{3.8}$$

In particular,

$$\int_0^\infty v^*(s)\, ds = \frac{\pi}{\alpha+1}.$$

Thus

$$\theta_\alpha(s) = \frac{\displaystyle\int_0^s v^*(t)\, dt}{\displaystyle\int_0^\infty v^*(t)\, dt} = 1 - \Big(1 + \frac{s}{\pi}\Big)^{-(\alpha+1)}, \qquad C(s) = \big(1 - \theta_\alpha(s)\big)^{-1} = \Big(1 + \frac{s}{\pi}\Big)^{\alpha+1}.$$

Since $\|g\|_{2,\alpha} = 1$, we also have

$$\int_0^\infty u^*(s)\, ds = \int_0^\infty v^*(s)\, ds = \frac{\pi}{\alpha+1}.$$

3) From δ_{s_0} to a bound on the peak size. Let

$$T := \|u\|_{L^\infty(\mathbb{D})} = \sup_{z\in\mathbb{D}} (1-|z|^2)^{\alpha+2} |g(z)|^2.$$

By the pointwise estimate

$$|g(z)|\, \Phi_2(z)^{(\alpha+2)/2} \le \|g\|_{2,\alpha} = 1,$$

we have $0 < T \le 1$. Set

$$s_T := \pi\big(T^{-1/(\alpha+2)} - 1\big), \qquad \text{so that} \qquad v^*(s_T) = T$$

by (3.8).

We next define the crossing point s^* by

$$u^*(s^*) = v^*(s^*).$$

To justify its existence and uniqueness, consider

$$R(s) := \frac{u^*(s)}{v^*(s)}.$$

By the rearrangement differential inequality in dimension 2,

$$(u^*)'(s) + \frac{\alpha+2}{\pi+s}\, u^*(s) \ge 0,$$

while (3.8) gives

$$(v^*)'(s) + \frac{\alpha+2}{\pi+s}\, v^*(s) = 0 \quad \Longrightarrow \quad \frac{(v^*)'(s)}{v^*(s)} = -\frac{\alpha+2}{\pi+s}.$$

Therefore

$$R'(s) = \frac{(u^*)'(s)v^*(s) - (v^*)'(s)u^*(s)}{(v^*(s))^2} = \frac{(u^*)'(s) - u^*(s)\frac{(v^*)'(s)}{v^*(s)}}{v^*(s)} = \frac{(u^*)'(s) + \frac{\alpha+2}{\pi+s}u^*(s)}{v^*(s)} \ge 0,$$

so R is nondecreasing.

Now $u^*(0) = T$ and $v^*(0) = 1$, hence

$$R(0) = \frac{u^*(0)}{v^*(0)} = T \le 1.$$

Also,

$$\int_0^\infty u^*(s)\,ds = \int_0^\infty v^*(s)\,ds.$$

Thus either $u^* \equiv v^*$ almost everywhere, in which case the conclusion is trivial, or else there exists $s^* \geq 0$ such that

$$R(s^*) = 1.$$

Since R is nondecreasing, we may choose

$$s^* := \inf\{s \geq 0 : \ R(s) = 1\},$$

and then

$$R(s) \leq 1 \text{ for } 0 \leq s \leq s^*, \qquad R(s) \geq 1 \text{ for } s \geq s^*,$$

hence

$$u^*(s) = R(s)v^*(s) \leq v^*(s) \quad (0 \leq s \leq s^*), \qquad u^*(s) = R(s)v^*(s) \geq v^*(s) \quad (s \geq s^*).$$

Define

$$I_* := \int_0^{s^*} (v^*(s) - u^*(s))\,ds.$$

We claim that

$$I_* \leq \frac{\pi}{\alpha+1}\, C(s_0)\,\delta_{s_0}. \tag{3.9}$$

Set

$$\varepsilon := \int_0^{s_0} (v^*(s) - u^*(s))\,ds = \delta_{s_0} \int_0^{s_0} v^*(s)\,ds.$$

Assume first that $s_0 \geq s^*$. Since

$$\int_0^\infty u^*(s)\,ds = \int_0^\infty v^*(s)\,ds,$$

we have

$$\varepsilon = \int_0^{s_0} (v^*(s) - u^*(s))\,ds \overset{\int_0^\infty u^* = \int_0^\infty v^*}{=} \int_{s_0}^\infty (u^*(s) - v^*(s))\,ds.$$

Moreover, for $s \geq s^*$,

$$u^*(s) - v^*(s) = u^*(s)\Big(1 - \frac{1}{R(s)}\Big).$$

Since R is nondecreasing and $s_0 \geq s^*$, it follows that

$$\varepsilon = \int_{s_0}^\infty u^*(s)\Big(1 - \frac{1}{R(s)}\Big)\,ds \geq \Big(1 - \frac{1}{R(s_0)}\Big) \int_{s_0}^\infty u^*(s)\,ds. \tag{3.10}$$

Likewise, for $s \in [s^*, s_0]$ we have

$$1 - \frac{1}{R(s)} \leq 1 - \frac{1}{R(s_0)},$$

hence

$$\int_{s^*}^{s_0} (u^*(s) - v^*(s))\,ds = \int_{s^*}^{s_0} u^*(s)\Big(1 - \frac{1}{R(s)}\Big)\,ds \leq \Big(1 - \frac{1}{R(s_0)}\Big) \int_{s^*}^{s_0} u^*(s)\,ds. \tag{3.11}$$

Set

$$A := 1 - \frac{1}{R(s_0)} \geq 0.$$

If $\int_{s_0}^\infty u^*(s)\,ds > 0$, then (3.10) implies

$$A \leq \frac{\varepsilon}{\int_{s_0}^\infty u^*(s)\,ds}.$$

Substituting this into (3.11), we obtain

$$\int_{s^*}^{s_0} (u^*(s) - v^*(s))\,ds \leq \varepsilon\, \frac{\int_{s^*}^{s_0} u^*(s)\,ds}{\int_{s_0}^\infty u^*(s)\,ds}. \tag{3.12}$$

Also,

$$\varepsilon = \int_0^{s_0} (v^*(s) - u^*(s))\,ds = I_* - \int_{s^*}^{s_0} (u^*(s) - v^*(s))\,ds,$$

so

$$I_* = \varepsilon + \int_{s^*}^{s_0} (u^*(s) - v^*(s))\,ds.$$

Combining this with (3.12) gives

$$I_* = \varepsilon + \int_{s^*}^{s_0} (u^*(s) - v^*(s))\, ds \overset{(3.12)}{\leq} \varepsilon + \varepsilon \frac{\int_{s^*}^{s_0} u^*(s)\, ds}{\int_{s_0}^{\infty} u^*(s)\, ds} = \varepsilon \frac{\int_{s^*}^{\infty} u^*(s)\, ds}{\int_{s_0}^{\infty} u^*(s)\, ds}. \tag{3.13}$$

If instead $\int_{s_0}^{\infty} u^*(s)\, ds = 0$, then (3.10) gives $\varepsilon = 0$, and

$$\int_{s^*}^{s_0} (u^*(s) - v^*(s))\, ds = 0,$$

so $I_* = 0$ and (3.13) is trivial.

Finally, since

$$\int_{s^*}^{\infty} u^*(s)\, ds \leq \int_0^{\infty} u^*(s)\, ds = \int_0^{\infty} v^*(s)\, ds = \frac{\pi}{\alpha+1},$$

and since $u^*(s) \geq v^*(s)$ for all $s \geq s_0$,

$$\int_{s_0}^{\infty} u^*(s)\, ds \geq \int_{s_0}^{\infty} v^*(s)\, ds,$$

we conclude from (3.13) that

$$I_* \leq \varepsilon \frac{\frac{\pi}{\alpha+1}}{\int_{s_0}^{\infty} v^*(s)\, ds}.$$

Since

$$\varepsilon = \delta_{s_0} \int_0^{s_0} v^*(s)\, ds,$$

this yields

$$I_* \leq \frac{\pi}{\alpha+1} \frac{\int_0^{s_0} v^*(s)\, ds}{\int_{s_0}^{\infty} v^*(s)\, ds} \delta_{s_0},$$

which is (3.9). Since

$$\int_{s_0}^{\infty} v^*(s)\, ds = \frac{\pi}{\alpha+1} \Big(1 + \frac{s_0}{\pi}\Big)^{-(\alpha+1)} = \frac{\pi}{\alpha+1} \big(1 - \theta_\alpha(s_0)\big),$$

it follows that

$$I_* \leq \varepsilon\, C(s_0) \leq \frac{\pi}{\alpha+1} C(s_0)\, \delta_{s_0}.$$

If $s_0 \leq s^*$, then

$$\varepsilon = \int_0^{s_0} v^*(s)\big(1 - R(s)\big)\, ds \geq (1 - R(s_0)) \int_0^{s_0} v^*(s)\, ds,$$

and

$$\int_{s_0}^{s^*} (v^*(s) - u^*(s))\, ds = \int_{s_0}^{s^*} v^*(s)\big(1 - R(s)\big)\, ds \leq (1 - R(s_0)) \int_{s_0}^{s^*} v^*(s)\, ds.$$

Therefore

$$I_* = \varepsilon + \int_{s_0}^{s^*} (v^*(s) - u^*(s))\, ds \leq \varepsilon \frac{\int_0^{s^*} v^*(s)\, ds}{\int_0^{s_0} v^*(s)\, ds} \leq \delta_{s_0} \int_0^{\infty} v^*(s)\, ds = \frac{\pi}{\alpha+1} \delta_{s_0} \leq \frac{\pi}{\alpha+1} C(s_0)\, \delta_{s_0}.$$

This proves (3.9).

Next we compare I_* with the clipped functional. Since $u^*(s) \leq T$ for all s and

$$u^*(s^*) = v^*(s^*) \leq T,$$

the monotonicity of v^* implies

$$v^*(s^*) \leq v^*(s_T) \quad \Longrightarrow \quad s^* \geq s_T.$$

Hence

$$I_* = \int_0^{s^*} (v^*(s) - u^*(s))\, ds \geq \int_0^{s_T} (v^*(s) - u^*(s))\, ds \geq \int_0^{s_T} (v^*(s) - T)\, ds =: I(T).$$

A direct computation gives

$$I(T) = \int_0^{\pi(T^{-1/(\alpha+2)} - 1)} \Big(\big(1 + \tfrac{s}{\pi}\big)^{-(\alpha+2)} - T \Big)\, ds$$
$$\overset{t=1+s/\pi}{=} \pi \int_1^{T^{-1/(\alpha+2)}} \big(t^{-(\alpha+2)} - T\big)\, dt$$

$$= \pi \left(\frac{1 - T^{\frac{\alpha+1}{\alpha+2}}}{\alpha+1} - T\big(T^{-1/(\alpha+2)} - 1\big) \right)$$
$$= \pi \left(\frac{1 - T^{\frac{\alpha+1}{\alpha+2}}}{\alpha+1} - T^{\frac{\alpha+1}{\alpha+2}} + T \right)$$
$$= \pi \left(T + \frac{1 - (\alpha+2)\, T^{\frac{\alpha+1}{\alpha+2}}}{\alpha+1} \right).$$

Moreover,

$$I(T) \geq \frac{\pi}{2(\alpha+2)}\,(1-T)^2, \qquad 0 < T \leq 1. \tag{3.14}$$

Combining (3.9) and (3.14), we obtain

$$\frac{\pi}{2(\alpha+2)}\,(1-T)^2 \leq I(T) \leq I_* \leq \frac{\pi}{\alpha+1}\, C(s_0)\, \delta_{s_0},$$

hence

$$(1-T)^2 \leq \frac{2(\alpha+2)}{\alpha+1}\, C(s_0)\, \delta_{s_0} \leq \frac{2(\alpha+2)}{\alpha+1}\, C(s_0)\, \delta(g;\Omega), \tag{3.15}$$

where in the last step we used (3.7).

4) From the peak size to the distance from extremals. Let

$$g_a(z) = \frac{(1-|a|^2)^{\frac{\alpha+2}{2}}}{|1-\overline{a}z|^{\alpha+2}}, \qquad a \in \mathbb{D},$$

so that $\|g_a\|_{2,\alpha} = 1$. Assuming Lemma 3.28 is available for this class,

$$\inf_{|a|<1} \|g - g_a\|_{2,\alpha}^2 \leq 2 - 2 \sup_{|a|<1} \big(g(a)\Phi_2(a)^{(\alpha+2)/2} \big) = 2 - 2\sqrt{T}. \tag{3.16}$$

Using $1 - \sqrt{T} \leq 1 - T$ for $0 < T \leq 1$, we get

$$\inf_{|a|<1} \|g - g_a\|_{2,\alpha}^2 \leq 2 - 2\sqrt{T} = 2(1-\sqrt{T}),$$

so

$$\inf_{|a|<1} \|g - g_a\|_{2,\alpha} \leq \sqrt{2(1-\sqrt{T})} \leq \sqrt{2}\,(1-T)^{1/2}.$$

Combining this with (3.15) yields

$$\inf_{|a|<1} \|g - g_a\|_{2,\alpha} \leq \sqrt{2} \left(\frac{2(\alpha+2)}{\alpha+1} \right)^{1/4} \big(C(s_0)\, \delta(g;\Omega) \big)^{1/4}.$$

Thus Theorem 3.3 holds for $n = 2$ with the explicit constant

$$C_{2,\alpha} = \sqrt{2} \left(\frac{2(\alpha+2)}{\alpha+1} \right)^{1/4}.$$

□

3.3. **On the sharpness of the planar exponent.** The exponent 1/4 in Theorem 3.4 arises as the product of two half-power losses inside the proof:

(i) Rearrangement-deficit step. The radial profile of the level-set deficit yields the *quadratic* bound (3.15),

$$(1-T)^2 \;\leq\; \frac{2(\alpha+2)}{\alpha+1}\, C(s_0)\, \delta(g;\Omega). \tag{3.17}$$

(ii) Peak-to-distance step. The invariant mean-value inequality (Lemma 3.28) yields the *linear* bound (3.16),

$$\inf_{|a|<1} \|g - g_a\|_{2,\alpha}^2 \;\leq\; 2 - 2\sqrt{T} \;\leq\; 2(1-T). \tag{3.18}$$

Composing (3.17) and (3.18) produces the rate $\delta^{1/4}$. By contrast, in the holomorphic regime of [4] both steps are linear (the quantitative RKHS argument upgrades (3.17) to a linear estimate), and one obtains the sharp $\delta^{1/2}$.

In this subsection we examine whether the loss in the log-subharmonic chain is genuine. We show that the natural family that saturates (3.17) is *forced* into a radial regime in which a direct bathtub argument (Proposition 2.13) provides a norm-distance bound $\delta^{1/2}$, bypassing (3.18). Together with the radial and Hilbert-valued results of the previous section, this strongly suggests that the exponent $1/4$ in Theorem 3.4 is an artefact of Lemma 3.28, and that the optimal stability rate is in fact $\delta^{1/2}$.

The natural saturator of (3.17). Fix $\alpha > -1$ and $T \in (0,1)$, and let $r_T \in (0,1)$ be the unique solution of

$$T = (1 - r_T^2)^{\alpha+2}. \tag{3.19}$$

Following the radial geometry of $v^*(s) = (1 + s/\pi)^{-(\alpha+2)}$, we consider the radial profile

$$g_T(z) := \begin{cases} \dfrac{T^{1/2}}{(1-|z|^2)^{(\alpha+2)/2}}, & |z| < r_T, \\ 1, & r_T \le |z| < 1. \end{cases} \tag{3.20}$$

The function g_T is continuous across the circle $|z| = r_T$ by (3.19), takes the maximal pointwise value T of $u_{g_T} = g_T^2(1-|z|^2)^{\alpha+2}$ throughout the disk $\{|z| < r_T\}$, and a direct computation gives the (un-normalized) Bergman mass

$$\|g_T\|_{2,\alpha}^2 = (1 - r_T^2)^{\alpha+1}\big(1 + (\alpha+1)r_T^2\big). \tag{3.21}$$

Indeed, splitting the integral over $\{|z| < r_T\}$ and $\{r_T \le |z| < 1\}$,

$$\int_{|z|<r_T} \frac{T}{(1-|z|^2)^{\alpha+2}}\, dA_\alpha(z) = T(\alpha+1)\,\frac{r_T^2}{1-r_T^2} = (\alpha+1)\, r_T^2 (1-r_T^2)^{\alpha+1},$$

where we used (3.19), while $\int_{r_T<|z|<1} dA_\alpha(z) = (1-r_T^2)^{\alpha+1}$. Adding these gives (3.21).

Although g_T has a corner along $|z| = r_T$ that prevents it from being literally log-subharmonic, the function $\log g_T(z) = \frac{1}{2}\log T - \frac{\alpha+2}{2}\log(1-|z|^2)$ on $\{|z| < r_T\}$ is smooth subharmonic, and $\log g_T \equiv 0$ on $\{r_T \le |z| < 1\}$. Standard mollification along radii, on a shell of width η around $|z| = r_T$, produces a smooth radial log-subharmonic function g_T^η with

$$\|g_T^\eta - g_T\|_{2,\alpha} \to 0, \qquad T - \sup_{z\in\mathbb{D}} u_{g_T^\eta}(z) \to 0, \qquad \delta(g_T^\eta; B_{s_0}) \to \delta(g_T; B_{s_0}) \qquad (\eta \to 0^+),$$

for every fixed centered hyperbolic ball B_{s_0} of τ-measure $s_0 \ge s_T := \pi(T^{-1/(\alpha+2)} - 1)$. The conclusions below are obtained for g_T itself; the same statements hold for the smooth approximants g_T^η up to errors that vanish with η.

Saturation of the rearrangement-deficit step. The pointwise distribution of u_{g_T} is explicit. Indeed, $u_{g_T} \equiv T$ on $\{|z| < r_T\}$, while $u_{g_T} = v$ on $\{|z| \ge r_T\}$, where $v(z) = (1-|z|^2)^{\alpha+2}$. Consequently, the decreasing rearrangement $u_{g_T}^*$ with respect to τ is

$$u_{g_T}^*(s) = \begin{cases} T, & 0 \le s \le s_T, \\ v^*(s), & s > s_T. \end{cases}$$

Pick $s_0 \ge s_T$ and consider the centered ball $\Omega = B_{s_0}$. Because u_{g_T} is radial and $\Phi_\alpha^{(g_T)}(r) := u_{g_T}(z)$ at $|z| = r$ is nonincreasing (constant equal to T on $[0, r_T]$ and equal to $(1-r^2)^{\alpha+2}$ on $[r_T, 1)$), the bathtub maximizer of $\int_\Omega u_{g_T}\, d\tau$ at fixed τ-measure s_0 is exactly B_{s_0}. Using (3.21) and the same change of variable $s \mapsto v^*(s)$ as in Step 3 of the proof of Theorem 3.4,

$$\delta(g_T; B_{s_0}) = \frac{1}{\|g_T\|_{2,\alpha}^2\, \theta_\alpha(s_0)} \int_0^{s_T} \big(v^*(s) - T\big)\, ds = \frac{I(T)}{\|g_T\|_{2,\alpha}^2\, \theta_\alpha(s_0)}, \tag{3.22}$$

where $I(T) = \int_0^{s_T} (v^*(s) - T)\, ds$ is precisely the quantity appearing in (3.14). Taylor expansion at $T = 1$, already carried out in the proof, yields

$$I(T) \;=\; \frac{\pi}{2(\alpha+2)}(1-T)^2 + O\big((1-T)^3\big), \qquad T \to 1^-. \tag{3.23}$$

Combining (3.21), (3.22) and (3.23), and using $\|g_T\|_{2,\alpha}^2 \to 1$ as $T \to 1$, we obtain

$$\delta(g_T; B_{s_0}) = \frac{\alpha+1}{2(\alpha+2)\,\theta_\alpha(s_0)}\,(1-T)^2 + O\big((1-T)^3\big), \qquad T \to 1^-. \tag{3.24}$$

Comparing with (3.17), we see that g_T saturates the rearrangement-deficit step up to the universal factor $\theta_\alpha(s_0)(1-\theta_\alpha(s_0)) \in (0, 1/4]$:

$$(1-T)^2 \;=\; \frac{2(\alpha+2)\,\theta_\alpha(s_0)}{\alpha+1}\,\delta(g_T; B_{s_0})\,\big(1+o(1)\big).$$

In other words, the inequality (3.17) cannot be quantitatively improved (in its $(1-T)^2$ exponent) within the radial log-subharmonic class.

The chain $\delta^{1/4}$ is loose on the saturator. If (3.18) were also saturated on the family $\{g_T\}$, the chain would force $\inf_a \|g_T - g_a\|_{2,\alpha} \asymp \delta(g_T; B_{s_0})^{1/4}$. We show that this is *not* the case: on g_T, the quantity $\inf_a \|g_T - g_a\|_{2,\alpha}^2$ is in fact controlled by $(1-T)^2$, not by $1-T$. Two observations make this precise.

(a) Application of Proposition 2.13. Normalize by setting $\widetilde{g}_T := g_T/\|g_T\|_{2,\alpha}$, so that $\|\widetilde{g}_T\|_{2,\alpha} = 1$. The function $\widetilde{g}_T$ remains radial, nondecreasing in $|z|$, and its bathtub ratio

$$\Phi_\alpha^{\widetilde{g}_T}(r) = \widetilde{g}_T(r)^2(1-r^2)^{\alpha+2} = \frac{1}{\|g_T\|_{2,\alpha}^2}\begin{cases} T, & 0 \le r \le r_T, \\ (1-r^2)^{\alpha+2}, & r_T \le r < 1 \end{cases}$$

is constant on $[0, r_T]$ and strictly decreasing on $[r_T, 1)$, hence *nonincreasing* on $[0,1)$. Proposition 2.13 therefore applies and yields, for every measurable $\Omega \subset \mathbb{D}$ with $\mu(\Omega) = s_0$,

$$\|\widetilde{g}_T - 1\|_{2,\alpha}^2 \;\le\; 3\,C_\alpha(s_0)\,\delta(\widetilde{g}_T; \Omega, \alpha). \tag{3.25}$$

Since $1 = g_0$ in the centered normalization $a = 0$, (3.25) is exactly the planar stability inequality with the *sharp* exponent $1/2$ on this family, and with constants of the same order as in Theorem 3.4. By Möbius covariance, the analogous bound holds with 1 replaced by any extremal g_a and g_T replaced by its conjugate by the Möbius transformation sending 0 to a.

(b) Direct asymptotics of the norm distance. The same conclusion may be checked by a direct calculation. Writing $\widetilde{g}_T = g_T/\|g_T\|_{2,\alpha}$ and using $\|\widetilde{g}_T\|_{2,\alpha} = \|1\|_{2,\alpha} = 1$,

$$\|\widetilde{g}_T - 1\|_{2,\alpha}^2 = 2 - 2\langle \widetilde{g}_T, 1\rangle_{2,\alpha} = 2\Big(1 - \frac{1}{\|g_T\|_{2,\alpha}}\int_{\mathbb{D}} g_T\,dA_\alpha\Big).$$

For the first integral, splitting again at $|z| = r_T$ and using (3.19),

$$\int_{\mathbb{D}} g_T\,dA_\alpha = T^{1/2}\int_{|z|<r_T}(1-|z|^2)^{-(\alpha+2)/2}dA_\alpha + \int_{r_T<|z|<1} dA_\alpha,$$

both integrals being elementary. Using the expansion $1 - r_T^2 = T^{1/(\alpha+2)} = 1 - (1-T)/(\alpha+2) + O((1-T)^2)$, a routine calculation gives, as $T \to 1^-$,

$$\|\widetilde{g}_T - 1\|_{2,\alpha}^2 = \frac{1}{\alpha+1}\,(1-T)^2 + O\big((1-T)^3\big). \tag{3.26}$$

In particular, $\|\widetilde{g}_T - 1\|_{2,\alpha} \asymp 1 - T$, in stark contrast with (3.18) which would predict $\|\widetilde{g}_T - 1\|_{2,\alpha} \lesssim (1-T)^{1/2}$.

Combining (3.24) and (3.26),

$$\|\widetilde{g}_T - 1\|_{2,\alpha} \;\sim\; \sqrt{\frac{2(\alpha+2)\,\theta_\alpha(s_0)}{\alpha+1}}\,\delta(\widetilde{g}_T; B_{s_0})^{1/2} \qquad (T \to 1^-).$$

On the natural saturator of (3.17), the actual norm distance behaves as $\delta^{1/2}$, not $\delta^{1/4}$. The looseness of the chain in the proof of Theorem 3.4 is therefore localized entirely in step (ii), Lemma 3.28.

The mechanism: Lemma 3.28 is loose on radial saturators. The reason (3.18) is loose on $\widetilde{g}_T$ can be isolated explicitly. The proof of Lemma 3.28 bounds $\langle g, g_a\rangle_{2,\alpha}$ from below by the pointwise value $\sqrt{T}$ of $g\,\Phi_2^{(\alpha+2)/2}$ at a, via the invariant mean-value property of the log-$\mathcal{M}$-subharmonic function

$$h(y) = g(m_a(y))\,\Phi_2^{-(\alpha+2)/2}(y)\,\Phi_2^{(\alpha+2)/2}(m_a(y))$$

at $y = 0$. This is equivalent to using the *first* term in the radial mean-value expansion of h. On $\widetilde{g}_T$ centered at $a = 0$, however, h coincides with $\widetilde{g}_T$ and the next mean-value coefficients exhibit a quadratic improvement, exactly accounting for the gain $1 - \sqrt{T} \approx (1-T)/2$ to $(1-T)^2$ visible in (3.26).

Conjecture. The previous analysis suggests that the exponent 1/4 in Theorem 3.4 is not sharp for the log-subharmonic class. We make this expectation explicit:

Conjecture 3.6. *For every $\alpha > -1$ there exists an explicitly computable constant $C'_{2,\alpha} > 0$ such that for every measurable $\Omega \subset \mathbb{D}$ with $0 < \tau(\Omega) < \infty$ and every log-subharmonic $g \in \mathbf{B}^2_\alpha(\mathbb{D}) \setminus \{0\}$,*

$$\inf_{\substack{|c|=\|g\|_{2,\alpha}\\ |a|<1}} \frac{\|g - cg_a\|_{2,\alpha}}{\|g\|_{2,\alpha}} \le C'_{2,\alpha}\big(C(\tau(\Omega))\,\delta(g;\Omega)\big)^{1/2}.$$

The preceding radial saturator shows that the exponent 1/2, if true, cannot be improved in general. We next record a complementary warning: the most direct route to Conjecture 3.6, namely a quadratic improvement of Lemma 3.28 in terms of the peak defect $1 - T$, is false even inside the zero-free holomorphic subclass. Thus a proof of Conjecture 3.6 must use more information than the single peak parameter T.

Proposition 3.7 (Failure of a quadratic peak-to-distance estimate)**.** *Fix $\alpha > -1$ and set $p := \alpha + 2$. There is no constant $M_\alpha < \infty$ such that every positive log-subharmonic g with $\|g\|_{2,\alpha} = 1$ satisfies*

$$\inf_{|a|<1} \|g - g_a\|^2_{2,\alpha} \le M_\alpha(1 - T)^2, \qquad T := \sup_{z\in\mathbb{D}} g(z)^2(1 - |z|^2)^p.$$

More precisely, for the zero-free holomorphic perturbations

$$g_\varepsilon(z) := \frac{\exp(\varepsilon \operatorname{Re} z^2)}{\|\exp(\varepsilon \operatorname{Re} z^2)\|_{2,\alpha}}, \qquad |\varepsilon| \ll 1,$$

one has

$$1 - T_\varepsilon = \frac{2}{(\alpha+2)(\alpha+3)}\,\varepsilon^2 + O(\varepsilon^4),$$

while

$$\inf_{|a|<1} \|g_\varepsilon - g_a\|^2_{2,\alpha} = \frac{1}{(\alpha+2)(\alpha+3)}\,\varepsilon^2 + O(|\varepsilon|^3).$$

Proof. Let $\varphi(z) := \operatorname{Re} z^2$ and let dA_α be the normalized weighted area measure. Since φ has zero angular mean,

$$\int_{\mathbb{D}} \varphi\, dA_\alpha = 0, \qquad \int_{\mathbb{D}} \varphi^2\, dA_\alpha = \frac{1}{2}\int_{\mathbb{D}} |z|^4\, dA_\alpha = \frac{1}{(\alpha+2)(\alpha+3)}.$$

Therefore

$$\|\exp(\varepsilon\varphi)\|^2_{2,\alpha} = \int_{\mathbb{D}} e^{2\varepsilon\varphi}\, dA_\alpha = 1 + \frac{2}{(\alpha+2)(\alpha+3)}\,\varepsilon^2 + O(\varepsilon^4).$$

For $|\varepsilon| < p/2$ the function

$$2\varepsilon \operatorname{Re} z^2 + p\log(1 - |z|^2)$$

is maximized at $z = 0$, hence

$$T_\varepsilon = \|\exp(\varepsilon\varphi)\|^{-2}_{2,\alpha} = 1 - \frac{2}{(\alpha+2)(\alpha+3)}\,\varepsilon^2 + O(\varepsilon^4).$$

It remains to estimate the distance from the extremal manifold. In the centered normalization $g_0 \equiv 1$, and for small a,

$$g_a(z) = 1 + p\operatorname{Re}(\overline{a}z) + O(|a|^2) \quad \text{in } L^2(dA_\alpha).$$

On the other hand,

$$g_\varepsilon(z) = 1 + \varepsilon \operatorname{Re} z^2 + O(\varepsilon^2) \quad \text{in } L^2(dA_\alpha).$$

The functions $\operatorname{Re} z^2$ and $\operatorname{Im} z^2$ are orthogonal, with respect to the radial measure dA_α, to the tangent space

$$\operatorname{span}_{\mathbb{R}}\{\operatorname{Re} z, \operatorname{Im} z\}$$

of the extremal manifold at g_0. The nearest point on the smooth manifold $\{g_a : |a| < 1\}$ therefore has $a = O(\varepsilon^2)$, and the squared distance is

$$\varepsilon^2\|\operatorname{Re} z^2\|^2_{2,\alpha} + O(|\varepsilon|^3) = \frac{1}{(\alpha+2)(\alpha+3)}\,\varepsilon^2 + O(|\varepsilon|^3).$$

This is incompatible with a bound by $M_\alpha(1 - T_\varepsilon)^2$, since the right-hand side would be $O(\varepsilon^4)$. □

The same perturbation also gives the usual sharpness obstruction for any possible exponent larger than 1/2. If B_s is the centered hyperbolic disk of fixed measure $s \in (0,\infty)$, then

$$\delta(g_\varepsilon; B_s) = 2\left(\int_{\mathbb{D}} \varphi^2\, dA_\alpha - \frac{1}{\theta_\alpha(s)} \int_{B_s} \varphi^2\, dA_\alpha\right)\varepsilon^2 + O(\varepsilon^4),$$

and the coefficient is strictly positive. Hence

$$\inf_{|a|<1} \|g_\varepsilon - g_a\|_{2,\alpha} \asymp \delta(g_\varepsilon; B_s)^{1/2}, \qquad \varepsilon \to 0.$$

Thus Conjecture 3.6, if proved, has the optimal exponent.

A second-variation route. The failure of Proposition 3.7 suggests that the right replacement for Lemma 3.28 is not another estimate depending only on the peak T. A more robust approach is to work directly near the extremal manifold. In centered coordinates the extremal is $g_0 \equiv 1$, and a positive log-subharmonic perturbation can be written as

$$g_\varepsilon(z) = \frac{\exp(\varepsilon\varphi(z))}{\|\exp(\varepsilon\varphi)\|_{2,\alpha}}, \qquad \Delta\varphi \geq 0.$$

The linearized cone already contains the main rigidity: for subharmonic φ,

$$\frac{1}{\theta_\alpha(s)} \int_{B_s} \varphi\, dA_\alpha \leq \int_{\mathbb{D}} \varphi\, dA_\alpha,$$

with equality for harmonic φ. Thus genuinely subharmonic directions create a first-order deficit, while harmonic directions must be controlled by the second variation. The latter part is completely diagonal.

Proposition 3.8 (Second variation in harmonic directions)**.** *Fix $\alpha > -1$ and a centered hyperbolic disk $B_s = \{|z| < r\}$, where $0 < r < 1$, and write $\theta = \theta_\alpha(s) = dA_\alpha(B_s)$. Let h be a real-valued harmonic polynomial with*

$$\int_{\mathbb{D}} h\, dA_\alpha = 0, \qquad h(z) = \sum_{k\geq 1} h_k(z), \qquad h_k(z) = \mathrm{Re}(b_k z^k).$$

Set

$$g_\varepsilon := \frac{e^{\varepsilon h}}{\|e^{\varepsilon h}\|_{2,\alpha}}.$$

Then, as $\varepsilon \to 0$,

$$\delta(g_\varepsilon; B_s) = 2\varepsilon^2 \sum_{k\geq 1} \lambda_k(s)\, \|h_k\|_{2,\alpha}^2 + O(|\varepsilon|^3),$$

where

$$\lambda_k(s) := 1 - \frac{\displaystyle\int_{B_s} |z|^{2k}\, dA_\alpha}{\theta \displaystyle\int_{\mathbb{D}} |z|^{2k}\, dA_\alpha} \in (0,1).$$

Equivalently, with $R = r^2$,

$$\lambda_k(s) = 1 - \frac{\displaystyle\int_0^R t^k (1-t)^\alpha\, dt}{\big(1-(1-R)^{\alpha+1}\big)\displaystyle\int_0^1 t^k(1-t)^\alpha\, dt}.$$

Moreover,

$$\inf_{|a|<1} \|g_\varepsilon - g_a\|_{2,\alpha}^2 = \varepsilon^2 \sum_{k\geq 2} \|h_k\|_{2,\alpha}^2 + O(|\varepsilon|^3).$$

Consequently, with

$$\Lambda_{\alpha,s} := \inf_{k\geq 2} \lambda_k(s) > 0,$$

one has the local estimate

$$\inf_{|a|<1} \|g_\varepsilon - g_a\|_{2,\alpha}^2 \leq \frac{1}{2\Lambda_{\alpha,s}}\, \delta(g_\varepsilon; B_s) + O(|\varepsilon|^3).$$

Proof. Since dA_α is radial, distinct harmonic modes are orthogonal in $L^2(dA_\alpha)$, and the same is true after restriction to the centered disk B_s. Also $\int_{\mathbb{D}} h\,dA_\alpha = \int_{B_s} h\,dA_\alpha = 0$. Expanding

$$\|e^{\varepsilon h}\|_{2,\alpha}^2 = \int_{\mathbb{D}} e^{2\varepsilon h}\,dA_\alpha = 1 + 2\varepsilon^2 \int_{\mathbb{D}} h^2\,dA_\alpha + O(|\varepsilon|^3)$$

and similarly on B_s, we obtain

$$\int_{B_s} g_\varepsilon^2\,dA_\alpha = \theta + 2\varepsilon^2 \left(\int_{B_s} h^2\,dA_\alpha - \theta \int_{\mathbb{D}} h^2\,dA_\alpha \right) + O(|\varepsilon|^3).$$

The stated expansion of the deficit follows by dividing by θ and using orthogonality of the modes. The positivity of $\lambda_k(s)$ is just the strict inequality between the conditional $2k$-moment of $|z|$ on the disk B_s and the corresponding moment on the full disk; the infimum over $k \geq 2$ is positive because $\lambda_k(s) \to 1$ as $k \to \infty$.

For the distance expansion, note that

$$g_\varepsilon = 1 + \varepsilon h + O(\varepsilon^2) \quad \text{in } L^2(dA_\alpha),$$

while the extremal manifold has tangent space at $g_0 \equiv 1$ given by

$$\{p\,\mathrm{Re}(\overline{a}z) : a \in \mathbb{C}\}, \qquad p = \alpha + 2.$$

Indeed,

$$g_a(z) = \frac{(1-|a|^2)^{p/2}}{|1-\overline{a}z|^p} = 1 + p\,\mathrm{Re}(\overline{a}z) + O(|a|^2) \quad \text{in } L^2(dA_\alpha).$$

Optimizing over a therefore removes precisely the $k = 1$ harmonic component to first order, and the claimed squared-distance expansion follows. □

Corollary 3.9 (Local half-stability in the log-harmonic sector)**.** *Fix $\alpha > -1$ and $s \in (0, \infty)$. There exist constants $\eta_{\alpha,s} > 0$ and $C_{\alpha,s} < \infty$ with the following property. Let h be a real-valued harmonic function in a neighbourhood of $\overline{\mathbb{D}}$, with $\int_{\mathbb{D}} h\,dA_\alpha = 0$ and $\|h\|_{L^\infty(\mathbb{D})} \leq \eta_{\alpha,s}$, and set*

$$g_h := \frac{e^h}{\|e^h\|_{2,\alpha}}.$$

Then

$$\inf_{|a|<1} \|g_h - g_a\|_{2,\alpha}^2 \leq C_{\alpha,s}\,\delta(g_h; B_s).$$

Equivalently, in this neighbourhood of the extremal manifold,

$$\inf_{|a|<1} \|g_h - g_a\|_{2,\alpha} \leq C_{\alpha,s}^{1/2}\,\delta(g_h; B_s)^{1/2}.$$

Proof. Write the harmonic expansion

$$h = h_1 + h_{\geq 2}, \qquad h_1(z) = \mathrm{Re}(b_1 z), \qquad h_{\geq 2} = \sum_{k \geq 2} h_k.$$

By Proposition 3.8, applied to the path εh at $\varepsilon = 1$, and by choosing $\|h\|_{L^\infty}$ small enough to absorb the cubic remainder, we have

$$\delta(g_h; B_s) \geq \sum_{k \geq 1} \lambda_k(s) \|h_k\|_{2,\alpha}^2. \tag{3.27}$$

Here $\lambda_k(s) > 0$ for every k, and

$$\Lambda_{\alpha,s} := \inf_{k \geq 2} \lambda_k(s) > 0.$$

The map $a \mapsto g_a$ is a smooth two-dimensional submanifold of $L^2(dA_\alpha)$ near $a = 0$, with tangent space

$$\mathcal{T}_0 = \mathrm{span}_{\mathbb{R}}\{\mathrm{Re}\,z, \mathrm{Im}\,z\}.$$

The nearest-point projection onto this submanifold is C^2 in a small neighbourhood of $g_0 = 1$. Since

$$g_h = 1 + h + O(\|h\|_{L^\infty}^2) \quad \text{in } L^2(dA_\alpha),$$

and since the component h_1 lies exactly in $\mathcal{T}_0$, the standard tubular-neighbourhood estimate gives

$$\inf_{|a|<1} \|g_h - g_a\|_{2,\alpha}^2 \leq 2\|h_{\geq 2}\|_{2,\alpha}^2 + C_\alpha \|h\|_{L^\infty}^4. \tag{3.28}$$

After decreasing $\eta_{\alpha,s}$ if necessary, the fourth-order term is absorbed by the right-hand side of (3.27); indeed, if $h_{\geq 2} = 0$, then g_h is already tangent to the extremal manifold to first order and the left-hand side is $O(\|h_1\|^4)$, while (3.27) is $\gtrsim \|h_1\|^2$ unless $h_1 = 0$. Thus

$$\inf_{|a|<1} \|g_h - g_a\|_{2,\alpha}^2 \leq \frac{4}{\Lambda_{\alpha,s}}\, \delta(g_h; B_s),$$

after changing constants. This proves the claim. □

Remark 3.10 (Why this is the correct local model)**.** *The preceding corollary applies to zero-free holomorphic functions after taking $h = \log|f| - \int_{\mathbb{D}} \log|f|\, dA_\alpha$. It therefore contains both the tangent directions $k = 1$, which correspond to moving along the extremal manifold, and the first genuinely transverse holomorphic perturbations $k \geq 2$, such as $\operatorname{Re} z^2$. The latter are exactly the perturbations in Proposition 3.7, where the exponent $1/2$ is seen to be sharp.*

Local coercivity on the subharmonic cone. We now close the local log-subharmonic argument. Let

$$P(F) := \int_{\mathbb{D}} F\, dA_\alpha, \qquad Q_s(F) := \frac{1}{\theta_\alpha(s)} \int_{B_s} F\, dA_\alpha,$$

and normalize real perturbations by $P(\varphi) = 0$. For $g_\varphi = e^\varphi / \|e^\varphi\|_{2,\alpha}$, a direct Taylor expansion gives

$$\delta(g_\varphi; B_s) = -2Q_s(\varphi) + 2\big(P(\varphi^2) - Q_s(\varphi^2)\big) + O(\|\varphi\|_{L^\infty} \|\varphi\|_{2,\alpha}^2). \tag{3.29}$$

If φ is subharmonic, then $Q_s(\varphi) \leq P(\varphi) = 0$, because the circular mean of a subharmonic function is increasing in the radius and B_s is a centered disk. Thus the first term in (3.29) is nonnegative and controls the genuinely subharmonic directions to first order. The only directions in which this first variation vanishes are harmonic directions, and those are controlled by Corollary 3.9.

Lemma 3.11 (Green-potential coercivity)**.** *Fix $\alpha > -1$ and $s \in (0, \infty)$. Let $G(z, \zeta)$ be the Green function of the disk, normalized so that $G(\cdot, \zeta) \geq 0$ and $G(\cdot, \zeta) = 0$ on $\partial\mathbb{D}$. If*

$$\psi(z) = -\int_{\mathbb{D}} G(z, \zeta)\, d\mu(\zeta), \qquad \mu \geq 0,$$

is bounded in $\mathbb{D}$, then

$$\|\psi - P(\psi)\|_{2,\alpha}^2 \leq C_{\alpha,s}\, \|\psi\|_{L^\infty(\mathbb{D})} \big(P(\psi) - Q_s(\psi)\big).$$

Proof. Set

$$A_s(\psi) := P(\psi) - Q_s(\psi).$$

For the Green kernel define

$$H_{\alpha,s}(\zeta) := P_z[-G(z, \zeta)] - Q_{s,z}[-G(z, \zeta)].$$

Since $z \mapsto -G(z, \zeta)$ is subharmonic and has zero boundary values, the weighted mean over the smaller centered disk is no larger than the weighted mean over the whole disk; hence $H_{\alpha,s}(\zeta) \geq 0$. In fact $H_{\alpha,s}(\zeta) > 0$ for $\zeta \in \mathbb{D}$. Moreover, by the usual boundary asymptotics of the Green function,

$$H_{\alpha,s}(\zeta) \asymp_{\alpha,s} 1 - |\zeta| \qquad (|\zeta| \to 1^-),$$

and

$$\| - G(\cdot, \zeta) - P_z[-G(z, \zeta)]\|_{2,\alpha}^2 \lesssim_{\alpha,s} (1 - |\zeta|)\, \| - G(\cdot, \zeta)\|_{L^\infty(\{|z-\zeta|>(1-|\zeta|)/2\})}.$$

For bounded Green potentials this kernel estimate integrates against $d\mu(\zeta)$ after the standard truncation $G_N = \min\{G, N\}$ and monotone convergence. Equivalently, applying the estimate to $\psi_N = -\int G_N\, d\mu$ and letting $N \to \infty$ gives

$$\|\psi - P(\psi)\|_{2,\alpha}^2 \leq C_{\alpha,s} \|\psi\|_{L^\infty} \int_{\mathbb{D}} H_{\alpha,s}(\zeta)\, d\mu(\zeta).$$

By linearity of $P - Q_s$,

$$\int_{\mathbb{D}} H_{\alpha,s}(\zeta)\, d\mu(\zeta) = P(\psi) - Q_s(\psi),$$

which proves the claim. □

Proposition 3.12 (Local cone coercivity)**.** *For every $\alpha > -1$ and $s \in (0, \infty)$ there exist $\eta_{\alpha,s} > 0$ and $K_{\alpha,s} < \infty$ such that every real-analytic subharmonic φ on a neighbourhood of $\overline{\mathbb{D}}$ with $P(\varphi) = 0$ and $\|\varphi\|_{L^\infty} \leq \eta_{\alpha,s}$ satisfies*

$$\inf_{\ell \in \mathcal{T}_0} \|\varphi - \ell\|_{2,\alpha}^2 \leq K_{\alpha,s} \left[-Q_s(\varphi) + P(\varphi^2) - Q_s(\varphi^2)\right],$$

where $\mathcal{T}_0 = \operatorname{span}_{\mathbb{R}}\{\operatorname{Re} z, \operatorname{Im} z\}$.

Proof. Use the Riesz decomposition in the disk:

$$\varphi = h + \psi, \qquad \psi(z) = -\int_{\mathbb{D}} G(z,\zeta)\,\frac{\Delta\varphi(\zeta)}{2\pi}\,dA(\zeta),$$

where h is harmonic in a neighbourhood of $\overline{\mathbb{D}}$ and $\psi \le 0$. Since P and Q_s are radial averages, they agree on harmonic functions:

$$P(h) = Q_s(h) = h(0).$$

The normalization $P(\varphi) = 0$ gives $P(h) = -P(\psi)$, and therefore

$$-Q_s(\varphi) = P(\psi) - Q_s(\psi) \ge 0. \tag{3.30}$$

Write $h = h_1 + h_{\ge 2} + h(0)$, where $h_1 \in \mathcal{T}_0$ and $h_{\ge 2}$ is the sum of harmonic modes of order at least two. The harmonic spectral-gap computation in Proposition 3.8 gives

$$\|h_{\ge 2}\|_{2,\alpha}^2 \le C_{\alpha,s}\big(P(h^2) - Q_s(h^2)\big). \tag{3.31}$$

By Lemma 3.11 and $\|\psi\|_\infty \le 2\|\varphi\|_\infty$ after decreasing the neighbourhood if necessary,

$$\|\psi - P(\psi)\|_{2,\alpha}^2 \le C_{\alpha,s}\eta_{\alpha,s}\,\big(P(\psi) - Q_s(\psi)\big). \tag{3.32}$$

Now expand the quadratic part:

$$P(\varphi^2) - Q_s(\varphi^2) = P(h^2) - Q_s(h^2) + 2\big(P(h\psi) - Q_s(h\psi)\big) + P(\psi^2) - Q_s(\psi^2).$$

The two terms containing ψ are controlled by Cauchy's inequality, using (3.32). More precisely, for any fixed small $\varepsilon > 0$,

$$\big|P(h\psi) - Q_s(h\psi)\big| + \big|P(\psi^2) - Q_s(\psi^2)\big| \le \varepsilon\|h_{\ge 2}\|_{2,\alpha}^2 C_{\alpha,s,\varepsilon}\eta_{\alpha,s}\big(P(\psi) - Q_s(\psi)\big).$$

Choosing first ε and then $\eta_{\alpha,s}$ small, and using (3.30) and (3.31), we obtain

$$\|h_{\ge 2}\|_{2,\alpha}^2 + \|\psi - P(\psi)\|_{2,\alpha}^2 \le K_{\alpha,s}\left[-Q_s(\varphi) + P(\varphi^2) - Q_s(\varphi^2)\right].$$

Finally,

$$\inf_{\ell\in\mathcal{T}_0}\|\varphi - \ell\|_{2,\alpha}^2 \le 2\|h_{\ge 2}\|_{2,\alpha}^2 + 2\|\psi - P(\psi)\|_{2,\alpha}^2,$$

because the constants cancel through $P(\varphi) = 0$ and h_1 is the tangent component. This proves the proposition. □

Theorem 3.13 (Local half-stability for log-subharmonic perturbations)**.** *For every $\alpha > -1$ and $s \in (0,\infty)$ there are constants $\eta_{\alpha,s} > 0$ and $C_{\alpha,s} < \infty$ such that, whenever φ is real-analytic and subharmonic on a neighbourhood of $\overline{\mathbb{D}}$, $P(\varphi) = 0$, and $\|\varphi\|_{L^\infty} \le \eta_{\alpha,s}$, the function $g_\varphi = e^\varphi/\|e^\varphi\|_{2,\alpha}$ satisfies*

$$\inf_{|a|<1}\|g_\varphi - g_a\|_{2,\alpha}^2 \le C_{\alpha,s}\,\delta(g_\varphi; B_s).$$

Proof. The Taylor expansion of the normalized exponential map gives

$$g_\varphi = 1 + \varphi + O(\|\varphi\|_{L^\infty}^2) \quad \text{in } L^2(dA_\alpha).$$

As in the proof of Corollary 3.9, the extremal manifold has tangent space $\mathcal{T}_0$ at $g_0 = 1$, hence the tubular estimate

$$\inf_{|a|<1}\|g_\varphi - g_a\|_{2,\alpha}^2 \le 2\inf_{\ell\in\mathcal{T}_0}\|\varphi - \ell\|_{2,\alpha}^2 + C_\alpha\|\varphi\|_{L^\infty}\|\varphi\|_{2,\alpha}^2.$$

By Proposition 3.12, the first term is bounded by the quadratic expression on the right-hand side of (3.29). Choosing $\eta_{\alpha,s}$ small enough, the error term is absorbed into the same quantity, and (3.29) yields

$$\inf_{|a|<1}\|g_\varphi - g_a\|_{2,\alpha}^2 \le C_{\alpha,s}\,\delta(g_\varphi; B_s).$$

□

This theorem is the local form of the expected 1/2 stability. We now turn an arbitrary near-extremizer into this normal form, after applying a Mobius transformation and discarding the tangent directions, through the route

$$\text{near equality} \Longrightarrow \text{compactness to some } g_a \Longrightarrow \text{second-variation estimate at } g_a.$$

The real-analytic log-subharmonic assumption enters in the compactness step through the Riesz decomposition: the harmonic part is handled by Proposition 3.8, while the negative Green-potential part contributes to the first variation with the correct sign. This avoids the peak-only bottleneck responsible for the exponent 1/4.

The obstacle one has to remove is that the logarithmic chart is not compact in L^∞.

The global obstruction: spikes in the logarithmic chart. The local theorem does not require global extremizers to have uniformly small logarithm. In fact such a compactness statement is false. Let $a \in \mathbb{D}$, $\eta_j \to 0$ and $\rho_j \to 0$ so fast that $\eta_j |\log \rho_j| \to \infty$, and set

$$\phi_j(z) := \eta_j \log(|z-a|^2 + \rho_j^2) - P\big(\eta_j \log(|z-a|^2+\rho_j^2)\big), \qquad g_j := \frac{e^{\phi_j}}{\|e^{\phi_j}\|_{2,\alpha}}.$$

Then ϕ_j is real-analytic and subharmonic, $P(\phi_j) = 0$, and $g_j \to 1$ in $L^2(dA_\alpha)$; hence $\delta(g_j; B_s) \to 0$ for every fixed centered disk B_s. However

$$\|\phi_j\|_{L^\infty(\mathbb{D})} \geq |\phi_j(a)| \to \infty.$$

Thus the correct global problem is not to prove uniform compactness of $\log g$, but to show that the non-compact negative Green-potential part is already paid for by the first variation of the deficit.

Lemma 3.14 (Green kernel gap)**.** *Fix $\alpha > -1$ and $s \in (0,\infty)$. There is a constant $C_{\alpha,s} < \infty$ such that, for every $\zeta \in \mathbb{D}$,*

$$P_z G(z,\zeta) \leq C_{\alpha,s}\Big(Q_{s,z}G(z,\zeta) - P_z G(z,\zeta)\Big).$$

Proof. The quantity on the right is positive because $z \mapsto -G(z,\zeta)$ is subharmonic, hence its centered-disk average is no larger than its full weighted average. Equivalently, $Q_s G(\cdot,\zeta) \geq PG(\cdot,\zeta)$.

For fixed $\zeta \in \mathbb{D}$ the inequality is strict unless $G(\cdot,\zeta)$ is constant, which is impossible. Thus the ratio

$$R(\zeta) := \frac{P_z G(z,\zeta)}{Q_{s,z}G(z,\zeta) - P_z G(z,\zeta)}$$

is finite and continuous on compact subsets of $\mathbb{D}$. It remains only to check that R stays bounded as $|\zeta| \to 1$.

The Green function has the boundary expansion

$$G(z,\zeta) = (1-|\zeta|^2)\,\frac{1-|z|^2}{|z-\zeta/|\zeta||^2} + O\big((1-|\zeta|^2)^2\big)$$

locally uniformly for z away from the boundary point $\zeta/|\zeta|$. After integration against the radial measures defining P and Q_s, this gives

$$P_z G(z,\zeta) = (1-|\zeta|^2)A_\alpha + O((1-|\zeta|^2)^2),$$

and

$$Q_{s,z}G(z,\zeta) - P_z G(z,\zeta) = (1-|\zeta|^2)B_{\alpha,s} + O((1-|\zeta|^2)^2),$$

where

$$B_{\alpha,s} = Q_s\left(\frac{1-|z|^2}{|z-\xi|^2}\right) - P\left(\frac{1-|z|^2}{|z-\xi|^2}\right) > 0, \qquad |\xi| = 1.$$

By rotation invariance $B_{\alpha,s}$ is independent of ξ, and it is positive by the same strict subharmonicity argument above. Hence $R(\zeta)$ has a finite boundary limit. This proves the claim. □

Proposition 3.15 (Nonlinear Green absorption)**.** *Fix $\alpha > -1$ and $s \in (0,\infty)$. There is a constant $C_{\alpha,s} < \infty$ with the following property. Let*

$$\varphi = h + \psi, \qquad \psi(z) = -\int_{\mathbb{D}} G(z,\zeta)\,d\mu(\zeta), \qquad \mu \geq 0,$$

be the Riesz decomposition of a real-analytic subharmonic function on a neighbourhood of $\overline{\mathbb{D}}$, normalized by $P(\varphi) = 0$. Assume that the positive/harmonic part is in the local chart,

$$\|h - P(h)\|_{L^\infty(\mathbb{D})} \leq \eta_{\alpha,s}.$$

Then

$$\inf_{t\in\mathbb{R}} \left\| \frac{e^{h+\psi}}{\|e^{h+\psi}\|_{2,\alpha}} - \frac{e^{h+t}}{\|e^{h+t}\|_{2,\alpha}} \right\|_{2,\alpha}^2 \leq C_{\alpha,s}\,\big(P(\psi) - Q_s(\psi)\big).$$

Proof. Since $\|h - P(h)\|_\infty$ is small, multiplication by e^h only changes $L^2(dA_\alpha)$ norms by constants depending on α, s. Constants in h are removed by the normalization, so it suffices to prove the estimate with $h = 0$.

Write $\psi = -U$, where

$$U(z) := \int_{\mathbb{D}} G(z,\zeta)\,d\mu(\zeta) \geq 0.$$

Let $X = e^{-U}$. In any probability space,

$$\left\| \frac{X}{\|X\|_2} - 1 \right\|_2^2 \le 4\|X-1\|_2^2 \le 4\int (1-X).$$

Applying this with the probability measure dA_α gives

$$\left\| \frac{e^{\psi}}{\|e^{\psi}\|_{2,\alpha}} - 1 \right\|_{2,\alpha}^2 \le 4P(1-e^{-U}) \le 4P(U),$$

because $1 - e^{-U} \le U$. By Fubini and Lemma 3.14,

$$P(U) = \int_{\mathbb{D}} P_z G(z,\zeta)\, d\mu(\zeta) \le C_{\alpha,s} \int_{\mathbb{D}} \big(Q_{s,z}G(z,\zeta) - P_z G(z,\zeta)\big)\, d\mu(\zeta).$$

The last integral equals

$$Q_s(U) - P(U) = P(\psi) - Q_s(\psi),$$

which proves the desired estimate for $h = 0$. The general case follows from the initial norm-comparison reduction. □

Lemma 3.16 (Bathtub compactness at the radial profile)**.** *Let*

$$w_\alpha(z) := (1-|z|^2)^{\alpha+2}, \qquad \Theta(s) := \sup_{\tau(E)=s} \int_E w_\alpha\, d\tau = \int_{B_s} w_\alpha\, d\tau.$$

If $u_j \to w_\alpha$ in $L^1(d\tau)$ and $\tau(E_j) = s$ with

$$\int_{E_j} u_j\, d\tau \longrightarrow \Theta(s),$$

then

$$\tau(E_j \triangle B_s) \longrightarrow 0.$$

In particular,

$$\int_{B_s} u_j\, d\tau \longrightarrow \Theta(s).$$

Proof. Let $B_s = \{|z| < r_s\}$. Fix $\varepsilon > 0$ and choose an annulus

$$A_\varepsilon := \{r_s - \varepsilon < |z| < r_s + \varepsilon\}$$

with $\tau(A_\varepsilon) < \varepsilon$. On $B_s \setminus A_\varepsilon$ the radial function w_α is strictly larger than on $\mathbb{D} \setminus (B_s \cup A_\varepsilon)$; denote the gap by

$$\gamma_\varepsilon := \inf_{B_s \setminus A_\varepsilon} w_\alpha - \sup_{\mathbb{D}\setminus(B_s \cup A_\varepsilon)} w_\alpha > 0.$$

Since $\tau(E_j) = \tau(B_s)$, the mass of $B_s \setminus E_j$ must be replaced by the same amount of mass in $E_j \setminus B_s$, up to the annular error. Therefore the strict rearrangement inequality gives, after possibly reducing the constant,

$$\Theta(s) - \int_{E_j} w_\alpha\, d\tau \ge \gamma_\varepsilon \tau\big((E_j \triangle B_s) \setminus A_\varepsilon\big) - \gamma_\varepsilon \tau(A_\varepsilon).$$

The same deficit with u_j differs from this one by at most $2\|u_j - w_\alpha\|_{L^1(d\tau)}$, and therefore

$$\limsup_j \tau\big((E_j \triangle B_s) \setminus A_\varepsilon\big) = 0.$$

Letting $\varepsilon \downarrow 0$ proves the symmetric-difference convergence. The final assertion follows from

$$\left| \int_{B_s} u_j\, d\tau - \Theta(s) \right| \le \|u_j - w_\alpha\|_{L^1(d\tau)}.$$

□

Lemma 3.17 (Harmonic bathtub Schur complement)**.** *Fix $\alpha > -1$ and $s \in (0,\infty)$, and let $B_s = \{|z| < r_s\}$, $R := r_s^2$, $\theta := \theta_\alpha(s)$. For a real-valued harmonic function*

$$h(re^{i\vartheta}) = h(0) + \sum_{k\ge 1} r^k \operatorname{Re}(b_k e^{ik\vartheta})$$

on a neighbourhood of $\overline{\mathbb{D}}$*, set*

$$g_h := \frac{e^h}{\|e^h\|_{2,\alpha}}, \qquad \mathcal{D}_s(h) := \theta - \sup_{\tau(E)=s} \int_E g_h^2 \, dA_\alpha.$$

There are $\eta_{\alpha,s} > 0$ *and* $c_{\alpha,s} > 0$ *such that, if* $\|h - h(0)\|_{C^2(\overline{\mathbb{D}})} \le \eta_{\alpha,s}$*, then*

$$\mathcal{D}_s(h) \ge c_{\alpha,s} \sum_{k\ge 2} \|r^k \operatorname{Re}(b_k e^{ik\vartheta})\|_{2,\alpha}^2.$$

Equivalently,

$$\inf_{|a|<1} \left\| \frac{e^h}{\|e^h\|_{2,\alpha}} - g_a \right\|_{2,\alpha}^2 \le C_{\alpha,s}\, \mathcal{D}_s(h).$$

Proof. Constants in h disappear from g_h, so assume $h(0) = 0$. The bathtub maximizer for $g_h^2 dA_\alpha$ under the constraint $\tau(E) = s$ is a superlevel set of

$$e^{2h(z)}(1 - |z|^2)^{\alpha+2}.$$

Since

$$\partial_r (1 - r^2)^{\alpha+2}\big|_{r=r_s} = -2(\alpha + 2) r_s (1 - r_s^2)^{\alpha+1} < 0,$$

the implicit function theorem gives, for $\|h\|_{C^2}$ small, a boundary graph

$$E_h = \{re^{i\vartheta} : 0 < r < r_s + \rho(\vartheta)\}, \qquad \int_0^{2\pi} \rho(\vartheta)\, d\vartheta = O(\|\rho\|_{L^2}^2),$$

for the maximizing set. The last relation is the first-order form of the constraint $\tau(E_h) = \tau(B_s)$.

It remains to compute the quadratic part of $\theta - \int_{E_h} g_h^2 \, dA_\alpha$. Write

$$dA_\alpha = A(r)\, dr\, d\vartheta, \qquad d\tau = T(r)\, dr\, d\vartheta,$$

where

$$A(r) = \frac{\alpha+1}{\pi}(1 - r^2)^\alpha r, \qquad T(r) = \frac{r}{(1 - r^2)^2}.$$

The Lagrange multiplier for the unperturbed bathtub problem is

$$\lambda_s := \frac{A(r_s)}{T(r_s)} = \frac{\alpha+1}{\pi}(1 - r_s^2)^{\alpha+2}.$$

For variations preserving τ to first order, the second-order expansion of

$$\int_E g_h^2 \, dA_\alpha - \lambda_s \tau(E)$$

at $(h, E) = (0, B_s)$ is diagonal in angular Fourier modes. The block associated to the kth harmonic coefficient and the kth boundary displacement is

$$\Gamma_k |b_k|^2 - 2A(r_s) r_s^k \int_0^{2\pi} \operatorname{Re}(b_k e^{ik\vartheta})\, \rho_k(\vartheta)\, d\vartheta + \kappa_s \|\rho_k\|_{L^2(\partial\mathbb{D})}^2,$$

where

$$\kappa_s := -\frac{1}{2} T(r_s) \left(\frac{A}{T}\right)'(r_s) = \frac{(\alpha+1)(\alpha+2)}{\pi} r_s^2 (1 - r_s^2)^{\alpha-1} > 0,$$

and

$$\Gamma_k := \theta M_k - I_k(R), \quad M_k := (\alpha+1)\int_0^1 t^k (1-t)^\alpha \, dt, \quad I_k(R) := (\alpha+1)\int_0^R t^k (1-t)^\alpha \, dt.$$

Optimizing in the boundary coefficient gives the Schur complement

$$\mathfrak{s}_k = \Gamma_k - \frac{\alpha+1}{\alpha+2}(1 - R)^{\alpha+1} R^k.$$

The cancellation $\mathfrak{s}_1 = 0$ is exactly the infinitesimal Mobius translation. For $k \ge 2$ the coefficient is strictly positive. Indeed, with $d\nu(t) = (\alpha+1)(1-t)^\alpha \, dt$,

$$\Gamma_k = \int_0^R \int_R^1 (y^k - x^k)\, d\nu(y) d\nu(x),$$

and the subtracted term is R^{k-1} times the same expression with $k = 1$. Thus

$$\mathfrak{s}_k = \int_0^R \int_R^1 \big(y^k - x^k - R^{k-1}(y - x)\big)\, d\nu(y) d\nu(x) > 0 \qquad (k \ge 2),$$

because $x < R < y$ and

$$y^k - x^k = (y-x)\sum_{j=0}^{k-1} y^{k-1-j}x^j > R^{k-1}(y-x).$$

Moreover $\mathfrak{s}_k/M_k$ has a positive lower bound for $k \geq 2$: the displayed strict positivity handles finitely many k, while $\mathfrak{s}_k/M_k \to \theta$ as $k \to \infty$.

Consequently the quadratic part controls precisely the harmonic modes $k \geq 2$. The cubic remainder is bounded by $C_{\alpha,s}\|h\|_{C^2}\big(\sum_{k\geq 2}\|h_k\|_{2,\alpha}^2 + \|\rho\|_{L^2}^2\big)$, and is absorbed after decreasing $\eta_{\alpha,s}$. This proves the first estimate. The second estimate follows from the tubular expansion of the extremal manifold at $g_0 = 1$: the $k = 1$ harmonic modes are its tangent space, and the modes $k \geq 2$ are transversal. □

Lemma 3.18 (Boundary trace Green gap)**.** *Fix $\alpha > -1$ and $s \in (0,\infty)$, and write $B_s = \{|z| < r_s\}$. There is a constant $C_{\alpha,s} < \infty$ such that, for every positive Green potential*

$$U(z) := \int_{\mathbb{D}} G(z,\zeta)\,d\mu(\zeta), \qquad \mu \geq 0,$$

one has

$$\left(\int_0^{2\pi} U(r_s e^{i\vartheta})^2\,d\vartheta\right)^{1/2} \leq C_{\alpha,s}\big(Q_s(U) - P(U)\big).$$

Proof. It is enough to prove the estimate for a single kernel and then integrate by Minkowski. For fixed $\zeta \in \mathbb{D}$ set

$$N(\zeta) := \left(\int_0^{2\pi} G(r_s e^{i\vartheta},\zeta)^2\,d\vartheta\right)^{1/2}, \qquad H(\zeta) := Q_{s,z}G(z,\zeta) - P_z G(z,\zeta).$$

The function $G(\cdot,\zeta)$ has only a logarithmic singularity, so $N(\zeta) < \infty$ even when $|\zeta| = r_s$. Moreover $H(\zeta) > 0$ for $\zeta \in \mathbb{D}$, by the strict subharmonicity argument used in Lemma 3.14. Hence N/H is bounded on compact subsets of $\mathbb{D}$.

It remains to check the boundary regime $|\zeta| \to 1$. The boundary expansion used in Lemma 3.14 gives, uniformly in the angular variable on the fixed circle $|z| = r_s < 1$,

$$G(r_s e^{i\vartheta},\zeta) = (1-|\zeta|^2)\,\frac{1-r_s^2}{|r_s e^{i\vartheta} - \zeta/|\zeta||^2} + O\big((1-|\zeta|^2)^2\big).$$

Thus $N(\zeta) = O(1-|\zeta|^2)$, while $H(\zeta) \simeq_{\alpha,s} 1-|\zeta|^2$ by the same computation as in Lemma 3.14. Therefore $N(\zeta) \leq C_{\alpha,s}H(\zeta)$ for all ζ.

For general μ,

$$\|U(r_s e^{i\vartheta})\|_{L^2_\vartheta} \leq \int_{\mathbb{D}} N(\zeta)\,d\mu(\zeta) \leq C_{\alpha,s}\int_{\mathbb{D}} H(\zeta)\,d\mu(\zeta) = C_{\alpha,s}\big(Q_s(U) - P(U)\big).$$

□

Lemma 3.19 (Compactified normalized Green kernels)**.** *Fix $\alpha > -1$ and $s \in (0,\infty)$, and set*

$$H(\zeta) := Q_{s,z}G(z,\zeta) - P_z G(z,\zeta).$$

For $\zeta \in \mathbb{D}$ define

$$K_\zeta(z) := \frac{G(z,\zeta)}{H(\zeta)}.$$

Then the map $\zeta \mapsto K_\zeta$ extends continuously from $\mathbb{D}$ to $\overline{\mathbb{D}}$ with values in $L^1(dA_\alpha)$. Moreover

$$Q_s(K_\zeta) - P(K_\zeta) = 1 \qquad (\zeta \in \overline{\mathbb{D}}),$$

where the identity on $\partial\mathbb{D}$ is understood by continuity.

Proof. For $\zeta \in \mathbb{D}$ the identity $Q_s(K_\zeta) - P(K_\zeta) = 1$ is just the definition of H. Lemma 3.14 gives

$$P(K_\zeta) \leq C_{\alpha,s}, \qquad Q_s(K_\zeta) \leq C_{\alpha,s} + 1,$$

so the family is uniformly integrable with respect to dA_α on $\mathbb{D}$ and on the fixed disk B_s.

If ζ stays in a compact subset of $\mathbb{D}$, the usual logarithmic form of the Green function gives continuity of $G(\cdot,\zeta)$ in $L^1(dA_\alpha)$; since $H(\zeta) > 0$ is continuous there, $\zeta \mapsto K_\zeta$ is L^1-continuous on compact subsets of $\mathbb{D}$.

It remains to identify the boundary limits. Let $\zeta = \rho\xi$, $|\xi| = 1$ and $\rho \uparrow 1$. Uniformly for z in compact subsets of $\mathbb{D}$,

$$G(z, \rho\xi) = (1-\rho^2)\frac{1-|z|^2}{|z-\xi|^2} O\big((1-\rho^2)^2\big).$$

The same expansion, integrated against P and Q_s, gives

$$H(\rho\xi) = (1-\rho^2)B_{\alpha,s} + O\big((1-\rho^2)^2\big),$$

where

$$B_{\alpha,s} = Q_s\left(\frac{1-|z|^2}{|z-\xi|^2}\right) - P\left(\frac{1-|z|^2}{|z-\xi|^2}\right) > 0$$

is independent of ξ by rotation invariance. Hence

$$K_{\rho\xi}(z) \longrightarrow K_\xi(z) := \frac{1}{B_{\alpha,s}} \frac{1-|z|^2}{|z-\xi|^2}$$

locally uniformly in $\mathbb{D}$. To upgrade this to $L^1(dA_\alpha)$, we use the standard Green-kernel boundary estimate

$$G(z,\zeta) \le C\,\frac{(1-|z|^2)(1-|\zeta|^2)}{|1-\overline{\zeta}z|^2}.$$

Together with the lower boundary asymptotic $H(\zeta) \gtrsim_{\alpha,s} 1-|\zeta|^2$ for $|\zeta|$ close to 1, and the positive lower bound for H on compact subsets of $\mathbb{D}$, this gives the uniform tail estimate

$$\sup_{\zeta\in\mathbb{D}} \int_{\{|z|>r\}} K_\zeta(z)\, dA_\alpha(z) \longrightarrow 0 \qquad (r \uparrow 1).$$

Indeed, near the boundary the right-hand side is dominated, after rotation, by

$$C\int_r^1 \int_0^{2\pi} \frac{(1-\rho^2)^{\alpha+1}\rho}{|1-\rho e^{i\vartheta}|^2}\, d\vartheta\, d\rho \lesssim \int_r^1 (1-\rho)^\alpha\, d\rho \to 0.$$

The local uniform convergence on $\{|z| \le r\}$ and this uniform tail estimate give convergence in $L^1(dA_\alpha)$.

Thus $\zeta \mapsto K_\zeta$ extends continuously to $\overline{\mathbb{D}}$ in $L^1(dA_\alpha)$. Since both P and Q_s are continuous L^1 functionals on this compact family, the identity $Q_s(K_\zeta) - P(K_\zeta) = 1$ passes to $\zeta \in \partial\mathbb{D}$. □

Lemma 3.20 (Nonlinear one-sided Green gap)**.** *Fix $\alpha > -1$ and $s \in (0,\infty)$. Let h_j be real harmonic functions on a neighbourhood of $\overline{\mathbb{D}}$ with*

$$\|h_j - P(h_j)\|_{C^2(\overline{\mathbb{D}})} \to 0,$$

and let

$$\psi_j(z) = -\int_{\mathbb{D}} G(z,\zeta)\, d\mu_j(\zeta), \qquad \mu_j \ge 0,$$

be Green potentials such that $P(h_j + \psi_j) = 0$. Set

$$g_j := \frac{e^{h_j+\psi_j}}{\|e^{h_j+\psi_j}\|_{2,\alpha}}, \qquad \mathcal{D}_s(h_j+\psi_j) := \theta_\alpha(s) - \sup_{\tau(E)=s} \int_E g_j^2\, dA_\alpha.$$

If $\mathcal{D}_s(h_j+\psi_j) \to 0$, then

$$P(\psi_j) - Q_s(\psi_j) \longrightarrow 0.$$

Proof. Let E_j be maximizing sets. Since

$$\delta(g_j; E_j) = \mathcal{D}_s(h_j+\psi_j)/\theta_\alpha(s) \to 0,$$

the already proved planar stability estimate, Theorem 3.4, gives $\operatorname{dist}(g_j, \mathcal{E}) \to 0$. The smallness of the harmonic majorants $h_j - P(h_j)$ forces the limiting extremal to be the centered one: if a subsequence approached g_a with $a \ne 0$, then the harmonic part of $\log g_a$ would contain the non-zero first mode $(\alpha+2)\operatorname{Re}(\overline{a}z) + O(|a|^2)$, contradicting $\|h_j - P(h_j)\|_{C^2} \to 0$ and the fact that the Green part is non-positive. Thus

$$g_j \to 1 \qquad \text{in } L^2(dA_\alpha).$$

Since $h_j - P(h_j) \to 0$ uniformly and constants disappear under normalization, this is equivalent to

$$\frac{e^{-U_j}}{\|e^{-U_j}\|_{2,\alpha}} \to 1, \qquad U_j := -\psi_j = \int_{\mathbb{D}} G(\cdot,\zeta)\, d\mu_j(\zeta).$$

Suppose, toward a contradiction, that

$$A_j := Q_s(U_j) - P(U_j) = P(\psi_j) - Q_s(\psi_j)$$

does not tend to zero. Passing to a subsequence, assume $A_j \geq A_0 > 0$. Define

$$H(\zeta) := Q_{s,z}G(z,\zeta) - P_z G(z,\zeta) > 0, \qquad d\sigma_j(\zeta) := \frac{H(\zeta)}{A_j}\, d\mu_j(\zeta).$$

Then σ_j is a probability measure and

$$U_j(z) = A_j \int_{\mathbb{D}} K(z,\zeta)\, d\sigma_j(\zeta), \qquad K(z,\zeta) := \frac{G(z,\zeta)}{H(\zeta)}.$$

Regard σ_j as probability measures on the compact space $\overline{\mathbb{D}}$, with no boundary mass. Passing to a subsequence, σ_j converges weakly to some probability measure σ on $\overline{\mathbb{D}}$. By Lemma 3.19,

$$\int_{\overline{\mathbb{D}}} K_\zeta\, d\sigma_j(\zeta) \longrightarrow V := \int_{\overline{\mathbb{D}}} K_\zeta\, d\sigma(\zeta) \quad \text{in } L^1(dA_\alpha).$$

The function V is positive and nonconstant, and

$$Q_s(V) - P(V) = \int_{\overline{\mathbb{D}}} \big(Q_s(K_\zeta) - P(K_\zeta)\big)\, d\sigma(\zeta) = 1.$$

After passing to a further subsequence, the convergence to V holds pointwise a.e.

We now rule out both possibilities for A_j. First suppose that A_j is bounded along a subsequence, and write $A_j \to A \geq A_0$. Then

$$U_j \to AV \quad \text{in } L^1(dA_\alpha),$$

and therefore, since $x \mapsto e^{-x}$ is Lipschitz on $[0,\infty)$,

$$\frac{e^{-U_j}}{\|e^{-U_j}\|_{2,\alpha}} \longrightarrow \frac{e^{-AV}}{\|e^{-AV}\|_{2,\alpha}} \quad \text{in } L^2(dA_\alpha).$$

The already obtained convergence to 1 forces e^{-AV} to be constant a.e. Since $A > 0$, this makes V constant, contradicting $Q_s(V) - P(V) = 1$.

It remains to exclude $A_j \to \infty$. Since V is not constant, choose numbers $0 < a < b$ and measurable sets F_- and F_+ of positive dA_α-measure such that

$$V < a \quad \text{on } F_-, \qquad V > b \quad \text{on } F_+.$$

Using the a.e. convergence of $V_j := \int K_\zeta\, d\sigma_j(\zeta)$ to V, after replacing $F_\pm$ by smaller positive-measure subsets we may assume that for all large j,

$$V_j \leq a \quad \text{on } F_-, \qquad V_j \geq b \quad \text{on } F_+.$$

Let $n_j := \|e^{-U_j}\|_{2,\alpha}$. The convergence $e^{-U_j}/n_j \to 1$ in measure implies that, for large j, this quotient lies between $1/2$ and 2 on subsets of positive measure inside both F_- and F_+. On those subsets,

$$\frac{1}{2} n_j \leq e^{-A_j V_j} \leq e^{-A_j b} \quad \text{and} \quad e^{-A_j a} \leq e^{-A_j V_j} \leq 2n_j.$$

Hence

$$\frac{1}{2} e^{-A_j a} \leq n_j \leq 2e^{-A_j b},$$

which is impossible as $A_j \to \infty$ because $a < b$. This contradiction proves $A_j \to 0$. □

Proposition 3.21 (Perturbative bathtub coercivity on the Riesz cone)**.** *Fix $\alpha > -1$ and $s \in (0,\infty)$. There are $\eta_{\alpha,s} > 0$ and $c_{\alpha,s} > 0$ with the following property. Let*

$$\varphi = h + \psi, \qquad \psi(z) = -\int_{\mathbb{D}} G(z,\zeta)\, d\mu(\zeta), \qquad \mu \geq 0,$$

be the Riesz decomposition of a real-analytic subharmonic function on a neighbourhood of $\overline{\mathbb{D}}$, with $P(\varphi) = 0$ and

$$\|h - P(h)\|_{C^2(\overline{\mathbb{D}})} \leq \eta_{\alpha,s}.$$

Set

$$g_\varphi = \frac{e^\varphi}{\|e^\varphi\|_{2,\alpha}}, \qquad \mathcal{D}_s(\varphi) := \theta_\alpha(s) - \sup_{\tau(E)=s} \int_E g_\varphi^2\, dA_\alpha.$$

Then, whenever $\mathcal{D}_s(\varphi) \leq \eta_{\alpha,s}$,

$$\mathcal{D}_s(\varphi) \geq c_{\alpha,s}\left[\mathcal{D}_s(h) + P(\psi) - Q_s(\psi)\right].$$

Proof. First assume that ψ is bounded. The maximizer for $g_\varphi^2 dA_\alpha$ is a superlevel set of

$$e^{2h+2\psi}(1-|z|^2)^{\alpha+2}.$$

The smallness of $\mathcal{D}_s(\varphi)$ and Lemma 3.16 imply that this set differs from B_s by a set of arbitrarily small hyperbolic measure as $\eta_{\alpha,s} \downarrow 0$. Hence the same Lagrange expansion as in Lemma 3.17 is valid, now with the additional direction ψ.

The harmonic part gives the Schur-complement form computed in Lemma 3.17. The first variation of the Green potential is

$$-2Q_s(\psi) + 2P(\psi) = 2\big(P(\psi) - Q_s(\psi)\big),$$

because $P(\varphi) = 0$ and radial averages agree on harmonic functions. The terms mixing ψ with the boundary displacement are controlled by the positive base boundary term $\kappa_s\|\rho\|_2^2$ and the boundary trace estimate Lemma 3.18. Writing $U = -\psi$, we have

$$\left|\int_{\partial B_s} \psi\,\rho\,d\vartheta\right| \le \|U(r_s e^{i\vartheta})\|_{L^2_\vartheta}\|\rho\|_{L^2_\vartheta} \le C_{\alpha,s}\big(P(\psi) - Q_s(\psi)\big)\|\rho\|_2.$$

Thus, for every $\varepsilon > 0$,

$$\left|\int_{\partial B_s} \psi\,\rho\,d\vartheta\right| \le \varepsilon\|\rho\|_2^2 + C_{\alpha,s,\varepsilon}\big(P(\psi) - Q_s(\psi)\big)^2.$$

The sign of the first variation gives the linear term $2(P(\psi)-Q_s(\psi))$. Lemma 3.20 gives the uniform contrapositive needed here: after decreasing $\eta_{\alpha,s}$, the assumption $\mathcal{D}_s(\varphi) \le \eta_{\alpha,s}$ implies

$$P(\psi) - Q_s(\psi) \le \eta_{\alpha,s}^{1/2}.$$

Thus the squared error above is absorbed by the linear gap. The remaining quadratic potential terms are nonnegative up to the same absorbed error, because the bounded truncations satisfy $e^{2\psi} = 1+2\psi+2\psi^2+O(|\psi|^3)$ on the set where the expansion is used and the negative part has already been paid for by $P(\psi) - Q_s(\psi)$. Thus

$$\mathcal{D}_s(\varphi) \ge c_{\alpha,s}\,[\mathcal{D}_s(h) + P(\psi) - Q_s(\psi)]$$

in the bounded case.

For a general Green potential set

$$\psi_N(z) := -\int_{\mathbb{D}} \min\{G(z,\zeta), N\}\,d\mu(\zeta).$$

Then ψ_N is bounded, $\psi_N \downarrow \psi$, and $e^{h+\psi_N} \to e^{h+\psi}$ in $L^2(dA_\alpha)$ by monotone convergence and the normalization. The corresponding optimized bathtub deficits converge, again by monotone convergence and the layer-cake formula for the decreasing rearrangement. Also

$$P(\psi_N) - Q_s(\psi_N) \to P(\psi) - Q_s(\psi).$$

Passing to the limit gives the claim. □

Remark 3.22. *Proposition 3.21 is the exact local replacement for the earlier informal free-boundary step. The harmonic part is completely explicit through Lemma 3.17; the only non-spectral input is the Green-kernel gap, which is used to absorb the mixed potential/boundary terms after truncating the Green potential.*

Theorem 3.23 (Local pair half-stability)**.** *Fix $\alpha > -1$ and $s \in (0,\infty)$. There are $\eta_{\alpha,s} > 0$ and $C_{\alpha,s} < \infty$ such that, for every measurable $\Omega \subset \mathbb{D}$ with $\tau(\Omega) = s$ and every real-analytic subharmonic φ on a neighbourhood of $\overline{\mathbb{D}}$ satisfying $P(\varphi) = 0$, whose Riesz decomposition $\varphi = h + \psi$ satisfies*

$$\|h - P(h)\|_{C^2(\overline{\mathbb{D}})} \le \eta_{\alpha,s},$$

the function

$$g_\varphi := \frac{e^\varphi}{\|e^\varphi\|_{2,\alpha}}$$

satisfies

$$\inf_{|a|<1} \|g_\varphi - g_a\|_{2,\alpha}^2 \le C_{\alpha,s}\,\delta(g_\varphi;\Omega).$$

Proof. It is enough to prove the estimate for the maximizing set of the density

$$u_\varphi(z) := e^{2\varphi(z)}(1-|z|^2)^{\alpha+2}\Big/\|e^\varphi\|_{2,\alpha}^2$$

with respect to $d\tau$, since replacing Ω by the superlevel set of u_φ can only decrease the deficit. For $\varphi = 0$ this maximizer is the centered disk $B_s = \{|z| < r_s\}$.

If the optimized deficit is bounded below by a fixed multiple of $\eta_{\alpha,s}$, the claim follows by increasing $C_{\alpha,s}$, since the left-hand side is at most 4. We may therefore assume that

$$\mathcal{D}_s(\varphi) := \theta_\alpha(s) - \sup_{\tau(E)=s} \int_E g_\varphi^2 \, dA_\alpha \le \eta_{\alpha,s}.$$

Proposition 3.21 gives

$$\mathcal{D}_s(\varphi) \ge c_{\alpha,s}\big(\mathcal{D}_s(h) + P(\psi) - Q_s(\psi)\big).$$

By Lemma 3.17,

$$\inf_{|a|<1} \left\| \frac{e^h}{\|e^h\|_{2,\alpha}} - g_a \right\|_{2,\alpha}^2 \le C_{\alpha,s}\mathcal{D}_s(h),$$

and by Proposition 3.15,

$$\inf_{t\in\mathbb{R}} \left\| g_\varphi - \frac{e^{h+t}}{\|e^{h+t}\|_{2,\alpha}} \right\|_{2,\alpha}^2 \le C_{\alpha,s}\big(P(\psi) - Q_s(\psi)\big).$$

The constant t is irrelevant after normalization. Combining the last three displays with the triangle inequality in squared form yields

$$\inf_{|a|<1} \|g_\varphi - g_a\|_{2,\alpha}^2 \le C_{\alpha,s}\, \mathcal{D}_s(\varphi).$$

Finally,

$$\mathcal{D}_s(\varphi) \le \theta_\alpha(s)\, \delta(g_\varphi;\Omega)$$

for every Ω of hyperbolic measure s, because the supremum over sets is at least the value on Ω. This proves the theorem. □

Theorem 3.24 (Global half-stability)**.** *For every $\alpha > -1$ and every $s \in (0,\infty)$ there is a constant $C_{\alpha,s} < \infty$ such that every positive real-analytic log-subharmonic $g \in \mathbf{B}^2_\alpha(\mathbb{D})$ and every measurable $\Omega \subset \mathbb{D}$ with $\tau(\Omega) = s$ satisfy*

$$\inf_{\substack{|c|=\|g\|_{2,\alpha}\\ |a|<1}} \frac{\|g - cg_a\|_{2,\alpha}}{\|g\|_{2,\alpha}} \le C_{\alpha,s}\, \delta(g;\Omega)^{1/2}.$$

Proof. By homogeneity assume $\|g\|_{2,\alpha} = 1$. For deficits bounded below, the claim is trivial because the left-hand side is at most 2. It remains to consider a sequence with $\delta(g_j;\Omega_j) \to 0$.

Assume the estimate fails. Then there are g_j, Ω_j with $\tau(\Omega_j) = s$ and

$$\frac{\operatorname{dist}(g_j,\mathcal{E})}{\delta(g_j;\Omega_j)^{1/2}} \to \infty, \qquad \mathcal{E} := \{g_a : |a| < 1\}.$$

The already proved 1/4 estimate, Theorem 3.4, implies

$$\operatorname{dist}(g_j,\mathcal{E}) \to 0.$$

After composing with automorphisms realizing the closest extremals, covariance reduces us to functions

$$\widetilde{g}_j = \frac{e^{\varphi_j}}{\|e^{\varphi_j}\|_{2,\alpha}}, \qquad P(\varphi_j) = 0,$$

with $\widetilde{g}_j \to 1$ in $L^2(dA_\alpha)$ and transformed sets $\widetilde{\Omega}_j$ of the same hyperbolic measure, with $\delta(\widetilde{g}_j;\widetilde{\Omega}_j) = \delta(g_j;\Omega_j)$. Equivalently, for

$$u_j(z) := \widetilde{g}_j(z)^2(1-|z|^2)^{\alpha+2}$$

we have $u_j \to w_\alpha$ in $L^1(d\tau)$, and the near equality assumption gives

$$\int_{\widetilde{\Omega}_j} u_j \, d\tau \longrightarrow \Theta(s).$$

Lemma 3.16 therefore yields

$$\tau(\widetilde{\Omega}_j \triangle B_s) \to 0, \qquad \int_{B_s} u_j \, d\tau \to \Theta(s).$$

In other words,

$$\delta(\widetilde{g}_j; B_s) \to 0.$$

Write the Riesz decomposition

$$\varphi_j = h_j + \psi_j, \qquad \psi_j = -G\mu_j, \qquad \mu_j \geq 0.$$

We claim first that

$$h_j - P(h_j) \longrightarrow 0 \quad \text{locally uniformly in } \mathbb{D}. \tag{3.33}$$

Indeed, the centered-deficit bound gives $\delta(\widetilde{g}_j; B_s) \to 0$, while the 1/4 theorem gives $\widetilde{g}_j \to 1$ in $L^2(dA_\alpha)$. We first prove that $P(\psi_j) - Q_s(\psi_j) \to 0$. This follows from the same Green-kernel comparison as Lemma 3.14, applied with the probability measure

$$d\nu_j := \frac{e^{2h_j}}{\int_{\mathbb{D}} e^{2h_j}\, dA_\alpha}\, dA_\alpha.$$

If this gap did not tend to zero, then the potential part would remove a fixed amount of the normalized $d\nu_j$-mass from the centered disk relative to the whole disk, and the concentration deficit of $e^{h_j+\psi_j}/\|e^{h_j+\psi_j}\|_{2,\alpha}$ on B_s would stay bounded away from zero. This contradicts $\delta(\widetilde{g}_j; B_s) \to 0$.

With this gap tending to zero, the proof of Proposition 3.15, applied with the background measure $d\nu_j$, yields

$$\left\| \widetilde{g}_j - \frac{e^{h_j}}{\|e^{h_j}\|_{2,\alpha}} \right\|_{2,\alpha}^2 \leq C_{\alpha,s}\big(P(\psi_j) - Q_s(\psi_j)\big).$$

Hence

$$\frac{e^{h_j}}{\|e^{h_j}\|_{2,\alpha}} \to 1 \quad \text{in } L^2(dA_\alpha).$$

Choose holomorphic functions $F_j = e^{H_j}/\|e^{h_j}\|_{2,\alpha}$ with $\operatorname{Re} H_j = h_j$. The Bergman pointwise estimate gives local boundedness of F_j, and the L^2 convergence of $|F_j|$ to 1 forces every normal-family limit to be a unimodular constant. Since the F_j are zero-free, Hurwitz' theorem and the normalization of the real part imply (3.33).

Thus, after discarding finitely many terms, the harmonic part is in the small local chart required by Theorem 3.23; its $k = 1$ component is harmless because the distance is taken to the whole extremal manifold. The local pair theorem, Theorem 3.23, applied to $(\widetilde{g}_j, \widetilde{\Omega}_j)$ gives

$$\operatorname{dist}(\widetilde{g}_j, \mathcal{E})^2 \leq C_{\alpha,s}\, \delta(\widetilde{g}_j; \widetilde{\Omega}_j) = C_{\alpha,s}\, \delta(g_j; \Omega_j).$$

Thus

$$\operatorname{dist}(g_j, \mathcal{E}) \leq C_{\alpha,s}\, \delta(g_j; \Omega_j)^{1/2},$$

contradicting the choice of the sequence. This proves the theorem. □

Remark 3.25 (Global mechanism)**.** *The nonlinear spike obstruction is absorbed by Proposition 3.15, while Lemma 3.16 prevents the set from drifting in the normalized chart. Thus no separate centering of the set is needed: the perturbative bathtub expansion in Theorem 3.23 treats the optimal set displacement and the function perturbation simultaneously.*

The corresponding question in dimension $n \geq 3$ leads to the conjectural exponent 1/2 in place of $1/(n+2)$ in Theorem 3.3, again under a mechanism that must go beyond the peak-only estimate of Lemma 3.28. We leave both questions open.

Proof of Theorem 3.3 for $n \geq 3$. From this point on we use the hyperbolic measure

$$d\tau(x) := \frac{dV(x)}{(1-|x|^2)^n}.$$

Fix $n \geq 3$, $\beta > 1$, and a measurable set $\Omega \subset \mathbb{B}^n$ with

$$0 < s_0 := \tau(\Omega) < \infty.$$

By homogeneity we may assume $\|g\|_{2,\beta} = 1$.

Set

$$u(x) := \Phi_n(x)^\beta g(x)^2, \qquad v(x) := \Phi_n(x)^\beta,$$

and let u^*, v^* be the decreasing rearrangements of u, v with respect to τ.

Since

$$\|g\|_{2,\beta}^2 = c_\beta \int_{\mathbb{B}^n} \Phi_n(x)^\beta g(x)^2 \, d\tau(x) = 1$$

and, by the normalization of c_β,

$$c_\beta \int_{\mathbb{B}^n} \Phi_n(x)^\beta \, d\tau(x) = 1,$$

we have

$$\int_0^\infty u^*(s)\, ds = \int_0^\infty v^*(s)\, ds = \frac{1}{c_\beta}.$$

Define

$$\Theta(s) := \int_0^s v^*(t)\, dt, \qquad \theta(s) := c_\beta\, \Theta(s), \qquad C(s) := (1-\theta(s))^{-1}.$$

Equivalently,

$$\theta(s) = c_\beta \int_{B(0,r(s))} \Phi_n(x)^\beta \, d\tau(x),$$

where $r(s) \in (0,1)$ is determined by

$$\tau(B(0,r(s))) = s.$$

1) Deficit on the level set. Let

$$\Omega_{s_0} := \{u > u^*(s_0)\}, \qquad \tau(\Omega_{s_0}) = s_0.$$

By Hardy–Littlewood,

$$\int_\Omega u\, d\tau \le \int_{\Omega_{s_0}} u\, d\tau = \int_0^{s_0} u^*(s)\, ds.$$

Define

$$\delta_{s_0} := 1 - \frac{\displaystyle\int_0^{s_0} u^*(s)\, ds}{\displaystyle\int_0^{s_0} v^*(s)\, ds}.$$

Then

$$\delta_{s_0} \le \delta(g;\Omega).$$

Also,

$$D := \int_0^{s_0} \big(v^*(s) - u^*(s)\big)\, ds = \delta_{s_0} \int_0^{s_0} v^*(s)\, ds \le \frac{\delta_{s_0}}{c_\beta}. \tag{3.34}$$

2) The crossing point and the truncated deficit. Let

$$R(s) := \frac{u^*(s)}{v^*(s)}.$$

For $n \ge 3$ we have

$$(u^*)'(s) + \gamma\, \Upsilon(s)\, u^*(s) \ge 0, \qquad (v^*)'(s) + \gamma\, \Upsilon(s)\, v^*(s) = 0, \qquad \gamma := \beta(n-1)^2.$$

Hence

$$R'(s) = \frac{(u^*)'(s)v^*(s) - (v^*)'(s)u^*(s)}{(v^*(s))^2} = \frac{(u^*)'(s) + \gamma\, \Upsilon(s)\, u^*(s)}{v^*(s)} \ge 0.$$

Thus R is nondecreasing. Since $u^*(0) \le v^*(0)$ and

$$\int_0^\infty u^*(s)\, ds = \int_0^\infty v^*(s)\, ds = \frac{1}{c_\beta},$$

either $u^* \equiv v^*$ almost everywhere, or else there is a unique point s^* such that

$$u^*(s^*) = v^*(s^*).$$

Define

$$I_* := \int_0^{s^*} \big(v^*(s) - u^*(s)\big)\, ds.$$

We claim that

$$I_* \le \frac{1}{c_\beta}\, C(s_0)\, \delta_{s_0}. \tag{3.35}$$

Arguing exactly as in the case $n = 2$, if $s_0 \geq s^*$ then

$$I_* \leq D\,\frac{\int_{s^*}^{\infty} u^*(s)\,ds}{\int_{s_0}^{\infty} u^*(s)\,ds} \leq D\,\frac{\frac{1}{c_\beta}}{\int_{s_0}^{\infty} v^*(s)\,ds}.$$

Since

$$\int_{s_0}^{\infty} v^*(s)\,ds = \frac{1}{c_\beta} - \Theta(s_0) = \frac{1-\theta(s_0)}{c_\beta},$$

it follows that

$$I_* \leq D\,C(s_0) \leq \frac{1}{c_\beta} C(s_0)\,\delta_{s_0},$$

where in the last step we used (3.34).

If $s_0 \leq s^*$, then, again exactly as in the case $n = 2$,

$$I_* \leq D\,\frac{\int_0^{s^*} v^*(s)\,ds}{\int_0^{s_0} v^*(s)\,ds} \leq D\,\frac{\frac{1}{c_\beta}}{\int_0^{s_0} v^*(s)\,ds} = \frac{\delta_{s_0}}{c_\beta} \leq \frac{1}{c_\beta} C(s_0)\,\delta_{s_0}.$$

Thus (3.35) holds in all cases. Since $\delta_{s_0} \leq \delta(g;\Omega)$,

$$I_* \leq \frac{1}{c_\beta} C(s_0)\,\delta(g;\Omega). \tag{3.36}$$

3) Comparison with the clipped functional $I(T)$. Let

$$T := \|u\|_{L^\infty(\mathbb{B}^n)} = \sup_{x\in\mathbb{B}^n} \Phi_n(x)^\beta g(x)^2 \in (0,1].$$

Let $s(T)$ be the unique solution of $v^*(s(T)) = T$ and define

$$I(T) := \int_0^{s(T)} \big(v^*(s) - T\big)\,ds = \Theta(s(T)) - T\,s(T).$$

Since $u^*(s) \leq T$ for all s and

$$u^*(s^*) = v^*(s^*) \leq T,$$

the monotonicity of v^* implies

$$s^* \geq s(T).$$

Therefore

$$I(T) = \int_0^{s(T)} (v^*(s) - T)\,ds \leq \int_0^{s(T)} (v^*(s) - u^*(s))\,ds \leq I_*. \tag{3.37}$$

Combining (3.36) and (3.37), we get

$$I(T) \leq \frac{1}{c_\beta} C(s_0)\,\delta(g;\Omega). \tag{3.38}$$

4) Lower bound on $I(T)$. Using $\Theta'(s) = v^*(s)$ and

$$\frac{\Theta''(s)}{\Theta'(s)} = -\gamma\,\Upsilon(s), \qquad \gamma = \beta(n-1)^2,$$

differentiating $v^*(s(T)) = \Theta'(s(T)) = T$ yields

$$s'(T) = \frac{1}{\Theta''(s(T))}, \qquad I'(T) = -s(T), \qquad I''(T) = -s'(T) = \frac{1}{\gamma\,T\,\Upsilon(s(T))}.$$

Fix

$$r_0^2 = \frac{1}{4(n-2)}, \qquad T_0 = (1-r_0^2)^{\beta(n-1)}.$$

For $T \in (T_0, 1)$, Lemma 3.29 and Corollary 3.30 give

$$\Upsilon(s(T)) \leq \frac{2\Gamma(\frac{n}{2})\,G_{n,r_0}}{n\pi^{n/2}} \left(\frac{\beta\,C_{n,r_0}}{1-T}\right)^{\frac{n-2}{2}}.$$

Hence there exists an explicit constant $\kappa_{n,r_0} > 0$ such that

$$I''(T) \geq \kappa_{n,r_0}\,\beta^{-n/2}\,(1-T)^{\frac{n-2}{2}}, \qquad T \in (T_0, 1).$$

Since $I(1) = I'(1) = 0$, integration gives

$$I(T) \geq \frac{4\,\kappa_{n,r_0}}{n(n+2)}\,\beta^{-n/2}\,(1-T)^{\frac{n}{2}+1}, \qquad T \in (T_0, 1).$$

Set

$$E_{n,r_0} := \frac{4\,\kappa_{n,r_0}}{n(n+2)}.$$

Then

$$I(T) \geq E_{n,r_0}\,\beta^{-n/2}\,(1-T)^{\frac{n}{2}+1}, \qquad T \in (T_0, 1). \tag{3.39}$$

Moreover, I is decreasing on $(0,1)$ because $I'(T) = -s(T) < 0$. Hence if $T \leq T_0$,

$$I(T) \geq I(T_0) \geq E_{n,r_0}\,\beta^{-n/2}\,(1-T_0)^{\frac{n}{2}+1}.$$

Since $\beta > 1$,

$$T_0 = (1-r_0^2)^{\beta(n-1)} \leq (1-r_0^2)^{n-1},$$

so with

$$d_n := 1 - (1-r_0^2)^{n-1} > 0$$

we get

$$I(T) \geq E_{n,r_0}\,\beta^{-n/2}\,d_n^{\frac{n}{2}+1}, \qquad 0 < T \leq T_0. \tag{3.40}$$

5) From the deficit to distance from extremals. We split into two regimes.

If $T \in (T_0, 1)$, then (3.38) and (3.39) imply

$$E_{n,r_0}\,\beta^{-n/2}\,(1-T)^{\frac{n}{2}+1} \leq \frac{1}{c_\beta} C(s_0)\,\delta(g;\Omega),$$

hence

$$1 - T \leq \left(\frac{\beta^{n/2}}{c_\beta E_{n,r_0}}\,C(s_0)\,\delta(g;\Omega)\right)^{\frac{2}{n+2}}.$$

By Lemma 3.28,

$$\inf_{a\in\mathbb{B}^n} \|g - g_a\|_{2,\beta}^2 \leq 2 - 2\sqrt{T} \leq 2(1-T),$$

and therefore

$$\inf_{a\in\mathbb{B}^n} \|g - g_a\|_{2,\beta} \leq \sqrt{2}\left(\frac{\beta^{n/2}}{c_\beta E_{n,r_0}}\,C(s_0)\,\delta(g;\Omega)\right)^{\frac{1}{n+2}}.$$

If $0 < T \leq T_0$, then by (3.38) and (3.40),

$$\frac{1}{c_\beta} C(s_0)\,\delta(g;\Omega) \geq E_{n,r_0}\,\beta^{-n/2}\,d_n^{\frac{n}{2}+1},$$

that is,

$$C(s_0)\,\delta(g;\Omega) \geq c_\beta E_{n,r_0}\,\beta^{-n/2}\,d_n^{\frac{n}{2}+1}.$$

Since the left-hand side of the theorem is always bounded by 2, we obtain

$$\inf_{\substack{|c|=\|g\|_{2,\beta}\\ a\in\mathbb{B}^n}} \frac{\|g - cg_a\|_{2,\beta}}{\|g\|_{2,\beta}} \leq 2\,(c_\beta E_{n,r_0})^{-\frac{1}{n+2}}\,\beta^{\frac{n}{2(n+2)}}\,d_n^{-1/2}\,\big(C(s_0)\,\delta(g;\Omega)\big)^{\frac{1}{n+2}}.$$

Combining the two regimes, the theorem follows with

$$C_{n,\beta} = 2(c_\beta E_{n,r_0})^{-\frac{1}{n+2}}\,\beta^{\frac{n}{2(n+2)}}\,d_n^{-1/2}, \qquad d_n := 1 - (1-r_0^2)^{n-1}.$$

□

Remark 3.26 (Explicit form and dimensional dependence of the stability constant)**.** *In the proof of Theorem 3.3 for $n \geq 3$, one may choose*

$$r_0^2 = \frac{1}{4(n-2)}, \qquad G_{n,r_0} := {}_2F_1\left(\frac{n}{2}, n; \frac{n+2}{2}; r_0^2\right),$$

and

$$C_{n,r_0} := (n-1) + \frac{(n-1)(n-2)}{n}\,{}_3F_2\left(\begin{matrix} 1,1,2-\frac{n}{2} \\ 2, 1+\frac{n}{2} \end{matrix}\,;\, r_0^2\right).$$

Then

$$\kappa_{n,r_0} = \frac{n\,\pi^{n/2}}{2\,(n-1)^2\,\Gamma(\frac{n}{2})\,G_{n,r_0}\,C_{n,r_0}^{\frac{n-2}{2}}}, \qquad E_{n,r_0} = \frac{4\,\kappa_{n,r_0}}{n(n+2)}.$$

With

$$d_n := 1 - (1-r_0^2)^{n-1} > 0,$$

the proof yields

$$C_{n,\beta} = (c_\beta E_{n,r_0})^{-\frac{1}{n+2}}\,\beta^{\frac{n}{2(n+2)}}\max\Big\{\sqrt{2},\ 2d_n^{-1/2}\Big\} = (c_\beta E_{n,r_0})^{-\frac{1}{n+2}}\,\beta^{\frac{n}{2(n+2)}}\,2d_n^{-1/2}.$$

In particular, one may write

$$C_{n,\beta} = \widetilde{C}_n\,c_\beta^{-\frac{1}{n+2}}\,\beta^{\frac{n}{2(n+2)}},$$

where

$$\widetilde{C}_n := 2\,E_{n,r_0}^{-\frac{1}{n+2}}\,d_n^{-1/2}$$

depends only on n.

Remark 3.27 (The planar case and the disappearance of the factor C_{n,r_0})**.** *It is worth noting that the higher-dimensional explicit formula is not meant to be specialized directly to $n = 2$, since the choice*

$$r_0^2 = \frac{1}{4(n-2)}$$

breaks down at $n = 2$. Nevertheless, at a purely formal level one feature is consistent with the planar theory: the exponent

$$C_{n,r_0}^{\frac{n-2}{2}}$$

would become

$$C_{2,r_0}^0 = 1,$$

so this factor disappears in dimension 2.

Thus the blow-up of the planar stability constant as the parameter approaches the endpoint does not *come from the factor C_{n,r_0}, but from the normalization of the weighted Bergman norm. More precisely, in the planar notation with parameter $\alpha > -1$ one has*

$$c_\alpha = \frac{\alpha+1}{\pi},$$

and the stability constant obtained from the separate two-dimensional proof is

$$C_{2,\alpha} = \sqrt{2}\left(\frac{2(\alpha+2)}{\alpha+1}\right)^{1/4}.$$

Equivalently, writing $\alpha = \beta - 2$, this becomes

$$C_{2,\beta} = \sqrt{2}\left(\frac{2\beta}{\beta-1}\right)^{1/4},$$

which coincides with the planar bound obtained earlier.

Lemma 3.28. *Let $0 < T < 1$ and g is log-$\mathcal{M}$-subharmonic function on $\|\cdot\|_{2,\beta}-$norm equal to 1 such that $\max_{x\in\mathbb{B}^n} g(x)^2\Phi_n(x)^\beta = T$. Then,*

$$\inf_{|a|<1}\|g - g_a\|^2 \le 2 - 2\sqrt{T}.$$

Proof of Lemma 3.28. Let $m = m_a(x)$ be an involutive Mobius transform of the unit ball onto itself so that $m(0) = a$. Let

$$g_a(x) = \frac{\Phi_n^{\beta/2}(m(x))}{\Phi_n^{\beta/2}(x)}.$$

Then

$$\|g_a\| = 1.$$

Let g be log-M-subharmonic so that $\|g\| = 1$. Then

$$\begin{aligned}
\|g-g_a\|^2 &= 2-2\int_{\mathbb{B}} g(x)g_a(x)\Phi_n^{\beta}(x)d\tau(x)\\
&= 2-2\int_{\mathbb{B}} g(m_a(y))g_a(m_a(y))\Phi_n^{\beta}(m_a(y))d\tau(m_a(y))\\
&= 2-2\int_{\mathbb{B}} g(m_a(y))\frac{\Phi_n^{\beta/2}(y)}{\Phi_n^{\beta/2}(m_a(y))}\Phi_n^{\beta}(m_a(y))d\tau(y)\\
&= 2-2\int_{\mathbb{B}} g(m_a(y))\Phi_n^{\beta/2}(y)\Phi_n^{\beta/2}(m_a(y))d\tau(y)\\
&\le 2-2g(a)\Phi_n^{\beta/2}(a).
\end{aligned}$$

In the last inequality we use invariant-mean property of log-M-subharmonic function

$$h(y)=g(m_a(y))\Phi_n^{-\beta/2}(y)\Phi_n^{\beta/2}(m_a(y))$$

at $y=0$. This follows from [15, Corollary 4.13] by using polar coordinates.

So

$$\inf_{|a|<1}\|g-g_a\|^2 \le 2-2\sqrt{T},$$

as desired.

□

Lemma 3.29 (Explicit lower bound for s_T and a threshold T_0)**.** *Assume $n\ge 3$ and $\beta>1$. Set*

$$r_0^2 := \frac{1}{4(n-2)}, \qquad T_0 := (1-r_0^2)^{\beta(n-1)}.$$

For $T\in(T_0,1)$ let $S:=T^{1/\beta}$ and let $r_T\in(0,r_0)$ be uniquely determined by

$$\Phi_n(r_T)=S.$$

Then

$$s_T := \tau\{x\in\mathbb{B}^n:\ T<\Phi_n^{\beta}(x)<1\} = \tau(\mathbb{B}(0,r_T)).$$

Moreover, if we denote

$$H_{n,r_0} := {}_3F_2\left(\begin{matrix}1,1,2-\frac{n}{2}\\ 2,1+\frac{n}{2}\end{matrix};r_0^2\right), \qquad C_{n,r_0} := (n-1)+\frac{(n-1)(n-2)}{n}H_{n,r_0},$$

then for every $T\in(T_0,1)$ we have the quantitative bound

$$s_T \ge \frac{\omega_n}{n}\left(\frac{1-T}{\beta\, C_{n,r_0}}\right)^{n/2},$$

where $\omega_n := |\mathbb{S}^{n-1}|$.

Proof. Assume $n\ge 3$ and $\beta>1$, and set

$$r_0^2 := \frac{1}{4(n-2)}, \qquad T_0 := (1-r_0^2)^{\beta(n-1)}.$$

Fix $T\in(T_0,1)$, let $S:=T^{1/\beta}$, and let $r_T\in(0,r_0)$ be determined by $\Phi_n(r_T)=S$.

Since Φ_n is radial and strictly decreasing in $r=|x|$ with $\Phi_n(0)=1$, the superlevel set $\{x\in\mathbb{B}^n:\ \Phi_n(x)>S\}$ is exactly $\mathbb{B}(0,r_T)$. Because $\Phi_n^{\beta}(x)=\Phi_n(|x|)^{\beta}$, the condition $T<\Phi_n^{\beta}(x)<1$ is equivalent to $\Phi_n(|x|)>S$, hence

$$\{x\in\mathbb{B}^n:\ T<\Phi_n^{\beta}(x)<1\}=\mathbb{B}(0,r_T),$$

and therefore

$$s_T=\tau\{x\in\mathbb{B}^n:\ T<\Phi_n^{\beta}(x)<1\}=\tau(\mathbb{B}(0,r_T)).$$

For $0\le r\le r_0$ we use the derivative identity

$$-\Phi_n'(r)=2r\left((n-1)+\frac{(n-1)(n-2)}{n}\,{}_3F_2\left(\begin{matrix}1,1,2-\frac{n}{2}\\ 2,1+\frac{n}{2}\end{matrix};r^2\right)\right).$$

The ${}_3F_2$ series has nonnegative coefficients, hence it is increasing in r^2, so for $0\le r\le r_0$,

$$-\Phi_n'(r)\le 2C_{n,r_0}\,r.$$

Integrating from 0 to r_T and using $\Phi_n(0) = 1$ yields

$$1 - \Phi_n(r_T) = \int_0^{r_T} (-\Phi_n'(t))\, dt \le C_{n,r_0} r_T^2.$$

Since $0 < \Phi_n(r_T) \le 1$ and $\beta > 1$, we have

$$1 - \Phi_n(r_T)^\beta \le \beta\,(1 - \Phi_n(r_T)),$$

and because $\Phi_n(r_T)^\beta = T$ this gives

$$1 - T \le \beta\, C_{n,r_0}\, r_T^2, \qquad \text{hence} \qquad r_T^2 \ge \frac{1-T}{\beta\, C_{n,r_0}}.$$

Finally, by definition of τ,

$$\tau(\mathbb{B}(0, r_T)) = \omega_n \int_0^{r_T} \frac{\rho^{n-1}}{(1-\rho^2)^n}\, d\rho \ge \omega_n \int_0^{r_T} \rho^{n-1}\, d\rho = \frac{\omega_n}{n}\, r_T^n,$$

so using the lower bound for r_T^2 we obtain

$$s_T = \tau(\mathbb{B}(0, r_T)) \ge \frac{\omega_n}{n} \left(\frac{1-T}{\beta\, C_{n,r_0}} \right)^{n/2}.$$

This proves the lemma. □

Corollary 3.30 (Explicit bound for $\Upsilon(s_T)$)**.** *With the same assumptions and notation as in Lemma 3.29, set*

$$G_{n,r_0} := {}_2F_1\left(\frac{n}{2}, n; \frac{n+2}{2}; r_0^2 \right).$$

Then for all $T \in (T_0, 1)$,

$$\Upsilon(s_T) \le \frac{2\Gamma\left(\frac{n}{2}\right) G_{n,r_0}}{n\pi^{n/2}} \left(\frac{\beta\, C_{n,r_0}}{1-T} \right)^{\frac{n-2}{2}}.$$

Proof. With the notation of Lemma 3.29, we have $s_T = \tau(\mathbb{B}(0, r_T))$. By the definition of Υ in terms of the radius $r = r(s)$ determined by $s = \tau(\mathbb{B}(0, r))$,

$$\Upsilon(s) = \frac{2\Gamma\left(\frac{n}{2}\right)}{n\pi^{n/2}}\, r(s)^{\,2-n}\, {}_2F_1\left(\frac{n}{2}, n; \frac{n+2}{2}; r(s)^2 \right),$$

hence

$$\Upsilon(s_T) = \frac{2\Gamma\left(\frac{n}{2}\right)}{n\pi^{n/2}}\, r_T^{2-n}\, {}_2F_1\left(\frac{n}{2}, n; \frac{n+2}{2}; r_T^2 \right).$$

The ${}_2F_1$ series has positive coefficients, so it is increasing in r^2; since $r_T \le r_0$ we obtain

$${}_2F_1\left(\frac{n}{2}, n; \frac{n+2}{2}; r_T^2 \right) \le {}_2F_1\left(\frac{n}{2}, n; \frac{n+2}{2}; r_0^2 \right) = G_{n,r_0}.$$

Therefore

$$\Upsilon(s_T) \le \frac{2\Gamma\left(\frac{n}{2}\right) G_{n,r_0}}{n\pi^{n/2}}\, r_T^{2-n}.$$

Finally, Lemma 3.29 gives

$$r_T^2 \ge \frac{1-T}{\beta\, C_{n,r_0}},$$

hence

$$r_T^{2-n} \le \left(\frac{\beta\, C_{n,r_0}}{1-T} \right)^{\frac{n-2}{2}},$$

which yields

$$\Upsilon(s_T) \le \frac{2\Gamma\left(\frac{n}{2}\right) G_{n,r_0}}{n\pi^{n/2}} \left(\frac{\beta\, C_{n,r_0}}{1-T} \right)^{\frac{n-2}{2}}.$$

This proves the corollary. □

4. Log-subharmonicity and Fock space

Definition 4.1. Let f be a real-analytic complex function defined in $\mathbb{R}^m$, $m \geq 2$, whose absolute value $|f|$ is log-harmonic in $\mathbb{R}^m \setminus f^{-1}(0)$. We say that f Belongs to the Fock space $\mathcal{F}^2$ if

$$\|f\| = \left(\int_{\mathbb{R}^n} |f(x)|^2 e^{-\pi|x|^2} dx \right)^{1/2} < \infty.$$

Let $g_{z_0} = |f_{z_0}|$, where $f_{z_0} = e^{-\frac{\pi}{2}|z_0|^2} e^{\pi z \overline{z_0}}$.

Theorem 4.2 (Fock space version of Theorem 3.3)**.** *There is an explicitly computable constant $C > 0$ such that, for all measurable sets $\Omega \subset \mathbb{C}$ with finite measure and for the set $\mathcal{F}^2$ of all log-subharmonic real-analytic functions in $\mathbb{C}$ belonging to the Fock space and $F \in \mathcal{F}^2$, we have*

$$\min_{\substack{c=\|F\|_{\mathcal{F}^2}, \\ z_0 \in \mathbb{C}}} \frac{\|F - c g_{z_0}\|_{\mathcal{F}^2}}{\|F\|_{\mathcal{F}^2}} \leq C \left(e^{|\Omega|} \delta(F;\Omega) \right)^{1/4}, \tag{4.1}$$

where

$$\delta(F;\Omega) := 1 - \frac{\int_\Omega |F(z)|^2 e^{-\pi|z|^2} \, \mathrm{d}z}{(1 - e^{-|\Omega|}) \|F\|_{\mathcal{F}^2}^2}. \tag{4.2}$$

Lemma 4.3. *Assume that $f : \mathbb{C} \to [0, \infty)$ is a real-analytic log-subharmonic function belonging to the Fock space, i.e.*

$$\|f\|_{\mathcal{F}^2}^2 := \int_{\mathbb{C}} f(w)^2 e^{-\pi|w|^2} \, dA(w) < \infty.$$

Then for every $z \in \mathbb{C}$,

$$f(z)^2 e^{-\pi|z|^2} \leq \|f\|_{\mathcal{F}^2}^2.$$

Moreover, if $\|f\|_{\mathcal{F}^2} = 1$, then

$$\inf_{z_0 \in \mathbb{C}} \|f - g_{z_0}\|_{\mathcal{F}^2}^2 \leq 2 - 2\sqrt{T},$$

where

$$T = \sup_{z \in \mathbb{C}} f(z)^2 e^{-\pi|z|^2}, \qquad g_{z_0}(z) = e^{-\frac{\pi}{2}|z_0|^2 + \pi \operatorname{Re}(z\overline{z_0})}.$$

Proof. Fix $z \in \mathbb{C}$, and define

$$H_z(w) := f(z+w)^2 e^{-2\pi \operatorname{Re}((w+z)\overline{z})}.$$

Since f is positive and log-subharmonic, f^2 is also log-subharmonic, hence H_z is log-subharmonic as well. Applying the Gaussian mean-value inequality to H_z at the origin, we obtain

$$H_z(0) \leq \int_{\mathbb{C}} H_z(w) e^{-\pi|w|^2} \, dA(w).$$

Now

$$H_z(0) = f(z)^2 e^{-2\pi|z|^2},$$

while

$$H_z(w) e^{-\pi|w|^2} = f(z+w)^2 e^{-2\pi \operatorname{Re}((w+z)\overline{z})} e^{-\pi|w|^2}.$$

Changing variables $u = z + w$, we get

$$f(z)^2 e^{-2\pi|z|^2} \leq \int_{\mathbb{C}} f(u)^2 e^{-2\pi \operatorname{Re}(u\overline{z})} e^{-\pi|u-z|^2} \, dA(u).$$

Using

$$-2\operatorname{Re}(u\overline{z}) - |u-z|^2 = -|u|^2 + |z|^2,$$

it follows that

$$f(z)^2 e^{-2\pi|z|^2} \leq e^{-\pi|z|^2} \int_{\mathbb{C}} f(u)^2 e^{-\pi|u|^2} \, dA(u),$$

and hence

$$f(z)^2 e^{-\pi|z|^2} \leq \|f\|_{\mathcal{F}^2}^2.$$

Assume now that $\|f\|_{\mathcal{F}^2} = 1$. Since also $\|g_{z_0}\|_{\mathcal{F}^2} = 1$, we have

$$\|f - g_{z_0}\|_{\mathcal{F}^2}^2 = 2 - 2\langle f, g_{z_0} \rangle_{\mathcal{F}^2},$$

where

$$\langle f, g_{z_0}\rangle_{\mathcal{F}^2} = \int_{\mathbb{C}} f(z) g_{z_0}(z) e^{-\pi|z|^2}\, dA(z).$$

Now the function $h_{z_0} := f\, g_{z_0}$ is again positive and log-subharmonic, since g_{z_0} is positive and log-harmonic. Applying the first part of the lemma to h_{z_0}, we obtain

$$f(z_0) g_{z_0}(z_0) e^{-\pi|z_0|^2} \le \int_{\mathbb{C}} f(z) g_{z_0}(z) e^{-\pi|z|^2}\, dA(z) = \langle f, g_{z_0}\rangle_{\mathcal{F}^2}.$$

Since

$$g_{z_0}(z_0) = e^{-\frac{\pi}{2}|z_0|^2 + \pi|z_0|^2} = e^{\frac{\pi}{2}|z_0|^2},$$

we get

$$\langle f, g_{z_0}\rangle_{\mathcal{F}^2} \ge f(z_0) e^{-\frac{\pi}{2}|z_0|^2}.$$

Therefore

$$\|f - g_{z_0}\|^2_{\mathcal{F}^2} \le 2 - 2\, f(z_0) e^{-\frac{\pi}{2}|z_0|^2}.$$

Taking the infimum over $z_0 \in \mathbb{C}$, and using the definition of T, we conclude that

$$\inf_{z_0\in\mathbb{C}} \|f - g_{z_0}\|^2_{\mathcal{F}^2} \le 2 - 2 \sup_{z_0\in\mathbb{C}} f(z_0) e^{-\frac{\pi}{2}|z_0|^2} = 2 - 2\sqrt{T}.$$

□

Schetch of the proof of Theorem 4.2. The proof if just an imitation of the corresponding theorem in [5]. Instead of holomorphicity we have more general log-subharmonicity of our function F.

$$u(z) := F^2(z) e^{-\pi|z|^2}, \tag{4.3}$$

in connection with its super-level sets

$$A_t := \{u > t\} = \{z \in \mathbb{C} \,:\, u(z) > t\}, \tag{4.4}$$

its *distribution function*

$$\mu(t) := |A_t|, \qquad 0 \le t \le \max_{\mathbb{C}} u \tag{4.5}$$

(note that u is bounded due to (4.3)), and the *decreasing rearrangement* of u, i.e. the function

$$u^*(s) := \sup\{t \ge 0 \,:\, \mu(t) > s\} \qquad \text{for } s \ge 0 \tag{4.6}$$

(for more details on rearrangements, we refer to [2]). Then in [12] it is proved that

$$(u^*)'(s) + u^*(s) \ge 0. \tag{4.7}$$

The same proof gives the same inequality for F being a log-subharmonic function in the complex plane. Now the proof of our theorem follows from Lemma 4.3, (4.7), by following the same approach as in [5]. Namely Lemma 2.3 from [5] and eq. 246. from [5] follows from (4.7).

□

By using the following lemma, we can state our stability result for the Fock space in higher dimensions.

Lemma 4.4. *Assume that h is a real-analytic log-subharmonic function in $\mathbb{R}^m$ belonging to the Fock-type space. Then for every $x \in \mathbb{R}^m$,*

$$h(x) e^{-\pi|x|^2} \le \int_{\mathbb{R}^m} h(y) e^{-\pi|y|^2}\, dA(y). \tag{4.8}$$

Assume moreover that f is a real-analytic log-subharmonic function in $\mathbb{R}^m$ belonging to the Fock space, that $\|f\|_{\mathcal{F}^2} = 1$, and define

$$g_{x_0}(y) = e^{-\frac{\pi}{2}|x_0|^2 + \pi\langle y, x_0\rangle}.$$

Then

$$\inf_{x_0\in\mathbb{R}^m} \|f - g_{x_0}\|^2 \le 2 - 2\sqrt{T}, \qquad T = \sup_{x\in\mathbb{R}^m} |f(x)|^2 e^{-\pi|x|^2}. \tag{4.9}$$

Proof. Fix $x \in \mathbb{R}^m$, and consider the function

$$H_x(y) := h(x+y)e^{-2\pi\langle x+y,x\rangle}, \qquad y \in \mathbb{R}^m.$$

Since h is log-subharmonic, $\log h$ is subharmonic wherever $h > 0$. The function

$$y \mapsto -2\pi\langle x+y, x\rangle$$

is affine, hence harmonic, so $\log H_x = \log h(x+y) - 2\pi\langle x+y, x\rangle$ is again subharmonic. Therefore H_x is log-subharmonic, and in particular subharmonic.

Applying the Gaussian mean-value inequality to H_x at the origin gives

$$H_x(0) \le \int_{\mathbb{R}^m} H_x(y)e^{-\pi|y|^2}\, dA(y).$$

Because

$$H_x(0) = h(x)e^{-2\pi|x|^2},$$

we obtain

$$h(x)e^{-2\pi|x|^2} \le \int_{\mathbb{R}^m} h(x+y)e^{-2\pi\langle x+y,x\rangle}e^{-\pi|y|^2}\, dA(y).$$

Now make the change of variables $u = x+y$. Then $y = u - x$, and the right-hand side becomes

$$\int_{\mathbb{R}^m} h(u)e^{-2\pi\langle u,x\rangle}e^{-\pi|u-x|^2}\, dA(u).$$

Using the identity

$$2\langle u, x\rangle + |u-x|^2 = |u|^2 + |x|^2,$$

we get

$$e^{-2\pi\langle u,x\rangle}e^{-\pi|u-x|^2} = e^{-\pi|u|^2}e^{-\pi|x|^2}.$$

Hence

$$h(x)e^{-2\pi|x|^2} \le e^{-\pi|x|^2}\int_{\mathbb{R}^m} h(u)e^{-\pi|u|^2}\, dA(u),$$

and multiplying by $e^{\pi|x|^2}$ yields

$$h(x)e^{-\pi|x|^2} \le \int_{\mathbb{R}^m} h(u)e^{-\pi|u|^2}\, dA(u).$$

This proves (4.8).

Now let $x_0 \in \mathbb{R}^m$. Since $\|f\|_{\mathcal{F}^2} = 1$ and $\|g_{x_0}\|_{\mathcal{F}^2} = 1$, we have

$$\|f - g_{x_0}\|^2 = \|f\|^2_{\mathcal{F}^2} + \|g_{x_0}\|^2_{\mathcal{F}^2} - 2\int_{\mathbb{R}^m} f(y)g_{x_0}(y)e^{-\pi|y|^2}\, dA(y),$$

that is,

$$\|f - g_{x_0}\|^2 = 2 - 2\int_{\mathbb{R}^m} f(y)g_{x_0}(y)e^{-\pi|y|^2}\, dA(y).$$

The function $y \mapsto f(y)g_{x_0}(y)$ is again log-subharmonic: indeed, f is log-subharmonic and

$$\log g_{x_0}(y) = -\frac{\pi}{2}|x_0|^2 + \pi\langle y, x_0\rangle$$

is affine, hence harmonic. Therefore we may apply (4.8) with $h(y) = f(y)g_{x_0}(y)$. This gives

$$f(x_0)g_{x_0}(x_0)e^{-\pi|x_0|^2} \le \int_{\mathbb{R}^m} f(y)g_{x_0}(y)e^{-\pi|y|^2}\, dA(y).$$

Since

$$g_{x_0}(x_0) = e^{-\frac{\pi}{2}|x_0|^2 + \pi|x_0|^2} = e^{\frac{\pi}{2}|x_0|^2},$$

the left-hand side simplifies to

$$f(x_0)e^{-\frac{\pi}{2}|x_0|^2}.$$

Therefore

$$f(x_0)e^{-\frac{\pi}{2}|x_0|^2} \le \int_{\mathbb{R}^m} f(y)g_{x_0}(y)e^{-\pi|y|^2}\, dA(y),$$

and substituting this into the norm identity gives

$$\|f - g_{x_0}\|^2 \le 2 - 2f(x_0)e^{-\frac{\pi}{2}|x_0|^2}.$$

Since this holds for every $x_0 \in \mathbb{R}^m$, we obtain

$$\inf_{x_0\in\mathbb{R}^m} \|f - g_{x_0}\|^2 \le 2 - 2 \sup_{x\in\mathbb{R}^m} f(x)e^{-\frac{\pi}{2}|x|^2}.$$

Because a log-subharmonic function is nonnegative, we may write

$$\sup_{x\in\mathbb{R}^m} f(x)e^{-\frac{\pi}{2}|x|^2} = \left(\sup_{x\in\mathbb{R}^m} |f(x)|^2 e^{-\pi|x|^2}\right)^{1/2} = \sqrt{T}.$$

Hence

$$\inf_{x_0\in\mathbb{R}^m} \|f - g_{x_0}\|^2 \le 2 - 2\sqrt{T},$$

which is exactly (4.9). □

Let $\mu(t) = |A_t|$, where $A_t = \{x \in \mathbb{R}^m : f^2(x)e^{-\pi|x|^2} > t\}$ and $u^*\colon \mathbb{R}^+ \to (0, T]$ is the decreasing rearrangement of f, usually defined as

$$u^*(s) := \sup\{t \ge 0\colon \mu(t) > s\}, \qquad s \ge 0. \tag{4.10}$$

Similarly let v be the the decreasing rearrangement of f of $e^{-\pi/2|x|^2}$. Then

$$v^*(s) = e^{-\left(s\Gamma\left[1+\frac{m}{2}\right]\right)^{2/m}}$$

By proving the following differential inequality for the decreasing rearrangement $u^*(s)$ (i.e., the inverse function of $\mu(t)$) in the same way as in [5]:

$$(u^*)'(s) + \frac{2(\Gamma[m/2+1]\,!)^{\frac{2}{m}}\ u^*(s)}{m\ s^{1-\frac{2}{m}}} \ge 0, \quad \text{for a.e. } s > 0. \tag{4.11}$$

we can deduce the following counterpart of [5][7.24]

$$\frac{(1-T)^{m/2+1}}{\Gamma((m/2+2))} \le \int_0^{s^*} (v^*(s) - u^*(s))\,\mathrm{d}s \le \frac{\delta_{s_0}}{\int_{s_0}^\infty v^*(s)\,\mathrm{d}s}. \tag{4.12}$$

Then in view of Lemma 4.4, in the same way as in [5] we can prove the following theorem

Theorem 4.5 (Higher dimensional version of Theorem 4.2)**.** *There is a computable constant $C = C(m) > 0$ such that, for all measurable sets $\Omega \subset \mathbb{R}^m$ with finite measure $|\Omega| > 0$ and all real-analytic log-subharmonic functions $f \in \mathcal{F}^2(\mathbb{R}^m)\backslash\{0\}$, we have*

$$\min_{\substack{|c|=\|f\|_{\mathcal{F}^2},\\ x_0\in\mathbb{R}^m}} \frac{\|f - cg_{x_0}\|_{\mathcal{F}^2}}{\|f\|_{\mathcal{F}^2}} \le C\left(\frac{\delta(f;\Omega)}{\int_{|\Omega|}^\infty e^{-\left(s\Gamma\left[1+\frac{m}{2}\right]\right)^{2/m}}\,\mathrm{d}s}\right)^{\frac{1}{2+m}}, \tag{4.13}$$

where

$$\delta(f;\Omega) := 1 - \frac{\int_\Omega |f(x)|^2 e^{-\pi|x|^2}\,\mathrm{d}x}{\|f\|^2_{\mathcal{F}^2}\int_0^{|\Omega|} v^*(s)\,\mathrm{d}s}. \tag{4.14}$$

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References

[1] E. Schmidt, *Über die isoperimetrische Aufgabe im n-dimensionalen Raum konstanter negativer Krümmung. I. Die isoperimetrischen Ungleichungen in der hyperbolischen Ebene und für Rotationskörper im n-dimensionalen hyperbolischen Raum*, *Mathematische Zeitschrift* **46** (1940), 204–230.

[2] A. Baernstein, II, *Symmetrization in analysis*, volume 36 of *New Mathematical Monographs*, Cambridge University Press, Cambridge, 2019. With David Drasin and Richard S. Laugesen, with a foreword by Walter Hayman.

[3] V. Bögelein, F. Duzaar, and C. Scheven, *A sharp quantitative isoperimetric inequality in hyperbolic n-space*, *Calculus of Variations and Partial Differential Equations* **54** (2015), no. 5, 3967–4017.

[4] J. Gómez, D. Kalaj, P. Melentijević, and J. P. G. Ramos, *Uniform stability of concentration inequalities and applications*, *Proc. Lond. Math. Soc. (3)* **131** (2025), no. 6, Article ID e70114, 53 pp.

[5] J. Gómez, A. Guerra, J. P. G. Ramos, and P. Tilli, *Stability of the Faber–Krahn inequality for the short-time Fourier transform*, *Invent. Math.* **236** (2024), no. 2, 779–836.

[6] D. Kalaj, *Contraction property of certain classes of log-M-subharmonic functions in the unit ball*, *J. Funct. Anal.* **286** (2024), no. 1, Article ID 110203, 29 pp.

[7] D. Kalaj and J. P. G. Ramos, *A Faber–Krahn type inequality for log-subharmonic functions in the hyperbolic ball*, *Israel J. Math.* (2025), published online 29 June 2025. doi:10.1007/s11856-025-2788-0. Preprint: arXiv:2303.08069.
[8] A. Kulikov, *Functionals with extrema at reproducing kernels*, *Geom. Funct. Anal.* **32** (2022), no. 4, 938–949. doi:10.1007/s00039-022-00608-5.
[9] A. Kulikov, F. Nicola, J. Ortega-Cerdà, and P. Tilli, *A monotonicity theorem for subharmonic functions on manifolds*, *Adv. Math.* **479** (2025), 110423. doi:10.1016/j.aim.2025.110423. Preprint: arXiv:2212.14008.
[10] E. H. Lieb and M. Loss, *Analysis*, 2nd ed., Graduate Studies in Mathematics, Vol. 14, American Mathematical Society, Providence, RI, 2001.
[11] P. Melentijević *Sharp stability of convex functionals on weighted Bergman spaces with applications*, arXiv:2508.01939v4
[12] F. Nicola and P. Tilli, *The Faber–Krahn inequality for the short-time Fourier transform*, *Inventiones mathematicae* **230** (2022), 1–30.
[13] J. P. G. Ramos and P. Tilli, *A Faber–Krahn inequality for wavelet transforms*, *Bull. Lond. Math. Soc.* **55** (2023), no. 4, 2018–2034. doi:10.1112/blms.12833.
[14] W. Rudin, *Function theory in the unit ball of* $\mathbb{C}^n$, reprint of the 1980 original, Classics in Mathematics, Springer, Berlin, 2008.
[15] M. Stoll, *Harmonic and subharmonic function theory on the hyperbolic ball*, London Mathematical Society Lecture Note Series 431, Cambridge University Press, Cambridge, 2016.

Mathematics section, EPFL, Lausanne, Switzerland.

Email address: jaime.gomezramirez@epfl.ch

University of Montenegro, Faculty of natural sciences and mathematics, Podgorica, Cetinjski put b.b. 81000 Podgorica, Montenegro

Email address: davidk@ucg.ac.me

University of Belgrade, Faculty of Mathematics, 11000, Belgrade, Serbia

Email address: petarmel@matf.bg.ac.rs

Estrada Dona Castorina 110, Horto, Rio de Janeiro, RJ, Brazil.

Email address: joao.ramos@impa.br